# آمار و احتمال مهندسی

# Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

**Complete Lecture Notes**

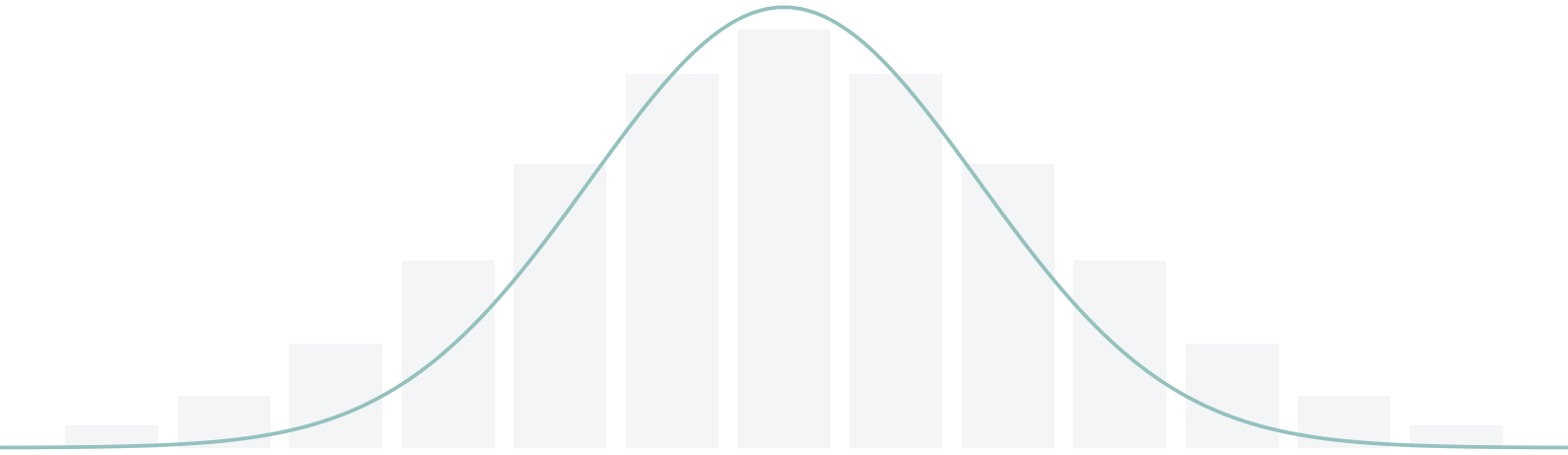
## Modules 1-16

جزوه کامل درس - ماژول‌های ۱ تا ۱۶

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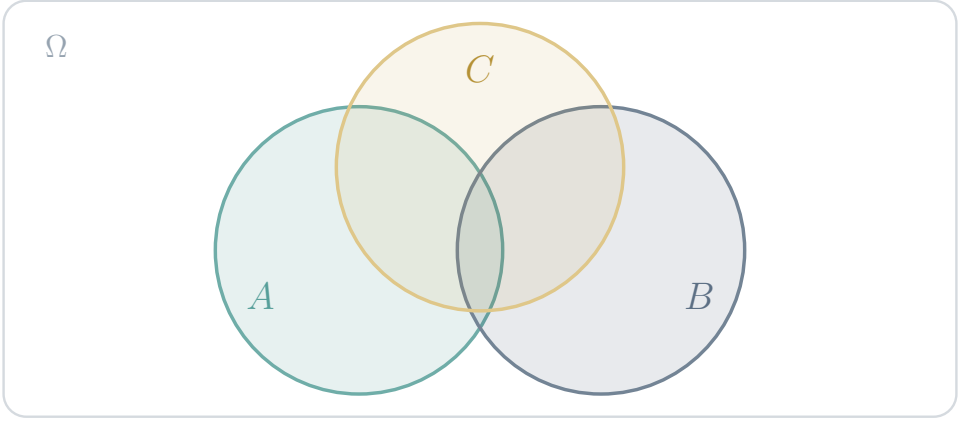
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Module 1

# Probability

احتمال

1



## In This Module


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**Learning Objectives**

After completing this module, students will be able to:

1. Define random experiments and identify sample spaces for engineering scenarios
2. Apply set operations to define complex events
3. Use probability axioms to derive probability rules
4. Compute conditional probabilities
5. Determine statistical independence
6. Apply Bayes' theorem to engineering inference problems
7. Use the total probability theorem for multi-path analysis
8. Analyze random experiments with probabilistic models

# 1.1 Introduction: Why Probability in Engineering?

## The Engineering Reality

No engineering system operates in a perfectly deterministic world. Consider:

- A digital communication system transmits bits, but **noise** corrupts some bits randomly
- A manufacturing line produces resistors, but their values **vary** around the nominal
- A sensor measures temperature, but each reading has **random error**
- A wireless signal travels to a receiver, but **fading** causes random attenuation

**Probability theory** provides the mathematical framework to model, analyze, and design systems that operate under uncertainty.

## Engineering Question

> "If I transmit 1000 bits over a noisy channel, how many errors should I expect? Can I guarantee fewer than 5 errors?"

To answer this, we need:

1. A model of the random phenomenon (bit errors)
2. Mathematical tools to compute probabilities
3. Methods to make engineering decisions based on those probabilities

# 1.2 Random Experiments

**Definition**

A **random experiment** is a procedure that:

1. Can be repeated under identical conditions
2. Has a well-defined set of possible outcomes
3. Has an outcome that cannot be predicted with certainty before the experiment is performed

## Engineering Examples

| Experiment | Context | Uncertainty Source |
|---|---|---|
| Transmit one bit over a noisy channel | Communications | Thermal noise |
| Measure a resistor's value | Electronics | Manufacturing variation |
| Count packets arriving in 1 second | Networking | Traffic randomness |
| Measure received signal power | Wireless | Fading, interference |
| Test a component until failure | Reliability | Material defects, wear |
| Sample ADC output voltage | Signal processing | Quantization + noise |

## Key Distinction

| Deterministic | Random |
|---|---|
| Output of a logic gate with known inputs | Output of a logic gate with noisy inputs |
| Resistance calculated from color code | Actual measured resistance |
| Free-space path loss at known distance | Actual received power with fading |

# 1.3 Sample Spaces

**Definition**

The **sample space** $S$ (or $\Omega$) is the set of all possible outcomes of a random experiment.

## Types of Sample Spaces

**Discrete (Countable):**

- Transmit one bit: $S = \{0, 1\}$ (received correctly or in error)
- Number of errors in a packet of $n$ bits: $S = \{0, 1, 2, \ldots, n\}$
- Number of arrivals at a server: $S = \{0, 1, 2, 3, \ldots\}$

**Continuous (Uncountable):**

- Measured voltage: $S = \mathbb{R}$ (or a practical range like $[0, 5]V$)
- Time until component failure: $S = [0, \infty)$
- Phase of received signal: $S = [0, 2\pi)$

**Engineering Example: Communication System**

A transmitter sends a bit (0 or 1) over a noisy channel. The receiver makes a decision.

Sample space for (transmitted, received) pairs:

$$S = \{(0,0), (0,1), (1,0), (1,1)\}$$

| | **received 0** | **received 1** |
|---|---|---|
| **transmitted 0** | $(0,0)$ correct | $(0,1)$ error |
| **transmitted 1** | $(1,0)$ error | $(1,1)$ correct |

# 1.4 Events

**Definition**

An **event** is a subset of the sample space. An event **occurs** if the outcome of the experiment belongs to that subset.

## Types of Events

- **Simple event**: Contains exactly one outcome. $E = \{(0, 1)\}$
- **Compound event**: Contains multiple outcomes. $E = \{(0, 1), (1, 0)\}$ (any error)
- **Certain event**: The entire sample space $S$ (always occurs)
- **Impossible event**: The empty set $\emptyset$ (never occurs)

## Set Operations on Events

| Operation | Notation | Meaning | Engineering Interpretation |
|---|---|---|---|
| Union | $A \cup B$ | A or $B$ (or both) occurs | Either failure mode A or $B$ triggers alarm |
| Intersection | $A \cap B$ | Both A and $B$ occur | Both sensors detect the signal |
| Complement | $A^c$ or $\bar{A}$ | A does not occur | Component does NOT fail |
| Difference | $A - B = A \cap B^c$ | A occurs but not $B$ | Sensor A triggers but sensor $B$ doesn't |
| Mutually exclusive | $A \cap B = \emptyset$ | A and $B$ cannot both occur | Component can't be both working and failed |

**Engineering Example: Manufacturing Defects**

A circuit board inspection checks for three types of defects:

- A = solder bridge defect
- $B$ = missing component
- $C$ = wrong component value

Events:

- $A \cup B \cup C$ = "board has at least one defect"
- $A \cap B$ = "board has both a solder bridge AND a missing component"
- $(A \cup B \cup C)^c$ = "board is defect-free"

# 1.5 Probability Axioms

## Kolmogorov's Axioms

For a sample space $S$ with events defined on it, a probability function $P$ satisfies:

**Axiom 1 (Non-negativity):**

$$P(A) \geq 0 \quad \text{for any event } A$$

**Axiom 2 (Normalization):**

$$P(S) = 1$$

**Axiom 3 (Countable Additivity):**
For mutually exclusive events $A_1$, $A_2$, $A_3$, ...:

$$P(A_1 \cup A_2 \cup A_3 \cup \cdots) = P(A_1) + P(A_2) + P(A_3) + \cdots$$

## Properties Derived from Axioms

**Property 1: Complement Rule**

$$P(A^c) = 1 - P(A)$$

*Proof:* $S = A \cup A^c$ and $A \cap A^c = \emptyset$, so $P(S) = P(A) + P(A^c) = 1$.

*Engineering meaning:* $P(\text{component works}) = 1 - P(\text{component fails})$

**Property 2: Impossible Event**

$$P(\emptyset) = 0$$

**Property 3: Monotonicity**
If $A \subseteq B$, then $P(A) \leq P(B)$

**Property 4: Bound**

$$0 \leq P(A) \leq 1$$

**Property 5: Inclusion-Exclusion (Two Events)**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

*Engineering meaning:* When computing the probability of "failure $A$ OR failure $B$," we must subtract the overlap where both fail simultaneously.

# 1.6 Probability Rules

## Addition Rule

For any two events:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

For mutually exclusive events ($A \cap B = \emptyset$):

$$P(A \cup B) = P(A) + P(B)$$

## Complement Rule

$$P(A) = 1 - P(A^c)$$

This is often the easiest way to compute probabilities:

> $P(\text{at least one error in 100 bits}) = 1 - P(\text{zero errors in 100 bits})$

## Inclusion-Exclusion (Three Events)

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

**Engineering Example: Network Redundancy**

A data center has three independent network links. Each link fails with probability 0.05.

- A = link 1 fails, $B$ = link 2 fails, $C$ = link 3 fails
- $P(\text{at least one fails}) = P(A \cup B \cup C)$
- Using inclusion-exclusion:

$P(A \cup B \cup C) = 3(0.05) - 3(0.05)^2 + (0.05)^3 = 0.15 - 0.0075 + 0.000125 = 0.142625$
Alternatively: $P(\text{at least one fails}) = 1 - P(\text{none fail}) = 1 - (0.95)^3 = 0.142625$ ✔

# 1.7 Conditional Probability

**Definition**

The **conditional probability** of event A given that event $B$ has occurred:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

## Interpretation

Conditional probability **updates** our knowledge. Once we know $B$ occurred, the effective sample space shrinks from $S$ to $B$, and we ask: "Within $B$, how likely is $A$?"

## Key Distinction

| Concept | Notation | Meaning |
|---|---|---|
| Probability of $A$ | $P(A)$ | Likelihood of A with no additional information |
| Conditional probability | $P(A \mid B)$ | Likelihood of $A$ **given** that $B$ has occurred |
| Joint probability | $P(A \cap B)$ | Likelihood that **both** A and $B$ occur |

These are related: $P(A \cap B) = P(A \mid B) \cdot P(B) = P(B \mid A) \cdot P(A)$

**Engineering Example: Detection Theory**

A radar system detects aircraft:

- $H_1$ = aircraft is present (target)
- $H_0$ = aircraft is absent (no target)
- $D$ = detector says "target present"

Important conditional probabilities:

- $P(D \mid H_1)$ = **probability of detection** (sensitivity)
- $P(D \mid H_0)$ = **probability of false alarm**
- $P(D^c \mid H_1)$ = **probability of miss** $= 1 - P(D \mid H_1)$

Suppose: $P(H_1) = 0.01$, $P(D \mid H_1) = 0.95$, $P(D \mid H_0) = 0.02$
What is $P(\text{target actually present} \mid \text{detector says present})$?
This requires **Bayes' theorem** (Section 1.9).

**Engineering Example: Communication Channel**

A binary symmetric channel has crossover probability $p = 0.01$:

- $P(\text{receive } 1 \mid \text{sent } 0) = p = 0.01$
- $P(\text{receive } 0 \mid \text{sent } 1) = p = 0.01$
- $P(\text{receive } 0 \mid \text{sent } 0) = 1 - p = 0.99$
- $P(\text{receive } 1 \mid \text{sent } 1) = 1 - p = 0.99$

If bits are equally likely to be 0 or 1:

- $P(\text{sent } 0 \cap \text{receive } 1) = P(\text{receive } 1 \mid \text{sent } 0) \cdot P(\text{sent } 0) = 0.01 \times 0.5 = 0.005$
- $P(\text{error}) = P(\text{sent } 0 \cap \text{receive } 1) + P(\text{sent } 1 \cap \text{receive } 0) = 0.005 + 0.005 = 0.01$

## Properties of Conditional Probability

Conditional probability is itself a valid probability measure:

1. $P(A \mid B) \geq 0$
2. $P(S \mid B) = 1$
3. $P(A_1 \cup A_2 \mid B) = P(A_1 \mid B) + P(A_2 \mid B)$ if $A_1 \cap A_2 = \emptyset$

## Multiplication Rule (Chain Rule)

$$P(A \cap B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

For three events:

$$P(A \cap B \cap C) = P(A) \cdot P(B|A) \cdot P(C|A \cap B)$$

# 1.8 Independence

**Definition**

Events A and $B$ are **statistically independent** if:

$$P(A \cap B) = P(A) \cdot P(B)$$

Equivalently: $P(A \mid B) = P(A)$ — knowing $B$ occurred gives no information about $A$.

## Engineering Significance

Independence means **one event provides no information about the other**. This is a modeling assumption we often make:

- Noise samples at different times are independent
- Bit errors in different time slots are independent (memoryless channel)
- Component failures in different subsystems are independent

## Pairwise vs. Mutual Independence

For three events $A$, $B$, $C$:

**Pairwise independence** (all three pairs):

- $P(A \cap B) = P(A)P(B)$
- $P(A \cap C) = P(A)P(C)$
- $P(B \cap C) = P(B)P(C)$

**Mutual independence** (all combinations):

- All pairwise conditions above, AND
- $P(A \cap B \cap C) = P(A)P(B)P(C)$

⚠ Pairwise independence does NOT imply mutual independence!

**Engineering Example: Independent Component Failures**

A system has three independent modules, each with failure probability 0.1.
$P(\text{all three fail}) = P(A)P(B)P(C) = (0.1)^3 = 0.001$
$P(\text{system works}) = P(\text{at least one works}) = 1 - P(\text{all fail}) = 1 - 0.001 = 0.999$
This calculation is ONLY valid because failures are independent.

## Testing Independence

To verify independence from data:

1. Compute $P(A)$, $P(B)$, and $P(A \cap B)$ empirically
2. Check if $P(A \cap B) \approx P(A) \cdot P(B)$

If not approximately equal, the events are **dependent**.

## Common Misconception

> ✘ "Mutually exclusive events are independent"

This is **FALSE**. If $A \cap B = \emptyset$ and $P(A) > 0,\ P(B) > 0$, then:

- $P(A \cap B) = 0$
- $P(A)P(B) > 0$
- Therefore $P(A \cap B) \neq P(A)P(B)$

Mutually exclusive events are actually **maximally dependent** — if one occurs, the other cannot.

# 1.9 Bayes' Theorem

## Derivation

From the definition of conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A) \cdot P(A)}{P(B)}$$

## Full Statement

If $B_1, B_2, \ldots, B_n$ partition the sample space (mutually exclusive and exhaustive):

$$P(B_i|A) = \frac{P(A|B_i) \cdot P(B_i)}{\sum_{j=1}^{n} P(A|B_j) \cdot P(B_j)}$$

## Terminology

| Term | Name | Engineering Meaning |
|---|---|---|
| $P(B_i)$ | Prior probability | What we believed before observing data |
| $P(A \mid B_i)$ | Likelihood | How likely is the observation given each hypothesis |
| $P(B_i \mid A)$ | Posterior probability | Updated belief after observing data |
| $P(A)$ | Evidence | Total probability of the observation |

### Engineering Example: Fault Diagnosis

A circuit board can have a fault in one of three subsystems:

- $B_1$ = power supply fault, $P(B_1) = 0.3$
- $B_2$ = processor fault, $P(B_2) = 0.5$
- $B_3$ = memory fault, $P(B_3) = 0.2$

A diagnostic test produces symptom $A$ (overheating):

- $P(A \mid B_1) = 0.8$ (power supply faults usually cause overheating)
- $P(A \mid B_2) = 0.3$ (processor faults sometimes cause overheating)
- $P(A \mid B_3) = 0.1$ (memory faults rarely cause overheating)

**Question:** Given that overheating is observed, what is the most likely fault?

**Solution**

$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2) + P(A \mid B_3)P(B_3)$

$= (0.8)(0.3) + (0.3)(0.5) + (0.1)(0.2)$

$= 0.24 + 0.15 + 0.02 = 0.41$

$P(B_1 \mid A) = \frac{(0.8)(0.3)}{0.41} = 0.585$ (most likely!)

$$P(B_2 \mid A) = \frac{(0.3)(0.5)}{0.41} = 0.366$$

$$P(B_3 \mid A) = \frac{(0.1)(0.2)}{0.41} = 0.049$$

**Engineering Decision:** Investigate the power supply first.

### Engineering Example: Detection with Bayes

Returning to the radar example from Section 1.7:

- $P(H_1) = 0.01$, $P(D \mid H_1) = 0.95$, $P(D \mid H_0) = 0.02$

$P(D) = P(D \mid H_1)P(H_1) + P(D \mid H_0)P(H_0) = (0.95)(0.01) + (0.02)(0.99) = 0.0095 + 0.0198 = 0.0293$

$$P(H_1 \mid D) = \frac{(0.95)(0.01)}{0.0293} = 0.324$$

**Interpretation:** Even with a good detector (95% detection, 2% false alarm), when targets are rare (1%), most detections are false alarms! Only 32.4% of alarms correspond to real targets.

**Engineering Lesson:** System design must account for **prior probabilities**, not just detector performance.

# 1.10 Total Probability Theorem

## Statement

If $B_1, B_2, \ldots, B_n$ form a **partition** of $S$ (mutually exclusive, exhaustive):

$$P(A) = \sum_{i=1}^{n} P(A|B_i) \cdot P(B_i)$$

## Interpretation

To find the total probability of event A, decompose into all possible "paths" to A through the partition.

**Engineering Example: Multi-Path Communication**

A packet can be routed through three different paths:

- Path 1: chosen 50% of the time, error probability 0.01
- Path 2: chosen 30% of the time, error probability 0.05
- Path 3: chosen 20% of the time, error probability 0.10

$P(\text{packet error}) = P(\text{error} \mid \text{path 1})P(\text{path 1}) + P(\text{error} \mid \text{path 2})P(\text{path 2}) + P(\text{error} \mid \text{path 3})P(\text{path 3})$
$= (0.01)(0.50) + (0.05)(0.30) + (0.10)(0.20)$
$= 0.005 + 0.015 + 0.020 = 0.040$
**Overall error rate: 4%**

**Engineering Example: Manufacturing with Multiple Suppliers**

A company sources capacitors from two suppliers:

- Supplier A: provides 60% of capacitors, defect rate 2%
- Supplier $B$: provides 40% of capacitors, defect rate 5%

$P(\text{defective}) = P(\text{defective} \mid A)P(A) + P(\text{defective} \mid B)P(B)$
$= (0.02)(0.60) + (0.05)(0.40)$
$= 0.012 + 0.020 = 0.032$
If a defective capacitor is found, which supplier likely produced it?
$P(A \mid \text{defective}) = \dfrac{(0.02)(0.60)}{0.032} = 0.375$
$P(B \mid \text{defective}) = \dfrac{(0.05)(0.40)}{0.032} = 0.625$
Despite providing fewer units, Supplier $B$ is more likely responsible for a given defect.

## 1.11 Summary: Key Distinctions

| Concept | Definition | Engineering Meaning |
|---|---|---|
| **Event** | A subset of the sample space | A specific outcome or set of outcomes we care about |
| **Probability** | $P(A)$ = measure of likelihood | Long-run relative frequency of occurrence |
| **Conditional Probability** | $P(A \mid B)$ = updated probability given $B$ | How knowledge of one event changes another's likelihood |
| **Independence** | $P(A \cap B) = P(A)P(B)$ | Knowing one event tells nothing about the other |

## 1.12 Practice Problems

### Problem 1: Component Reliability

A system has two independent processors. Processor 1 fails with probability 0.02, Processor 2 fails with probability 0.03. The system fails only if BOTH processors fail.

(a) What is the probability that the system fails?
(b) What is the probability that at least one processor fails?
(c) Given that the system is still working, what is the probability that processor 1 has failed?

**Solution**

(a) $P(\text{system fails}) = P(\text{both fail}) = (0.02)(0.03) = 0.0006$
(b) $P(\text{at least one fails}) = 1 - P(\text{none fail}) = 1 - (0.98)(0.97) = 1 - 0.9506 = 0.0494$
(c) System working = NOT(both fail) = at least one works.

$$P(P1 \text{ failed} \mid \text{system works}) = \frac{P(P1 \text{ failed} \cap P2 \text{ works})}{P(\text{system works})}$$

$$= \frac{(0.02)(0.97)}{1 - 0.0006} = \frac{0.0194}{0.9994} \approx 0.0194$$

### Problem 2: Sensor Detection

A fire detection system has: $P(\text{fire}) = 0.001$, $P(\text{alarm} \mid \text{fire}) = 0.98$, $P(\text{alarm} \mid \text{no fire}) = 0.01$.

(a) What is $P(\text{alarm})$?
(b) If the alarm sounds, what is $P(\text{fire} \mid \text{alarm})$?
(c) Comment on the result.

**Solution**

(a) $P(\text{alarm}) = P(\text{alarm} \mid \text{fire})P(\text{fire}) + P(\text{alarm} \mid \text{no fire})P(\text{no fire})$
$= (0.98)(0.001) + (0.01)(0.999) = 0.000980 + 0.009990 = 0.010970$

(b) $$P(\text{fire} \mid \text{alarm}) = \frac{P(\text{alarm} \mid \text{fire})P(\text{fire})}{P(\text{alarm})}$$
$$= \frac{(0.98)(0.001)}{0.01097} = 0.0894$$

(c) Only $\sim 9\%$ of alarms correspond to real fires! This is the base rate fallacy. Even with an excellent detector, rare events produce many false alarms.

### Problem 3: Communication Error

A packet of 8 bits is transmitted. Each bit has independent error probability 0.05.

(a) What is $P(\text{no errors in the packet})$?
(b) What is $P(\text{exactly one error})$?
(c) What is $P(\text{at least one error})$?

**Solution**

(a) $P(\text{no errors}) = (0.95)^8 = 0.6634$
(b) $P(\text{exactly one error}) = C(8,\,1)(0.05)^1(0.95)^7 = 8 \times 0.05 \times 0.6983 = 0.2793$
(c) $P(\text{at least one error}) = 1 - P(\text{no errors}) = 1 - 0.6634 = 0.3366$

### Problem 4: Total Probability Application

A wireless system operates in three modes:

- Mode 1 (good channel): 60% of time, $P(\text{outage}) = 0.01$
- Mode 2 (moderate channel): 30% of time, $P(\text{outage}) = 0.10$
- Mode 3 (poor channel): 10% of time, $P(\text{outage}) = 0.50$

(a) What is the overall outage probability?
(b) Given an outage occurred, what is the probability the system was in Mode 3?

**Solution**

(a) $P(\text{outage}) = (0.01)(0.60) + (0.10)(0.30) + (0.50)(0.10) = 0.006 + 0.030 + 0.050 = 0.086$
(b) $P(\text{Mode 3} \mid \text{outage}) = \dfrac{(0.50)(0.10)}{0.086} = 0.581$

### Problem 5: Independence Analysis

Two sensors monitor a process. Sensor A has false alarm probability 0.05. Sensor $B$ has false alarm probability 0.03.

(a) If sensors are independent, what is $P(\text{both give false alarm})$?
(b) If sensors share a noise source and $P(\text{both false alarm}) = 0.01$, are they independent?
(c) What is $P(A \text{ false alarm} \mid B \text{ false alarm})$ in case (b)?

**Solution**

(a) If independent: $P(\text{both}) = (0.05)(0.03) = 0.0015$
(b) If $P(\text{both}) = 0.01 \neq 0.0015$, they are NOT independent. The shared noise source creates dependence.
(c) $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{0.01}{0.03} = 0.333$

This is much higher than $P(A) = 0.05$, confirming strong dependence.

## 1.13 Key Takeaways

1. **Probability quantifies uncertainty** — essential for engineering design under uncertainty
2. **Conditional probability updates beliefs** — the foundation of detection, estimation, and diagnosis
3. **Independence simplifies analysis** — but must be justified, not assumed
4. **Bayes' theorem inverts conditional probabilities** — from "likelihood of evidence given cause" to "likelihood of cause given evidence"
5. **Base rates matter** — even good detectors produce false alarms when events are rare

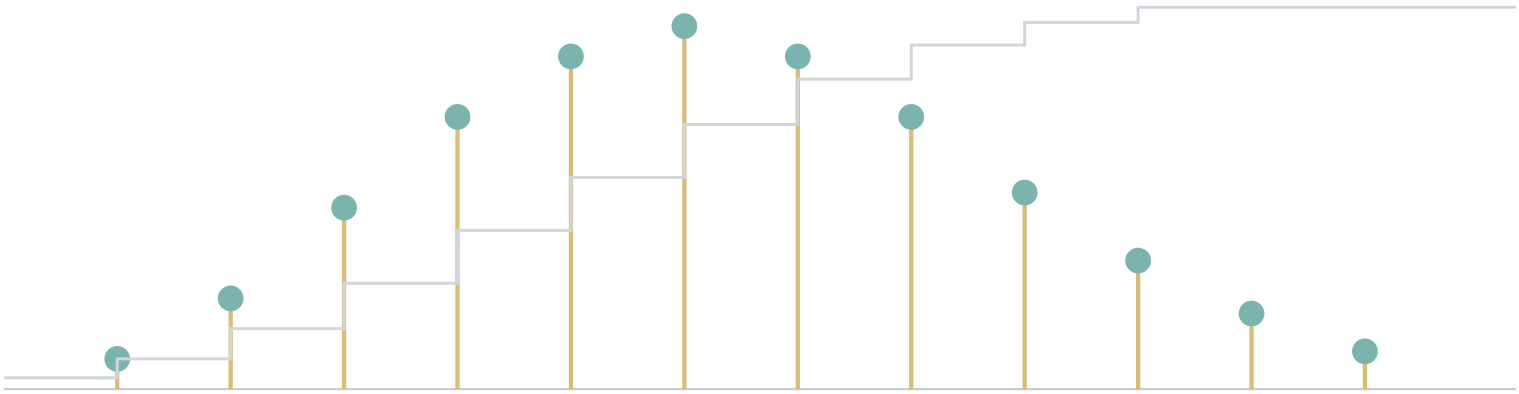
*Next Module: Module 2 — One Random Variable and Its Functions*

Module 2

# One Random Variable and Its Functions

یک متغیر تصادفی و توابع آن

2



## In This Module


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**Learning Objectives**

After completing this module, students will be able to:

1. Define random variables and explain their role in engineering analysis
2. Distinguish between discrete and continuous random variables
3. Compute and interpret PMF, PDF, and CDF
4. Clearly explain the difference between probability mass, density, and cumulative probability
5. Apply transformations to random variables
6. Analyze and visualize random variables

# 2.1 Introduction: Why Random Variables?

## The Problem

In Module 1, we worked with events — subsets of a sample space. But engineering analysis requires **numbers**:

- How much noise voltage is present?
- How many packets arrived?
- What is the received signal power?

A **random variable** maps outcomes of a random experiment to real numbers, enabling us to use all the tools of calculus and algebra.

## Engineering Motivation

| We want to analyze… | We define a random variable… |
|---|---|
| Noise in a circuit | $X$ = noise voltage (volts) |
| Bit errors in a frame | $N$ = number of bit errors |
| Component lifetime | $T$ = time until failure (hours) |
| Signal amplitude | A = received amplitude $(V)$ |
| Packet delay | $D$ = end-to-end delay $(ms)$ |

# 2.2 Definition of a Random Variable

## Formal Definition

A **random variable** $X$ is a function that maps each outcome $\omega$ in the sample space $S$ to a real number:

$$X : S \to \mathbb{R}$$

$$\omega \mapsto X(\omega)$$

## Interpretation

The random variable assigns a numerical value to each possible outcome of the experiment. The randomness comes from which outcome $\omega$ occurs — once $\omega$ is determined, $X(\omega)$ is a definite number.

## Engineering Example

**Experiment:** Transmit 5 bits over a noisy channel.
**Sample space:** $S$ = all possible sequences of correct/error for 5 bits ($2^5 = 32$ outcomes)
**Random variable:** $X$ = number of bit errors

| Outcome $\omega$ | $X(\omega)$ |
|---|---|
| $(C, C, C, C, C)$ | 0 |
| $(E, C, C, C, C)$ | 1 |
| $(E, E, C, C, C)$ | 2 |
| … | … |
| $(E, E, E, E, E)$ | 5 |

Now instead of working with 32 outcomes, we work with a single number $X \in \{0, 1, 2, 3, 4, 5\}$.

## 2.3 Discrete Random Variables

**Definition**

A random variable $X$ is **discrete** if it takes values from a **countable** set (finite or countably infinite).

### Characteristics

- Values can be listed: $\{x_1, x_2, x_3, \ldots\}$
- Each value has a specific probability (a "mass" of probability)
- Sum of all probabilities = 1

### Engineering Examples

| Random Variable | Possible Values | Context |
|---|---|---|
| Number of bit errors | $\{0, 1, 2, \ldots, n\}$ | Packet transmission |
| Number of retransmissions | $\{0, 1, 2, \ldots\}$ | ARQ protocol |
| Quantized voltage level | $\{0, 1, \ldots, 2^N - 1\}$ | N-bit ADC output |
| Number of arrivals per second | $\{0, 1, 2, \ldots\}$ | Network traffic |
| Number of defective chips | $\{0, 1, \ldots, n\}$ | Wafer testing |

# 2.4 Continuous Random Variables

**Definition**

A random variable $X$ is **continuous** if it takes values from an **uncountable** set (interval of real numbers).

## Characteristics

- $X$ can take any value in a range (e.g., $[0, \infty)$ or $(-\infty, \infty)$)
- Probability of any single exact value is ZERO: $P(X = x) = 0$
- Probabilities are defined for intervals: $P(a \leq X \leq b)$

## Engineering Examples

| Random Variable | Range | Context |
|---|---|---|
| Thermal noise voltage | $(-\infty, \infty)$ | Circuit noise |
| Time until failure | $[0, \infty)$ | Reliability |
| Phase of received signal | $[0, 2\pi)$ | Communications |
| Resistor value | $(R_0 - \delta, R_0 + \delta)$ | Manufacturing |
| Channel gain | $[0, \infty)$ | Wireless fading |

## Why $P(X = x) = 0$ for Continuous RVs

Imagine measuring a voltage. The probability that $V$ = exactly 3.14159265… volts (infinite precision) is zero. We can only ask about intervals: $P(3.14 < V < 3.15)$.

# 2.5 Probability Mass Function (PMF)

**Definition**

For a discrete random variable $X$, the **probability mass function** is:

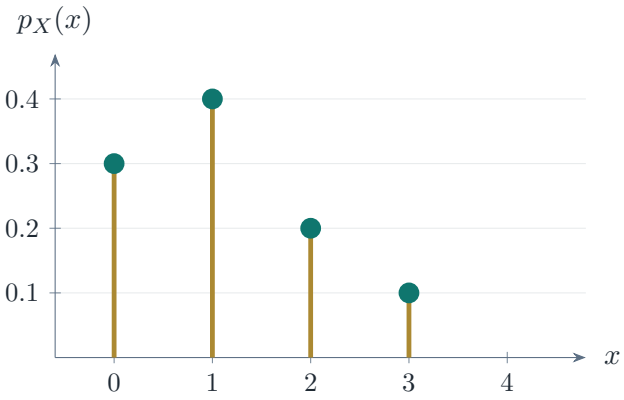
$$p_X(x) = P(X = x)$$

## Properties

1. **Non-negativity:** $p_X(x) \geq 0$ for all $x$
2. **Normalization:** $\sum p_X(x_i) = 1$ (sum over all possible values)
3. **Probability computation:** $P(X \in A) = \sum_{x \in A} p_X(x)$

## Visualization

The PMF is represented as **vertical lines** (impulses) at each possible value, with height equal to the probability.




## Engineering Example: Bit Errors in a Byte

Transmit 8 bits, each with independent error probability $p = 0.1$.
$X$ = number of bit errors.

$p_X(k) = C(8,\, k) \cdot (0.1)^k \cdot (0.9)^{8-k}$, for $k = 0,\, 1,\, \ldots,\, 8$

| $k$ | $p_X(k)$ |
|---|---|
| 0 | 0.4305 |
| 1 | 0.3826 |
| 2 | 0.1488 |
| 3 | 0.0331 |
| 4 | 0.0046 |
| 5+ | $< 0.001$ |

# 2.6 Probability Density Function (PDF)

**Definition**

For a continuous random variable $X$, the **probability density function** $f_X(x)$ is defined such that:

$$P(a \leq X \leq b) = \int_a^b f_X(x)\,dx$$

## Properties

1. **Non-negativity:** $f_X(x) \geq 0$ for all $x$
2. **Normalization:** $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$
3. **Probability = Area under curve** between two points

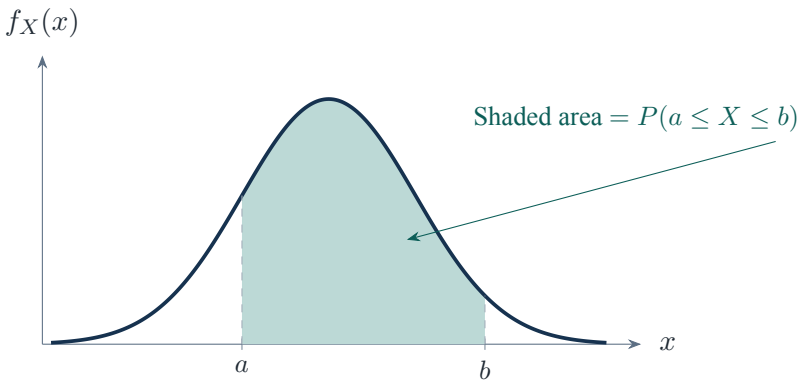
## Critical Point: $f_X(x)$ is NOT a Probability!

⚠ $f_X(x)$ can be GREATER than 1!

The PDF gives probability **density** — probability per unit length. Only the **integral** (area) gives probability.

**Analogy:** Mass density (kg/m$^3$) is not mass. You need to integrate density over volume to get mass. Similarly, probability density is not probability — you integrate over an interval to get probability.

## Visualization




Total area under curve $= 1$

**Engineering Example: Thermal Noise Voltage**

Thermal noise voltage in a circuit often follows a Gaussian distribution:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/(2\sigma^2)}$$

where $\sigma$ is the RMS noise voltage.

$P(\text{noise exceeds } 2\sigma) = P(|X| > 2\sigma) = 1 - P(-2\sigma \leq X \leq 2\sigma) \approx 1 - 0.9545 = 0.0455$

# 2.7 Cumulative Distribution Function (CDF)

**Definition**

For any random variable $X$ (discrete or continuous), the **cumulative distribution function** is:

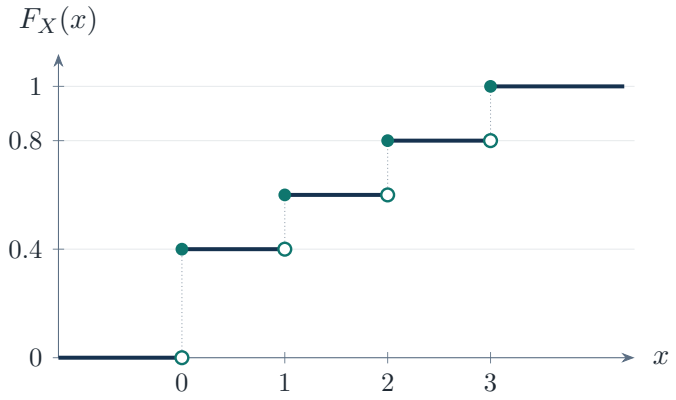
$$F_X(x) = P(X \leq x)$$

## Properties

1. **Non-decreasing:** if $a < b$, then $F_X(a) \leq F_X(b)$
2. **Limits:** $F_X(-\infty) = 0$, $F_X(+\infty) = 1$
3. **Right-continuous:** $F_X(x)$ is continuous from the right
4. **Probability of interval:** $P(a < X \leq b) = F_X(b) - F_X(a)$

## CDF for Discrete RVs

The CDF is a **staircase function** with jumps at each possible value:




Jump at $x = k$ has height $p_X(k)$.

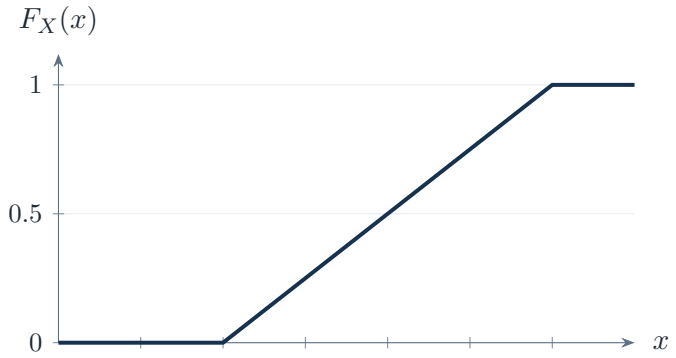
## CDF for Continuous RVs

The CDF is a **smooth, continuous** curve:

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt$$

And conversely:

$$f_X(x) = \frac{dF_X(x)}{dx}$$




## ■ Relationship: PDF ↔ CDF

| Direction | Formula | Meaning |
|---|---|---|
| PDF → CDF | $F_X(x) = \int_{-\infty}^{x} f_X(t)dt$ | Integrate density to get cumulative probability |
| CDF → PDF | $f_X(x) = \frac{dF_X}{dx}$ | Differentiate CDF to get density |

# 2.8 KEY DISTINCTION: Mass vs. Density vs. Cumulative

This is one of the most important conceptual distinctions in probability theory.

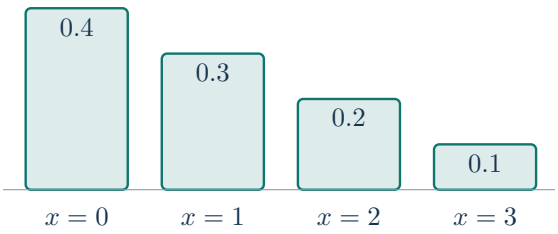
## Comparison Table

| Property | PMF $p_X(x)$ | PDF $f_X(x)$ | CDF $F_X(x)$ |
|---|---|---|---|
| **Applies to** | Discrete RVs only | Continuous RVs only | Both |
| **Gives directly** | $P(X = x)$ | Probability density at $x$ | $P(X \leq x)$ |
| **Range of values** | [0, 1] | $[0, \infty)$ — can exceed 1! | [0, 1] |
| **Sum/Integral** | $\sum p_X(x) = 1$ | $\int f_X(x)dx = 1$ | Starts at 0, ends at 1 |
| **Get probability** | $P(X = x) = p_X(x)$ | $P(a \leq X \leq b) = \int f_X(x)dx$ | $P(a < X \leq b) = F(b) - F(a)$ |

## Visual Analogy: The Warehouse

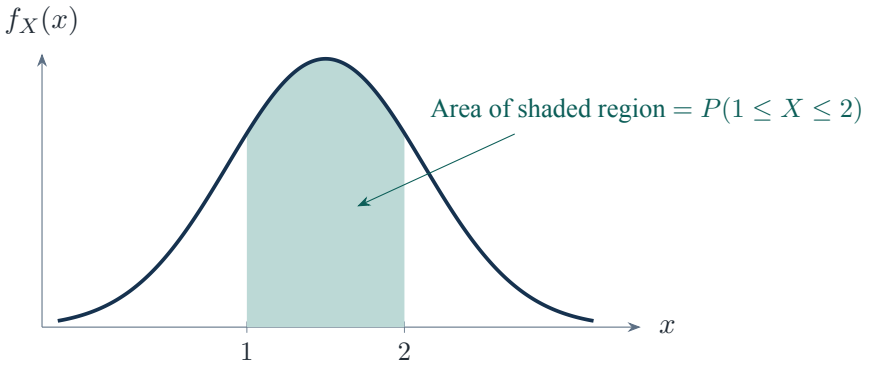
Imagine probability as "material" distributed across possible values:

**Discrete (PMF):** Like storing material in numbered boxes. Each box has a specific mass. The PMF tells you the mass in each box.




Total mass in all boxes $= 1$

**Continuous (PDF):** Like spreading material along a continuous shelf. The density tells you how thick the material is at each point. You find mass by computing area (thickness $\times$ width).




Total area under curve $= 1$

**Cumulative (CDF):** Like a running total — how much material have you accumulated up to this point?

## Why This Matters in Engineering

**Example:** A noise voltage $X$ has PDF $f_X(x) = 2$ for $x \in [0,\, 0.5]$.

- $f_X(0.3) = 2$ — this is NOT a probability! It means the density is $2\,\text{V}^{-1}$.
- $P(0.2 \le X \le 0.4) = \int_{0.2}^{0.4} 2\,dx = 2 \times 0.2 = 0.4$ — THIS is a probability.
- $P(X = 0.3) = 0$ — a single point has zero probability for continuous RVs.

## Common Errors to Avoid

| ✘ Wrong Statement | ✔ Correct Statement |
|---|---|
| "The probability at $x = 2$ is $f(2)$" | "The probability density at $x = 2$ is $f(2)$" |
| "$f(x)$ must be $\le 1$" | "$f(x)$ can be any non-negative value" |
| "$P(X = 3) = f(3)$ for continuous $X$" | "$P(X = 3) = 0$ for continuous $X$" |
| "PMF and PDF are the same thing" | "PMF gives mass; PDF gives density" |

# 2.9 Transformations of Random Variables

## Motivation

In engineering, we often know the distribution of one quantity but need the distribution of a **function** of that quantity:

- Know voltage $V \rightarrow$ need power $P = \frac{V^2}{R}$
- Know linear signal $X \rightarrow$ need dB value $Y = 10 \cdot \log_{10}(X)$
- Know noise $N \rightarrow$ need $|N|$ (rectified noise)

**Problem Statement**

Given: $X$ with known PDF $f_X(x)$ or PMF $p_X(x)$
Find: The distribution of $Y = g(X)$

## Case 1: Discrete RV Transformation

If $X$ is discrete with PMF $p_X(x)$, and $Y = g(X)$:

$$p_Y(y) = \sum_{x:g(x)=y} p_X(x)$$

Sum the probabilities of all $x$ values that map to the same $y$.

**Example:** $X \in \{-2, -1, 0, 1, 2\}$ with equal probability 0.2 each.
$Y = X^2$

| $Y = X^2$ | $X$ values mapping to $Y$ | $p_Y(y)$ |
|---|---|---|
| 0 | {0} | 0.2 |
| 1 | {-1, 1} | 0.4 |
| 4 | {-2, 2} | 0.4 |

## Case 2: Monotonic Transformation (Continuous)

If $Y = g(X)$ where $g$ is strictly monotonic and differentiable:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{dg^{-1}(y)}{dy} \right|$$

Or equivalently:

$$f_Y(y) = \left. \frac{f_X(x)}{|g'(x)|} \right|_{x=g^{-1}(y)}$$

**Derivation via CDF method:**

1. Start with CDF: $F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$
2. Invert: $= P(X \leq g^{-1}(y))$ if $g$ is increasing
3. Differentiate: $f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy}[g^{-1}(y)]$

## Case 3: Non-Monotonic Transformation

If $g$ is not monotonic, split the domain into intervals where $g$ IS monotonic:

$$f_Y(y) = \sum_i \frac{f_X(x_i)}{|g'(x_i)|}$$

where $x_1$, $x_2$, … are all solutions of $g(x) = y$.

**Engineering Example: Power from Voltage**

A noise voltage $X$ has PDF $f_X(x)$ (e.g., Gaussian with zero mean).
Power dissipated in a 1Ω resistor: $P = X^2$
Since $Y = X^2$ is NOT monotonic (both $+x$ and $-x$ give the same $y$):
For $y > 0$: $x = +\sqrt{y}$ and $x = -\sqrt{y}$ are both solutions.
$g(x) = x^2$, so $g'(x) = 2x$

$$f_Y(y) = \frac{f_X(\sqrt{y})}{2\sqrt{y}} + \frac{f_X(-\sqrt{y})}{2\sqrt{y}}, \quad y > 0$$

If $X$ is zero-mean Gaussian: $f_X(x) = f_X(-x)$, so:

$$f_Y(y) = \frac{f_X(\sqrt{y})}{\sqrt{y}}, \quad y > 0$$

**Engineering Example: Linear Amplification**

Signal $Y$ = aX $+b$ (amplifier with gain a and offset b).
$X$ has PDF $f_X(x)$.
$g^{-1}(y) = \frac{y-b}{a}$, and $|\frac{d}{dy}\left[\frac{y-b}{a}\right]| = \frac{1}{|a|}$

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

## 2.10 Practice Problems

### Problem 1: Identifying RV Type

Classify each as discrete or continuous and state the range:

(a) Number of dropped calls in one hour
(b) Received signal strength (dBm)
(c) Number of bits in error in a 1500-byte Ethernet frame
(d) Time between packet arrivals
(e) Quantizer output level (3-bit quantizer)

**Solution**

(a) Discrete, $X \in \{0, 1, 2, \ldots\}$
(b) Continuous, $X \in (-\infty, \infty)$ practically $\sim (-120, 0)$ dBm
(c) Discrete, $X \in \{0, 1, 2, \ldots, 12000\}$
(d) Continuous, $T \in [0, \infty)$
(e) Discrete, $X \in \{0, 1, 2, 3, 4, 5, 6, 7\}$

### Problem 2: PMF Properties

A random variable $X$ has PMF: $p_X(0) = 0.1$, $p_X(1) = 0.3$, $p_X(2) = c$, $p_X(3) = 0.2$.

(a) Find $c$.
(b) Find $P(X \geq 2)$.
(c) Find $F_X(1.5)$.
(d) Find $P(0.5 < X \leq 2.5)$.

**Solution**

(a) Sum = 1: $0.1 + 0.3 + c + 0.2 = 1 \rightarrow c = 0.4$
(b) $P(X \geq 2) = p_X(2) + p_X(3) = 0.4 + 0.2 = 0.6$
(c) $F_X(1.5) = P(X \leq 1.5) = p_X(0) + p_X(1) = 0.1 + 0.3 = 0.4$
(d) $P(0.5 < X \leq 2.5) = p_X(1) + p_X(2) = 0.3 + 0.4 = 0.7$

### Problem 3: PDF and CDF

A random variable has PDF:
$f_X(x) = \text{cx}$ for $0 \le x \le 4$, and 0 otherwise.

(a) Find $c$.
(b) Find the CDF $F_X(x)$.
(c) Find $P(1 \le X \le 3)$.
(d) Find the value $x_0$ such that $P(X \le x_0) = 0.5$.

**Solution**

(a) $\int_0^4 \text{cx}\, dx = c \cdot \left[\frac{x^2}{2}\right]_0^4 = c \cdot 8 = 1 \to c = \frac{1}{8}$

(b) $F_X(x) = \int_0^x \frac{t}{8} dt = \frac{x^2}{16}$, for $0 \le x \le 4$
$F_X(x) = 0$ for $x < 0$; $F_X(x) = 1$ for $x > 4$

(c) $P(1 \le X \le 3) = F_X(3) - F_X(1) = \dfrac{9}{16} - \dfrac{1}{16} = \dfrac{8}{16} = 0.5$

(d) $\dfrac{x_0^2}{16} = 0.5 \to x_0^2 = 8 \to x_0 = 2\sqrt{2} \approx 2.83$

### Problem 4: Transformation

A sensor measures temperature $X$ uniformly distributed on [20, 30] °C.
The output voltage is $Y = 0.1X - 2$ (linear sensor).

(a) Find the PDF of $Y$.
(b) Find the range of $Y$.
(c) Find $P(Y > 0.5)$.

**Solution**

(a) $X \sim \text{Uniform}[20, 30]$, so $f_X(x) = \frac{1}{10}$ for $20 \le x \le 30$.
$Y = 0.1X - 2$, so $a = 0.1$, $b = -2$.
$f_Y(y) = \frac{1}{|a|} \cdot f_X\left(\frac{y-b}{a}\right) = \frac{1}{0.1} \cdot \frac{1}{10} = 1$ for the valid range.

(b) When $X = 20$: $Y = 0.1(20) - 2 = 0$; When $X = 30$: $Y = 0.1(30) - 2 = 1$
Range of $Y$: $[0, 1]$. So $Y \sim \text{Uniform}[0, 1]$.

(c) $P(Y > 0.5) = 1 - F_Y(0.5) = 1 - 0.5 = 0.5$

**Problem 5: Non-Monotonic Transformation**

A voltage $X \sim \text{Uniform}[-1,\ 1]V$. The instantaneous power is $Y = X^2$.

(a) Find the CDF of $Y$.

(b) Find the PDF of $Y$.

**Solution**

(a) For $0 \leq y \leq 1$:

$$F_Y(y) = P(X^2 \leq y) = P\left(-\sqrt{y} \leq X \leq \sqrt{y}\right) = F_X\left(\sqrt{y}\right) - F_X\left(-\sqrt{y}\right)$$

Since $X \sim \text{Uniform}[-1,\ 1]$: $F_X(x) = \frac{x+1}{2}$

$$F_Y(y) = \frac{\sqrt{y}+1}{2} - \frac{-\sqrt{y}+1}{2} = \sqrt{y}$$

(b) $f_Y(y) = \frac{dF_Y}{dy} = \frac{1}{2\sqrt{y}}$ for $0 < y \leq 1$

## 2.11 Key Takeaways

1. **Random variables** convert experimental outcomes into numbers for mathematical analysis
2. **PMF** gives the actual probability for discrete RVs (bar heights sum to 1)
3. **PDF** gives probability density for continuous RVs (area gives probability, not height)
4. **CDF** works universally — gives cumulative probability $P(X \leq x)$
5. **Transformations** let us derive distributions of functions of random variables
6. The **CDF method** (find CDF first, then differentiate) is often the safest approach for transformations

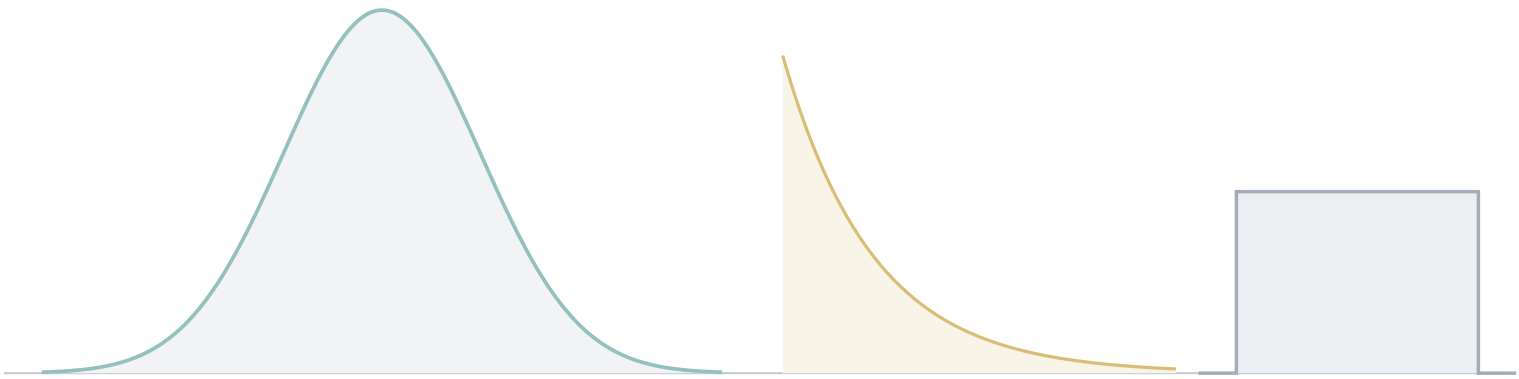
*Next Module: Module 3 — Important Probability Distributions*

Module 3

# Important Probability Distributions

توزیع‌های مهم احتمال

3



## In This Module


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**Learning Objectives**

After completing this module, students will be able to:

1. Identify which probability distribution models a given engineering scenario
2. State the $\frac{\text{PMF}}{\text{PDF}}$, CDF, mean, and variance for all major distributions
3. Explain the physical interpretation and engineering context for each distribution
4. Determine when to use (and when NOT to use) each distribution
5. Analyze and compare probability distributions
6. Recognize distribution patterns from data characteristics

# 3.1 Introduction: Choosing the Right Distribution

## The Engineering Challenge

When modeling a random phenomenon, the critical question is:

> "Which probability distribution best describes this uncertain quantity?"

Choosing the wrong distribution leads to incorrect predictions, improper designs, and system failures.

## How to Choose

The distribution should match the **physical mechanism** generating randomness:

- Counting successes → Binomial
- Waiting for events → Exponential/Geometric
- Sum of many small effects → Gaussian
- Signal envelope in fading → Rayleigh

This module provides the tools to make this choice correctly.

# 3.2 Bernoulli Distribution

## Physical Interpretation

Models a **single trial** with two possible outcomes: "success" (1) or "failure" (0).

## Engineering Context

Any binary random outcome:

- One bit transmitted: error or correct
- One component tested: pass or fail
- One packet sent: delivered or lost
- One detection attempt: target present or absent

**Mathematical Definition**

$X \sim$ Bernoulli$(p)$, where $p$ = probability of success.

## PMF

$$p_X(x) = \begin{cases} p & x = 1 \\ 1 - p = q & x = 0 \end{cases}$$

Compact form: $p_X(x) = p^x(1-p)^{1-x}$ for $x \in \{0, 1\}$

## CDF

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1 - p & 0 \le x < 1 \\ 1 & x \ge 1 \end{cases}$$

## Mean

$$E[X] = p$$

## Variance

$$\text{Var}(X) = p(1-p)$$

Maximum variance at $p = 0.5$ (maximum uncertainty).

## When to Use

✔ Single binary trial with fixed probability
✔ Building block for more complex distributions
✔ Modeling $\frac{ON}{OFF}$ states, pass/fail outcomes

## When NOT to Use

✘ Multiple trials (use Binomial)
✘ Outcome is not binary
✘ Probability changes between trials

# 3.3 Binomial Distribution

## Physical Interpretation

Models the **number of successes** in $n$ **independent** trials, each with the same success probability $p$.

## Engineering Context

- Number of bit errors in $n$ transmitted bits
- Number of defective components in a batch of $n$
- Number of successful packet transmissions out of $n$ attempts
- Number of sensors that detect a signal (out of $n$ sensors)

**Mathematical Definition**

$X \sim \text{Binomial}(n,\, p)$

## PMF

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, 2, \ldots, n$$

where $C(n,\, k) = \frac{n!}{k!(n-k)!}$

## CDF

$$F_X(k) = \sum_{i=0}^{\lfloor k \rfloor} \binom{n}{i} p^i (1-p)^{n-i}$$

No closed-form expression — use tables or numerical summation.

## Mean

$$E[X] = np$$

## Variance

$$\text{Var}(X) = np(1-p)$$

## When to Use

✔ Fixed number of trials $n$
✔ Each trial is independent
✔ Each trial has the same probability $p$
✔ Counting the number of "successes"

## When NOT to Use

✘ Trials are not independent (correlated failures)
✘ Probability changes between trials
✘ $n$ is not fixed (use Negative Binomial or Poisson)
✘ Very large $n$ with small $p$ (use Poisson approximation)

# 3.4 Geometric Distribution

## Physical Interpretation

Models the **number of trials until the first success** (or equivalently, the number of failures before the first success).

## Engineering Context

- Number of transmissions until a packet is successfully delivered
- Number of components tested until finding a defective one
- Number of login attempts until successful authentication
- Number of time slots until channel becomes available

**Mathematical Definition**

$X \sim \text{Geometric}(p)$
Convention: $X$ = number of the trial on which first success occurs ($X \in \{1, 2, 3, \ldots\}$).

## PMF

$$p_X(k) = (1-p)^{k-1}p, \quad k = 1, 2, 3, \ldots$$

($k-1$ failures, then one success)

## CDF

$$F_X(k) = 1 - (1-p)^k, \quad k = 1, 2, 3, \ldots$$

## Mean

$$E[X] = \frac{1}{p}$$

## Variance

$$\text{Var}(X) = \frac{1-p}{p^2}$$

## Memoryless Property

The geometric distribution is the ONLY discrete distribution with the memoryless property:

$$P(X > m+n \mid X > m) = P(X > n)$$

**Engineering meaning:** If a system hasn't succeeded in $m$ attempts, the probability of needing $n$ more attempts is the same as starting fresh. "Past failures don't affect future probability."

## When to Use

✔ Repeated independent trials until first success
✔ Each trial has the same probability
✔ Memoryless assumption is appropriate (e.g., independent retransmissions)

## When NOT to Use

✘ Success probability changes over time (aging, learning)
✘ Trials are not independent
✘ Looking for the $r-$ th success (use Negative Binomial)

# 3.5 Negative Binomial Distribution

## Physical Interpretation

Models the **number of trials until the** $r-$ th **success** (generalization of Geometric).

## Engineering Context

- Number of transmissions until $r$ packets are successfully delivered
- Number of components inspected until finding $r$ defectives
- Number of time slots until $r$ channels become available

**Mathematical Definition**

$X \sim \text{NegBin}(r,\, p)$ = number of trials until $r-$ th success.

## PMF

$$p_X(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r+1, r+2, \ldots$$

(Choose which $r-1$ of the first $k-1$ trials were successes, the $k-$ th is the $r-$ th success.)

## CDF

No simple closed form. Computed by summation or numerical methods.

## Mean

$$E[X] = \frac{r}{p}$$

## Variance

$$\text{Var}(X) = \frac{r(1-p)}{p^2}$$

## Relationship to Geometric

When $r = 1$, the Negative Binomial reduces to the Geometric distribution.

## When to Use

✔ Repeated independent trials until $r-$ th success
✔ Constant probability per trial
✔ Generalizing geometric to multiple required successes

## When NOT to Use

✘ Only need first success (use Geometric — simpler)
✘ Trials not independent
✘ Fixed number of trials (use Binomial)

# 3.6 Poisson Distribution

## Physical Interpretation

Models the **number of events occurring in a fixed interval** (time, space, area) when events occur independently at a constant average rate.

## Engineering Context

- Number of photons hitting a detector in $1\mu s$
- Number of packets arriving at a router in 1 second
- Number of defects per $cm^2$ on a silicon wafer
- Number of cosmic ray events in a sensor per hour
- Number of interference spikes in a measurement window

**Mathematical Definition**

$X \sim \text{Poisson}(\lambda)$, where $\lambda$ = average number of events per interval.

## PMF

$$p_X(k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \ldots$$

## CDF

$$F_X(k) = e^{-\lambda} \sum_{i=0}^{\lfloor k \rfloor} \frac{\lambda^i}{i!}$$

## Mean

$$E[X] = \lambda$$

## Variance

$$\text{Var}(X) = \lambda$$

**Unique property:** Mean equals variance! This is a quick check — if sample mean ≈ sample variance, data may be Poisson.

## Poisson as Binomial Approximation

When $n$ is large and $p$ is small, with $\lambda = np$:

$$\text{Binomial}(n, p) \approx \text{Poisson}(\lambda = np)$$

Rule of thumb: $n \geq 20$ and $p \leq 0.05$.

## When to Use

✔ Counting events in a fixed interval
✔ Events occur independently
✔ Events occur at a constant average rate
✔ Rare events with many opportunities (large $n$, small $p$)
✔ Mean $\approx$ Variance in data

## When NOT to Use

✘ Events are not independent (clustered arrivals)
✘ Rate varies with time (non-homogeneous process)
✘ Events are not rare ($p$ not small) — use Binomial
✘ Mean $\neq$ Variance (look at overdispersion/underdispersion)

# 3.7 Uniform Distribution (Continuous)

## Physical Interpretation

Models a quantity that is **equally likely** to take any value in an interval $[a, b]$. Represents **maximum uncertainty** within a known range.

## Engineering Context

- Phase of a received signal (unknown, equally likely in $[0, 2\pi)$)
- Round-off error in an ADC (uniform in $\left[-\frac{\Delta}{2}, \frac{\Delta}{2}\right]$ where $\Delta$ = quantization step)
- Arrival time within a time slot (unknown position)
- Random access protocol: choose a random backoff time

**Mathematical Definition**

$X \sim \text{Uniform}(a, b)$ or $X \sim U(a, b)$

## PDF

$$f_X(x) = \begin{cases} \frac{1}{b-a} & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

## CDF

$$F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \le x \le b \\ 1 & x > b \end{cases}$$

## Mean

$$E[X] = \frac{a+b}{2}$$

## Variance

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

## When to Use

✔ All values in a range are equally likely
✔ No information favors any particular value
✔ Quantization error modeling
✔ Phase of unmodulated carrier

## When NOT to Use

✘ Values near the center are more likely (use Gaussian)
✘ Values are non-negative with exponential decay (use Exponential)
✘ Range is unbounded

# 3.8 Exponential Distribution

## Physical Interpretation

Models the **time between events** in a Poisson process, or the **lifetime** of a memoryless component.

## Engineering Context

- Time between packet arrivals
- Time until component failure (constant failure rate)
- Time between cosmic ray events in a detector
- Inter-arrival time of calls at a switch
- Duration of a fading dip in a wireless channel

**Mathematical Definition**

$X \sim$ Exponential$(\lambda)$ where $\lambda$ = rate parameter (events per unit time).
Alternative parameterization: $X \sim \text{Exp}(\beta)$ where $\beta = \frac{1}{\lambda}$ = mean.

## PDF

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

Or with mean $\beta$: $f_X(x) = \frac{1}{\beta} e^{-x/\beta}$

## CDF

$$F_X(x) = 1 - e^{-\lambda x}, \quad x \geq 0$$

## Mean

$$E[X] = \frac{1}{\lambda} = \beta$$

## Variance

$$\text{Var}(X) = \frac{1}{\lambda^2} = \beta^2$$

**Note:** Standard deviation = mean for exponential distribution.

## Memoryless Property

$$P(X > s + t \mid X > s) = P(X > t)$$

The exponential is the ONLY continuous distribution with this property.

**Engineering meaning:** A component that has been working for $s$ hours is statistically "as good as new." Its remaining lifetime has the same distribution as a brand-new component.

## Relationship to Poisson

If events arrive as a Poisson process with rate $\lambda$:

- Number of events in time $T \sim \text{Poisson}(\lambda T)$
- Time between events $\sim \text{Exponential}(\lambda)$

## When to Use

✔ Modeling time between events in a Poisson process
✔ Component lifetime with constant failure rate
✔ Memoryless assumption is reasonable

## When NOT to Use

✘ Failure rate increases with age (wear-out) — use Weibull or Gamma
✘ Failure rate decreases with age (burn-in) — use Weibull
✘ Data shows increasing hazard — NOT memoryless

# 3.9 Gaussian (Normal) Distribution

## Physical Interpretation

Models quantities that result from the **sum of many small, independent random effects**. The most important distribution in engineering due to the Central Limit Theorem.

## Engineering Context

- Thermal noise voltage (sum of many electron contributions)
- Measurement errors (sum of many small error sources)
- Manufacturing variations (many independent process variations)
- Aggregate interference in communications
- Any quantity well-modeled by CLT

**Mathematical Definition**

$X \sim N(\mu,\, \sigma^2)$ where $\mu$ = mean, $\sigma^2$ = variance.

## PDF

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty$$

## CDF

$$F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{2}\left[1 + \text{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right]$$

No closed-form — computed numerically or via tables.

## Mean

$$E[X] = \mu$$

## Variance

$$\text{Var}(X) = \sigma^2$$

## Standard Normal

$Z \sim N(0,\, 1)$: standardized form.

Any Normal can be standardized: $Z = \frac{X-\mu}{\sigma}$

## Key Probability Rules ($\sigma$ − rules)

| Range | Probability |
|---|---|
| $\mu \pm 1\sigma$ | 68.27% |
| $\mu \pm 2\sigma$ | 95.45% |

| Range | Probability |
|---|---|
| $\mu \pm 3\sigma$ | 99.73% |
| $\mu \pm 4\sigma$ | 99.9937% |

## Properties

1. **Symmetric** about $\mu$
2. **Linear transformation:** If $X \sim N(\mu,\, \sigma^2)$, then aX $+b \sim N(a\mu + b,\, a^2\sigma^2)$
3. **Sum of Gaussians:** If $X \sim N(\mu_1,\, \sigma_1^2)$ and $Y \sim N(\mu_2,\, \sigma_2^2)$ are independent, then $X + Y \sim N(\mu_1 + \mu_2,\, \sigma_1^2 + \sigma_2^2)$
4. **Fully characterized** by mean and variance (all higher moments determined)

## When to Use

✔ Thermal noise in circuits
✔ Measurement errors
✔ Any sum of many independent random effects (CLT)
✔ Manufacturing variations
✔ When data histogram is bell-shaped and symmetric

## When NOT to Use

✘ Strictly positive quantities (Gaussian allows negative values)
✘ Heavily skewed data
✘ Bounded quantities (Gaussian has infinite support)
✘ Discrete counts
✘ Heavy-tailed phenomena (use t-distribution or Cauchy)

# 3.10 Gamma Distribution

## Physical Interpretation

Models the **total waiting time until the** $k-$ th **event** in a Poisson process, or any sum of $k$ independent exponential random variables.

## Engineering Context

- Total repair time for $k$ components (each with exponential repair time)
- Time until $k-$ th packet arrival
- Aggregate service time for $k$ queued jobs
- Rainfall amount modeling

**Mathematical Definition**

$X \sim$ Gamma$(\alpha,\ \beta)$ where $\alpha$ = shape parameter, $\beta$ = scale parameter (or rate $\lambda = \frac{1}{\beta}$).

## PDF

$$f_X(x) = \frac{x^{\alpha-1}e^{-x/\beta}}{\beta^{\alpha}\Gamma(\alpha)}, \quad x \geq 0$$

where $\Gamma(\alpha) = (\alpha-1)!$ for integer $\alpha$, and $\Gamma(\alpha) = \int_0^{\infty} t^{\alpha-1}e^{-t}dt$ in general.

## CDF

No closed form for general $\alpha$. Available via incomplete gamma function.

## Mean

$$E[X] = \alpha\beta$$

## Variance

$$\text{Var}(X) = \alpha\beta^2$$

## Special Cases

| Parameters | Distribution |
|---|---|
| $\alpha = 1$ | Exponential$(\beta)$ |
| $\alpha = \frac{n}{2},\ \beta = 2$ | Chi-square(n) |
| $\alpha =$ integer $k$ | Erlang$(k,\ \beta)$ |

## When to Use

✔ Sum of independent exponential lifetimes
✔ Waiting time for $k-$ th Poisson event
✔ Skewed, non-negative data
✔ Flexible shape (adjustable skewness)

## When NOT to Use

✘ Data can be negative
✘ Data is symmetric (use Gaussian)
✘ Simpler distribution fits (exponential for $k = 1$)

# 3.11 Rayleigh Distribution

## Physical Interpretation

Models the **magnitude (envelope)** of a $2D$ Gaussian random vector. When $X$ and $Y$ are independent $N(0,\, \sigma^2)$, then $R = \sqrt{X^2 + Y^2}$ follows a Rayleigh distribution.

## Engineering Context

- Envelope of narrowband noise (in-phase + quadrature)
- Magnitude of wireless fading channel coefficient (Rayleigh fading)
- Wind speed modeling
- Distance error in $2D$ positioning (when errors are Gaussian in $x$ and $y$)

**Mathematical Definition**

$R \sim$ Rayleigh$(\sigma)$, where $\sigma$ is the parameter (NOT the standard deviation of $R$).

## PDF

$$f_R(r) = \frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)}, \quad r \geq 0$$

## CDF

$$F_R(r) = 1 - e^{-r^2/(2\sigma^2)}, \quad r \geq 0$$

## Mean

$$E[R] = \sigma\sqrt{\frac{\pi}{2}} \approx 1.2533\sigma$$

## Variance

$$\text{Var}(R) = \frac{4-\pi}{2}\sigma^2 \approx 0.4292\sigma^2$$

## Relationship to Other Distributions

- $R^2 \sim$ Exponential$(2\sigma^2)$ — the power is exponentially distributed
- If $X,\, Y \sim N(0,\, \sigma^2)$ independent, then $\sqrt{X^2 + Y^2} \sim$ Rayleigh$(\sigma)$

## When to Use

✔ Envelope/magnitude of complex Gaussian signal
✔ Rayleigh fading channel modeling (no line-of-sight)
✔ $2D$ distance with Gaussian errors in each dimension
✔ RSS (root-sum-square) of two independent Gaussian components

## When NOT to Use

✘ Strong line-of-sight component present (use Rician)
✘ Not modeling a magnitude/envelope
✘ Components are not Gaussian or not equal-variance

# 3.12 Chi-Square Distribution

## Physical Interpretation

Models the **sum of squares of independent standard normal** random variables. Arises naturally in variance estimation and goodness-of-fit testing.

## Engineering Context

- Sample variance distribution (for Gaussian data)
- Power of Gaussian noise (sum of squared components)
- Goodness-of-fit test statistic
- Confidence intervals for variance
- Energy detection in spectrum sensing

**Mathematical Definition**

If $Z_1, Z_2, \ldots, Z_n$ are independent $N(0, 1)$, then:

$$\chi^2 = Z_1^2 + Z_2^2 + \cdots + Z_n^2 \sim \chi^2(n)$$

$n$ = degrees of freedom.

## PDF

$$f_X(x) = \frac{x^{n/2-1} e^{-x/2}}{2^{n/2}\Gamma(n/2)}, \quad x \geq 0$$

## CDF

No closed form for general $n$. Computed via incomplete gamma function.

## Mean

$$E[X] = n$$

## Variance

$$\text{Var}(X) = 2n$$

## Relationship to Other Distributions

- Chi-square(n) = Gamma $\left(\frac{n}{2}, 2\right)$
- Chi-square(1) = square of $N(0, 1)$
- Chi-square(2) = Exponential $\left(\frac{1}{2}\right)$
- For large $n$: $\chi^2(n) \approx N(n, 2n)$ by CLT

## When to Use

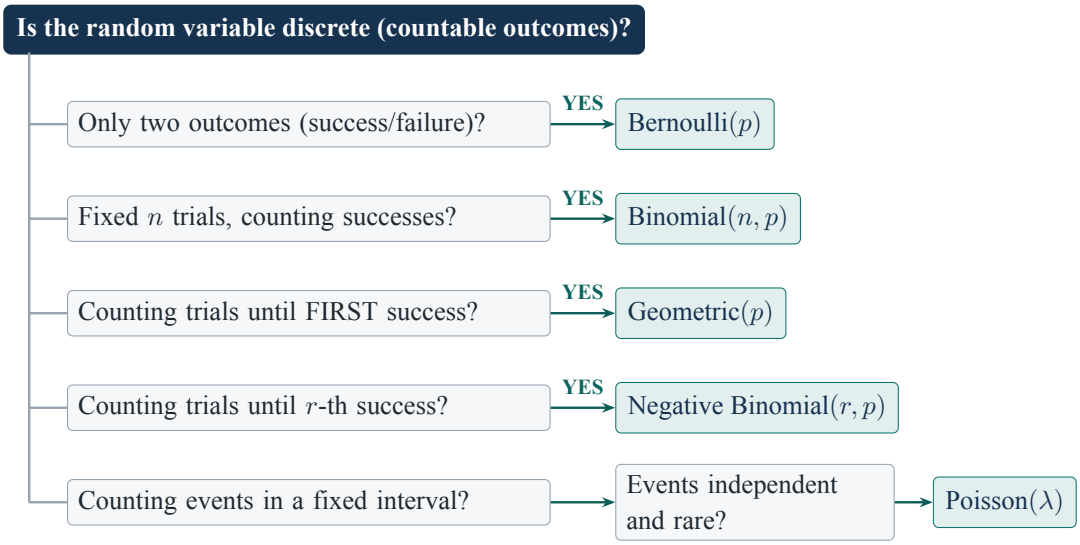
✔ Sum of squared Gaussian random variables
✔ Variance testing and confidence intervals
✔ Goodness-of-fit tests
✔ Energy detection in $N$ dimensions

## When NOT to Use

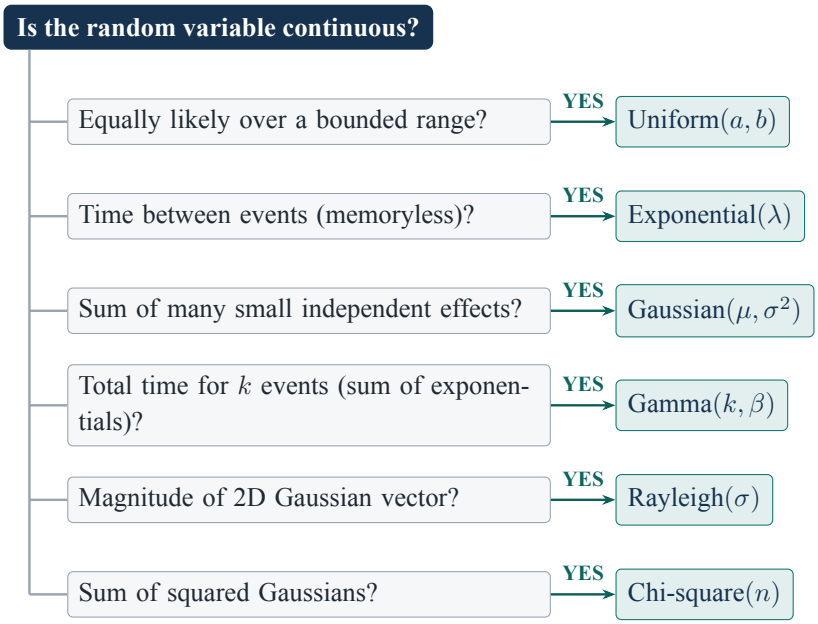
✘ Variables are not Gaussian
✘ Variables are not independent
✘ Variables are not standard normal (need scaling)

# 3.13 Distribution Selection Guide

## Decision Tree: Discrete Distributions




## Decision Tree: Continuous Distributions




## Quick Recognition Patterns

| If you see… | Think… |
|---|---|
| Binary outcome | Bernoulli |
| "$n$ trials, $k$ successes" | Binomial |

| If you see… | Think… |
|---|---|
| "How many until first…" | Geometric |
| "Events per interval" | Poisson |
| "Equal chance anywhere in $[a, b]$" | Uniform |
| "Time until next event" | Exponential |
| "Sum of many effects" or "noise" | Gaussian |
| "Total waiting time for $k$ events" | Gamma |
| "Signal envelope" or "fading" | Rayleigh |
| "Sum of squared normals" | Chi-square |

## 3.14 Summary Table

| Distribution | Type | Parameters | Mean | Variance |
|---|---|---|---|---|
| Bernoulli | Discrete | $p$ | $p$ | $p(1-p)$ |
| Binomial | Discrete | $n$, $p$ | $np$ | $np(1-p)$ |
| Geometric | Discrete | $p$ | $\frac{1}{p}$ | $\frac{1-p}{p^2}$ |
| Neg. Binomial | Discrete | $r$, $p$ | $\frac{r}{p}$ | $\frac{r(1-p)}{p^2}$ |
| Poisson | Discrete | $\lambda$ | $\lambda$ | $\lambda$ |
| Uniform | Continuous | a, $b$ | $\frac{a+b}{2}$ | $\frac{(b-a)^2}{12}$ |
| Exponential | Continuous | $\lambda$ (or $\beta = \frac{1}{\lambda}$) | $\frac{1}{\lambda}$ | $\frac{1}{\lambda^2}$ |
| Gaussian | Continuous | $\mu$, $\sigma^2$ | $\mu$ | $\sigma^2$ |
| Gamma | Continuous | $\alpha$, $\beta$ | $\alpha\beta$ | $\alpha\beta^2$ |
| Rayleigh | Continuous | $\sigma$ | $\sigma\sqrt{\frac{\pi}{2}}$ | $\frac{(4-\pi)\sigma^2}{2}$ |
| Chi-square | Continuous | $n$ | $n$ | $2n$ |

## 3.15 Practice Problems

**Problem 1: Distribution Identification**

Identify the appropriate distribution for each scenario:

(a) Number of dropped calls in a 10-minute window (average: 2 per 10 min)
(b) Voltage of thermal noise across a resistor
(c) Time until a server crashes (constant failure rate)
(d) Number of defective ICs out of a batch of 100 (defect rate 3%)
(e) Envelope of received signal through multipath with no LOS

**Solution** (a) Poisson($\lambda = 2$), (b) Gaussian($0$, $\sigma^2$), (c) Exponential($\lambda$), (d) Binomial($100$, $0.03$), (e) Rayleigh($\sigma$)

**Problem 2: Exponential vs Poisson**

Packets arrive at a router as a Poisson process with rate $\lambda = 5$ packets/second.

(a) What is $P(\text{more than 8 packets in 1 second})$?
(b) What is $P(\text{no packet arrives in the next 0.5 seconds})$?
(c) What distribution describes the inter-arrival time?
(d) What is the mean inter-arrival time?

**Solution**

(a) $X \sim \text{Poisson}(5)$: $P(X > 8) = 1 - \texttt{poisscdf}(8, 5) = 0.0681$
(b) $T \sim \text{Exp}(5)$: $P(T > 0.5) = e^{-5\times 0.5} = e^{-2.5} = 0.0821$
(c) Exponential with rate $\lambda = 5$
(d) Mean $= \frac{1}{\lambda} = 0.2$ seconds

**Problem 3: Gaussian Probability**

A resistor has nominal value $1000\Omega$ with manufacturing variation modeled as $N(1000,\, 25)(\sigma = 5\,\Omega)$.

(a) What fraction of resistors are within $\pm 10\,\Omega$ of nominal (spec: $990 - 1010\,\Omega$)?
(b) What fraction fail the $\pm 1\%$ tolerance test?
(c) What tolerance range contains 99.7% of resistors?

**Solution**

(a) $P(990 \le X \le 1010) = P(-2 \le Z \le 2) = 0.9545 \to 95.45\%$
(b) $\pm 1\% = \pm 10\,\Omega \to$ same as (a), fail rate $= 1 - 0.9545 = 4.55\%$
(c) $\pm 3\sigma = \pm 15\,\Omega \to$ range $[985,\, 1015]\Omega$

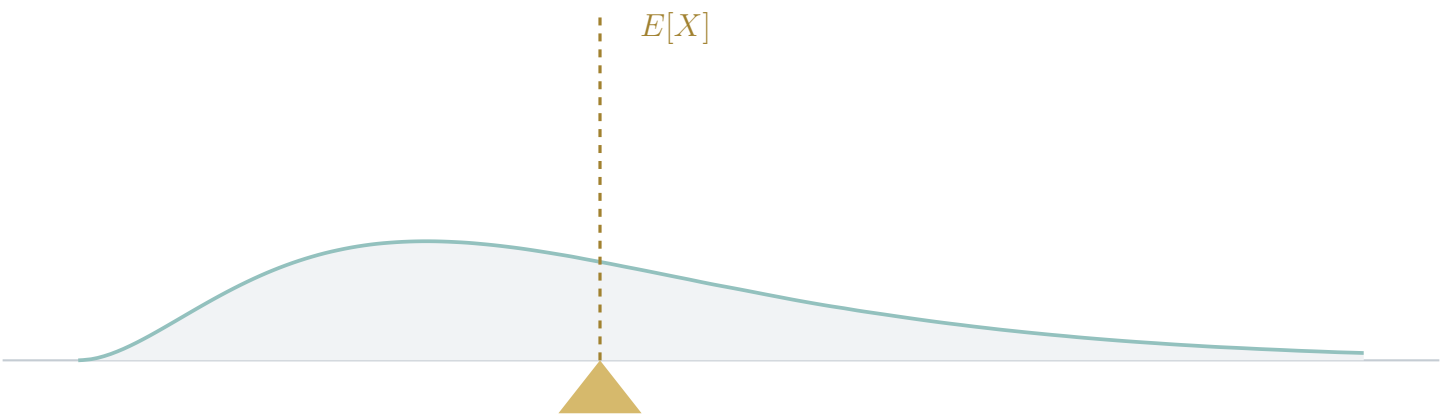
*Next Module: Module 4 — Expectation and Moments*

Module 4

# Expectation and Moments

امید ریاضی و گشتاورها

4



## In This Module


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**Learning Objectives**

After completing this module, students will be able to:

1. Compute expected values for discrete and continuous random variables
2. Calculate variance and standard deviation and interpret their engineering meaning
3. Apply properties of expectation and variance in engineering analysis
4. Compute and interpret higher-order moments
5. Use the moment generating function (MGF)
6. Relate statistical moments to physical engineering quantities

# 4.1 Introduction: Why Summary Statistics?

## The Engineering Need

A probability distribution contains ALL information about a random variable. But for engineering decisions, we often need concise summaries:

- **Signal processing:** What is the average power? $\rightarrow$ Mean of $X^2$
- **Quality control:** How much *do* components vary? $\rightarrow$ Variance
- **Communication:** What is the noise power? $\rightarrow E[N^2]$
- **Reliability:** What is the average lifetime? $\rightarrow$ Mean

Moments (mean, variance, etc.) are the essential **engineering-relevant summaries** of a distribution.

## 4.2 Expected Value (Mean)

**Definition — Discrete RV**

$$E[X] = \mu_X = \sum_x x \cdot p_X(x)$$

Sum of (each value $\times$ its probability).

**Definition — Continuous RV**

$$E[X] = \mu_X = \int_{-\infty}^{\infty} x \cdot f_X(x)\, dx$$

### Engineering Interpretation

The expected value is the **long-run average**: if you repeat the experiment many times and average the results, you get approximately $E[X]$.

| Engineering Quantity | Expected Value Meaning |
|---|---|
| Noise voltage $E[N]$ | DC bias/offset in noise |
| Signal amplitude $E[A]$ | Average signal level |
| Component lifetime $E[T]$ | Mean time to failure (MTTF) |
| Packet delay $E[D]$ | Average delay |
| Bit errors $E[X]$ | Average error count |

**Example: Average Noise Power**

For zero-mean Gaussian noise $N \sim N(0, \sigma^2)$:

- $E[N] = 0$ (no DC component)
- Average power $= E[N^2] = \sigma^2$ (this is the variance!)

**Example: Expected Number of Retransmissions**

ARQ protocol: each transmission succeeds with probability $p = 0.8$.
Number of attempts $X \sim$ Geometric$(0.8)$.
$E[X] = \frac{1}{p} = \frac{1}{0.8} = 1.25$ attempts on average.

# 4.3 Expected Value of a Function: $E[g(X)]$

## The LOTUS Theorem (Law of the Unconscious Statistician)

To find $E[g(X)]$, you *do* NOT need the distribution of $Y = g(X)$:

**Discrete:**

$$E[g(X)] = \sum_x g(x) \cdot p_X(x)$$

**Continuous:**

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f_X(x)\, dx$$

## Engineering Applications

| Function $g(X)$ | Engineering Meaning |
|---|---|
| $g(X) = X^2$ | Mean square value (power) |
| $g(X) = (X - \mu)^2$ | Variance (AC power) |
| $g(X) =$ | $X$ |
| $g(X) = e^{j\omega X}$ | Characteristic function |

**Example: Mean Power of a Signal**

Signal $X$ has PDF $f_X(x)$. Power $P = X^2$.
$E[P] = E[X^2] = \int x^2 f_X(x)\, dx$
For $X \sim N(0, \sigma^2)$: $E[X^2] = \sigma^2$ (the noise power equals the variance).
For $X \sim N(\mu, \sigma^2)$: $E[X^2] = \mu^2 + \sigma^2$ (total power = DC power + AC power).

# 4.4 Variance

**Definition**

$$\text{Var}(X) = E[(X - \mu_X)^2] = E[X^2] - (E[X])^2$$

The second formula (computational formula) is often easier to use.

## Derivation of Computational Formula

$$\begin{aligned}
&\text{Var}(X) = E[(X - \mu)^2] \\
&= E[X^2 - 2\mu X + \mu^2] \\
&= E[X^2] - 2\mu E[X] + \mu^2 \\
&= E[X^2] - 2\mu^2 + \mu^2 \\
&= E[X^2] - \mu^2
\end{aligned}$$

## Engineering Interpretation

The variance measures **spread/dispersion/uncertainty**:

| Context | Variance Meaning |
|---|---|
| Noise signal | Noise power (for zero-mean noise) |
| Manufacturing | How much components deviate from nominal |
| Measurements | Measurement precision (smaller = more precise) |
| Signal | AC power component |
| Lifetime | How predictable is the lifetime? |

## Key Insight: Variance = Power in Engineering

For a zero-mean signal $X$:

- $E[X^2] = \text{Var}(X) = \sigma^2$ = **total power**
- For $X \sim N(0, \sigma^2)$: noise power $= \sigma^2 = \text{Var}(X)$

For a signal with DC offset ($\mu \neq 0$):

- $E[X^2] = \mu^2 + \sigma^2$ = **total power = DC power + AC power**
- $\text{Var}(X) = \sigma^2$ = **AC power only**

# 4.5 Standard Deviation

**Definition**

$$\sigma_X = \sqrt{\mathrm{Var}(X)}$$

## Engineering Significance

The standard deviation has the **same units** as $X$, making it directly interpretable:

| Random Variable | Standard Deviation Unit | Interpretation |
|---|---|---|
| Voltage noise | Volts | RMS noise voltage |
| Timing jitter | Seconds | RMS jitter |
| Distance error | Meters | RMS position error |
| Temperature | °C | RMS temperature fluctuation |

The standard deviation is the **RMS (root-mean-square)** value for zero-mean signals.

# 4.6 Properties of Expectation

## Linearity (FUNDAMENTAL)

$$E[aX + b] = aE[X] + b$$

$$E[X + Y] = E[X] + E[Y] \quad \text{(ALWAYS, even if dependent!)}$$

$$E\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i E[X_i]$$

## Monotonicity

If $X \geq 0$ always, then $E[X] \geq 0$.

If $X \geq Y$ always, then $E[X] \geq E[Y]$.

## For Independent RVs

If $X$ and $Y$ are independent:

$$E[XY] = E[X] \cdot E[Y]$$

⚠ This does NOT hold for dependent variables!

**Engineering Example: Total Noise**

A received signal: $R = S + N_1 + N_2$ (signal plus two noise sources)
$E[R] = E[S] + E[N_1] + E[N_2] = E[S] + 0 + 0 = E[S]$
(Zero-mean noise doesn't bias the expected received signal.)

# 4.7 Properties of Variance

## Scaling and Shift

$$\text{Var}(aX + b) = a^2\text{Var}(X)$$

- Adding a constant $b$ does NOT change variance (shifting doesn't change spread)
- Scaling by $a$ multiplies variance by $a^2$

## Sum of Independent RVs

If $X_1, X_2, \ldots, X_n$ are **independent**:

$$\text{Var}(X_1 + X_2 + \cdots + X_n) = \text{Var}(X_1) + \text{Var}(X_2) + \cdots + \text{Var}(X_n)$$

⚠ Variances add for INDEPENDENT variables only!

**Engineering Example: Noise Power Addition**

Two independent noise sources ($\sigma_1^2 = 0.01\,\text{V}^2$, $\sigma_2^2 = 0.04\,\text{V}^2$):
Total noise power $= \sigma_1^2 + \sigma_2^2 = 0.05\,\text{V}^2$
Total RMS noise $= \sqrt{0.05} = 0.224\,\text{V}$
Note: RMS values *do* NOT add directly! $(0.1 + 0.2 \neq 0.224)$

## Variance of Sample Mean

For $n$ i.i.d. observations $X_1, \ldots, X_n$ with variance $\sigma^2$:

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n}\sum X_i\right) = \frac{\sigma^2}{n}$$

More measurements $\rightarrow$ lower uncertainty in the average.

# 4.8 Higher-Order Moments

## $n-$ th Moment

$$\mu_n' = E[X^n] = \int_{-\infty}^{\infty} x^n f_X(x)\, dx$$

## $n-$ th Central Moment

$$\mu_n = E[(X-\mu)^n]$$

- $\mu_1 = 0$ (always)
- $\mu_2 = \text{Var}(X)$
- $\mu_3$ relates to **skewness**
- $\mu_4$ relates to **kurtosis**

## Skewness ($3rd$ Central Moment, Normalized)

$$\gamma_1 = \frac{E[(X-\mu)^3]}{\sigma^3}$$

| Skewness | Shape | Example |
|---|---|---|
| $\gamma_1 = 0$ | Symmetric | Gaussian |
| $\gamma_1 > 0$ | Right-skewed (long right tail) | Exponential |
| $\gamma_1 < 0$ | Left-skewed (long left tail) | Negated exponential |

**Engineering significance:** Skewness tells us if extreme values are more likely in one direction. Important for reliability (failure time distributions are right-skewed).

## Kurtosis (4th Central Moment, Normalized)

$$\gamma_2 = \frac{E[(X-\mu)^4]}{\sigma^4} - 3$$

(Subtracting 3 makes Gaussian have kurtosis 0: "excess kurtosis.")

| Kurtosis | Interpretation | Meaning |
|---|---|---|
| $\gamma_2 = 0$ | Mesokurtic | Gaussian-like tails |
| $\gamma_2 > 0$ | Leptokurtic | Heavier tails than Gaussian |
| $\gamma_2 < 0$ | Platykurtic | Lighter tails than Gaussian |

**Engineering significance:** High kurtosis means more outliers — important for impulsive noise modeling.

# 4.9 Moment Generating Function (MGF)

**Definition**

$$M_X(t) = E[e^{tX}]$$

**Discrete:** $M_X(t) = \sum e^{tx} p_X(x)$
**Continuous:** $M_X(t) = \int e^{tx} f_X(x)\, dx$

## Why "Moment Generating"?

The $n-$ th moment is obtained by differentiating $n$ times at $t = 0$:

$$E[X^n] = \left.\frac{d^n M_X(t)}{dt^n}\right|_{t=0}$$

- $M'_X(0) = E[X]$
- $M''_X(0) = E[X^2]$
- $\text{Var}(X) = M''_X(0) - [M'_X(0)]^2$

## Properties

1. **Uniqueness:** If $M_X(t) = M_Y(t)$ for all $t$ in a neighborhood of 0, then $X$ and $Y$ have the same distribution.
2. **Sum of Independent RVs:** If $X$ and $Y$ are independent: $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$
3. **Linear transformation:** $M_{aX+b}(t) = e^{bt} \cdot M_X(at)$

## MGFs of Common Distributions

| Distribution | MGF $M_X(t)$ |
|---|---|
| Bernoulli($p$) | $1 - p + pe^t$ |
| Binomial($n$, $p$) | $(1 - p + pe^t)^n$ |
| Poisson($\lambda$) | $\exp(\lambda(e^t - 1))$ |
| Exponential($\lambda$) | $\frac{\lambda}{\lambda - t}$, $t < \lambda$ |
| Gaussian($\mu$, $\sigma^2$) | $\exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$ |

## Engineering Application: Sum of Independent Signals

If signal components $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent:

$$M_{X_1+X_2}(t) = M_{X_1}(t) \cdot M_{X_2}(t)$$
$$= \exp\left(\mu_1 t + \frac{\sigma_1^2 t^2}{2}\right) \cdot \exp\left(\mu_2 t + \frac{\sigma_2^2 t^2}{2}\right)$$
$$= \exp\left((\mu_1 + \mu_2)t + \frac{(\sigma_1^2 + \sigma_2^2)t^2}{2}\right)$$

This is the MGF of $N(\mu_1 + \mu_2,\, \sigma_1^2 + \sigma_2^2)$. Therefore $X_1 + X_2 \sim N(\mu_1 + \mu_2,\, \sigma_1^2 + \sigma_2^2)$.

# 4.10 Engineering Applications of Moments

## Application 1: Noise Power Characterization

A noise signal $N(t)$ is sampled. The samples have mean $\mu_N$ and variance $\sigma_N^2$.

- **DC component (bias):** $\mu_N$ — should be zero for ideal noise
- **AC noise power:** $\sigma_N^2$ — determines SNR
- **Total power:** $E[N^2] = \mu_N^2 + \sigma_N^2$

**SNR (Signal-to-Noise Ratio):**

$$\text{SNR} = \frac{E[S^2]}{\sigma_N^2} = \frac{\text{Signal Power}}{\text{Noise Power}}$$

## Application 2: Measurement Error

A sensor measures true value $\theta$ with error $E$:

- Measurement: $X = \theta + E$
- $E[X] = \theta + E[E]$ — bias in measurement
- $\text{Var}(X) = \text{Var}(E)$ — measurement precision

**Accuracy:** $|E[E]|$ (how close the average is to truth)
**Precision:** $\sigma_E$ (how repeatable measurements are)

## Application 3: Component Lifetime Statistics

Components have lifetime $T \sim \text{Exponential}(\lambda)$:

- Mean Time To Failure: $\text{MTTF} = E[T] = \frac{1}{\lambda}$
- Standard Deviation: $\sigma_T = \frac{1}{\lambda} = \text{MTTF}$
- For exponential: coefficient of variation $\text{CV} = \frac{\sigma}{\mu} = 1$ (always!)

## Application 4: Communication Signal Amplitude

Received signal $R = A + N$ where $A$ = signal amplitude, $N \sim N(0, \sigma^2)$:

- $E[R] = A$ (signal amplitude preserved on average)
- $\text{Var}(R) = \sigma^2$ (noise power determines variance)
- $\text{SNR} = \dfrac{A^2}{\sigma^2}$

## 4.11 Practice Problems

### Problem 1: Basic Computation

A discrete RV $X$ has PMF: $P(X=-1)=0.2$, $P(X=0)=0.5$, $P(X=1)=0.2$, $P(X=2)=0.1$.

(a) Find $E[X]$.
(b) Find $E[X^2]$.
(c) Find $\text{Var}(X)$.
(d) Find $E[3X+2]$.
(e) Find $\text{Var}(3X+2)$.

**Solution**

(a) $E[X] = (-1)(0.2) + (0)(0.5) + (1)(0.2) + (2)(0.1) = -0.2 + 0 + 0.2 + 0.2 = 0.2$
(b) $E[X^2] = (1)(0.2) + (0)(0.5) + (1)(0.2) + (4)(0.1) = 0.2 + 0 + 0.2 + 0.4 = 0.8$
(c) $\text{Var}(X) = E[X^2] - (E[X])^2 = 0.8 - 0.04 = 0.76$
(d) $E[3X+2] = 3E[X] + 2 = 3(0.2) + 2 = 2.6$
(e) $\text{Var}(3X+2) = 9 \cdot \text{Var}(X) = 9(0.76) = 6.84$

### Problem 2: Signal and Noise

A received signal $R = 2.5 + N$, where $N \sim N(0,\, 0.04)$.

(a) Find $E[R]$ and $\text{Var}(R)$.
(b) Find the SNR in dB.
(c) Find $P(R<0)$ — probability the signal goes negative.

**Solution**

(a) $E[R] = 2.5$, $\text{Var}(R) = \text{Var}(N) = 0.04$, $\sigma_R = 0.2$
(b) Signal power $= 2.5^2 = 6.25$, Noise power = 0.04
SNR $= \frac{6.25}{0.04} = 156.25 \rightarrow \text{SNR}_{dB} = 10 \cdot \log_{10}(156.25) = 21.94$ dB
(c) $P(R<0) = P(N<-2.5) = \Phi\left(-\frac{2.5}{0.2}\right) = \Phi(-12.5) \approx 0$ (extremely unlikely)

### Problem 3: Averaging Measurements

A sensor makes $n$ independent measurements of a constant voltage $V = 3.3\,\text{V}$.
Each measurement: $X_i = 3.3 + N_i$ where $N_i \sim N(0,\ 0.01)$.

(a) What is $E[\bar{X}]$ and $\text{Var}(\bar{X})$ for $n = 1$?
(b) How many measurements $n$ needed so that $\sigma_{\bar{X}} < 0.01\,\text{V}$?
(c) With $n = 100$, what is $P(|\bar{X} - 3.3| > 0.02)$?

**Solution**

(a) $E[\bar{X}] = 3.3,\ \text{Var}(\bar{X}) = \dfrac{0.01}{1} = 0.01,\ \sigma = 0.1\,\text{V}$

(b) $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} < 0.01 \rightarrow \sqrt{n} > 10 \rightarrow n > 100 \rightarrow$ need $n \geq 101$

(c) $\sigma_{\bar{X}} = \dfrac{0.1}{\sqrt{100}} = 0.01\,\text{V}$
$P(|\bar{X} - 3.3| > 0.02) = P(|Z| > 2) = 2(1 - \Phi(2)) = 0.0455$

### Problem 4: Component Lifetime

Components have lifetime $T \sim \text{Exp}(\lambda = 0.002$ per hour$)$.

(a) Find MTTF, $\sigma_T$, and coefficient of variation.
(b) Find $P(T > 1000\,\text{hours})$.
(c) Find $E[T^2]$ and interpret.

**Solution**

(a) MTTF $= \frac{1}{\lambda} = 500$ hours, $\sigma_T = 500$ hours, CV $= \frac{\sigma}{\mu} = 1$

(b) $P(T > 1000) = e^{-0.002 \times 1000} = e^{-2} = 0.1353$

(c) $E[T^2] = \text{Var}(T) + (E[T])^2 = 500^2 + 500^2 = 500000\,\text{hours}^2$
$\sqrt{E[T^2]} = 707$ hours (RMS lifetime)

**Problem 5: MGF Application**

$X$ has MGF $M_X(t) = \frac{1}{3}e^t + \frac{2}{3}e^{2t}$.

(a) What type of RV is $X$? Find its PMF.

(b) Find $E[X]$ using the MGF.

(c) Find $\text{Var}(X)$ using the MGF.

**Solution**

(a) $X$ is discrete. Comparing with $\sum p_X(x)e^{tx}$:

$$P(X=1) = \frac{1}{3},\ P(X=2) = \frac{2}{3}$$

(b) $M'(t) = \frac{1}{3}e^t + \frac{4}{3}e^{2t}$

$$E[X] = M'(0) = \frac{1}{3} + \frac{4}{3} = \frac{5}{3}$$

(c) $M''(t) = \frac{1}{3}e^t + \frac{8}{3}e^{2t}$

$$E[X^2] = M''(0) = \frac{1}{3} + \frac{8}{3} = 3$$

$$\text{Var}(X) = 3 - \frac{5}{3}^2 = 3 - \frac{25}{9} = \frac{2}{9}$$

## 4.12 Key Takeaways

1. $E[X]$ = **long-run average** — the center of the distribution, DC component
2. $\mathrm{Var}(X)$ = **spread/power** — for zero-mean signals, variance IS power
3. $\sigma$ **has same units as** $X$ — directly interpretable as RMS value
4. **Linearity of expectation** always holds; variance addition requires independence
5. **MGF uniquely identifies** a distribution and generates all moments
6. **In engineering:** mean $\rightarrow$ signal level, variance $\rightarrow$ noise power, std dev $\rightarrow$ RMS noise

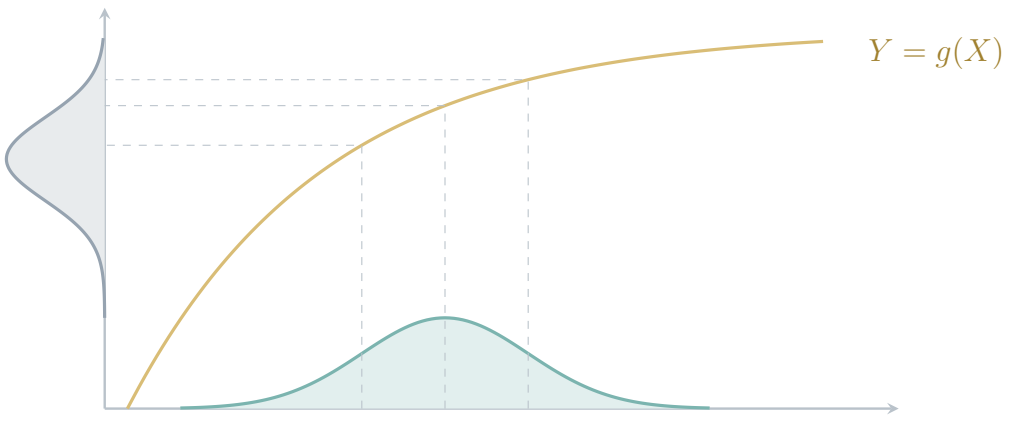
*Next Module: Module 5 — Important Random-Variable Functions*

Module 5

# Important Random-Variable Functions

توابع مهم متغیر تصادفی

5



## In This Module


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**Learning Objectives**

After completing this module, students will be able to:

1. Apply linear transformations to random variables
2. Derive PDFs of nonlinear transformations using the CDF and Jacobian methods
3. Apply transformations to engineering problems (power, SNR, envelope)
4. Verify analytical results with worked examples

## 5.1 Introduction: Why Transform Random Variables?

In engineering, we often know the statistics of one quantity but need the statistics of a **derived** quantity:

| Known | Needed | Transformation |
|---|---|---|
| Voltage $V$ | Power $P$ | $P = \frac{V^2}{R}$ |
| Linear SNR | SNR in dB | $Y = 10 \cdot \log_{10}(X)$ |
| Gaussian I, $Q$ | Envelope | $R = \sqrt{I^2 + Q^2}$ |
| Current I | Power | $P = I^2 R$ |
| Distance $d$ | Path loss | $L = d^{\alpha}$ |

## 5.2 Linear Transformations: $Y = aX + b$

### Result

If $X$ has PDF $f_X(x)$, then $Y = aX + b$ has:

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

### Mean and Variance

- $E[Y] = aE[X] + b$
- $\mathrm{Var}(Y) = a^2\,\mathrm{Var}(X)$

**Engineering Example: Amplifier**

Input signal $X \sim N(0,\, \sigma^2)$ passes through amplifier with gain $G = 10$ and DC offset $V_0 = 1.5\,\mathrm{V}$.
Output: $Y = 10X + 1.5$

- $Y \sim N(10 \cdot 0 + 1.5,\, 10^2 \cdot \sigma^2) = N(1.5,\, 100\sigma^2)$
- Output noise power increased by $G^2 = 100$

**Engineering Example: Temperature Sensor**

Sensor output $V = 0.01 \cdot T + 0.5$ (V in volts, $T$ in °C)
If $T \sim N(25,\, 4)$: $V \sim N(0.01 \times 25 + 0.5,\, 0.01^2 \times 4) = N(0.75,\, 0.0004)$

# 5.3 General Monotonic Transformations

## CDF Method (Most General)

For $Y = g(X)$, find $f_Y(y)$ by:

1. Write $F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$
2. Solve the inequality for $X$
3. Express in terms of $F_X$
4. Differentiate to get $f_Y(y)$

## Formula for Monotonic $g$

If $g$ is strictly increasing with inverse $g^{-1}$:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy}[g^{-1}(y)]$$

If $g$ is strictly decreasing:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left|\frac{d}{dy}[g^{-1}(y)]\right|$$

Combined (works for both):

$$f_Y(y) = \left.\frac{f_X(x)}{|g'(x)|}\right|_{x=g^{-1}(y)}$$

**Engineering Example: SNR in dB**

Linear SNR: $X \sim$ Exponential$(1)$ (normalized Rayleigh fading power).
dB SNR: $Y = 10 \cdot \log_{10}(X)$

- $g(x) = 10 \cdot \log_{10}(x)$, $g'(x) = \dfrac{10}{x \cdot \ln 10}$
- $g^{-1}(y) = 10^{y/10}$

$$f_Y(y) = \frac{f_X(10^{y/10})}{10/(10^{y/10} \cdot \ln 10)} = \frac{\ln 10}{10} \cdot 10^{y/10} \cdot e^{-10^{y/10}}$$

This is NOT Gaussian — it has a longer left tail (deep fades).

# 5.4 Nonlinear Transformations: Non-Monotonic Case

## When $Y = g(X)$ is Not One-to-One

If multiple x-values map to the same $y$, sum over all solutions:

$$f_Y(y) = \sum_i \frac{f_X(x_i)}{|g'(x_i)|}$$

where $x_1, x_2, \ldots$ are all roots of $g(x) = y$.

**Engineering Example: Power from Voltage ($Y = X^2$)**

Voltage $X \sim N(0, \sigma^2)$. Power: $Y = X^2$.
For $y > 0$, solutions are $x = +\sqrt{y}$ and $x = -\sqrt{y}$. $g'(x) = 2x$.

$$f_Y(y) = \frac{f_X(\sqrt{y})}{2\sqrt{y}} + \frac{f_X(-\sqrt{y})}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi}\sigma} \cdot \frac{e^{-y/(2\sigma^2)}}{\sqrt{y}}$$

This is a **Chi-squared distribution with 1 degree of freedom** (scaled by $\sigma^2$).

**Engineering Example: Full-Wave Rectifier ($Y = |X|$)**

Input $X \sim N(0, \sigma^2)$. Output $Y = |X|$.
For $y > 0$: $x = +y$ and $x = -y$ are solutions. $|g'(x)| = 1$.

$$f_Y(y) = f_X(y) + f_X(-y) = \frac{2}{\sigma\sqrt{2\pi}} e^{-y^2/(2\sigma^2)}, \quad y \geq 0$$

This is the **folded normal** (half-normal) distribution.

# 5.5 The Jacobian Method

## Procedure for $Y = g(X)$

1. Identify the transformation $y = g(x)$
2. Find the inverse: $x = g^{-1}(y)$
3. Compute the Jacobian: $J = |\frac{dx}{dy}| = |\frac{d[g^{-1}(y)]}{dy}|$
4. Apply: $f_Y(y) = f_X(g^{-1}(y)) \cdot |J|$

## Step-by-Step Example: Exponential of Gaussian

$X \sim N(\mu, \sigma^2)$. $Y = e^x$ (log-normal transformation).

1. $g(x) = e^x$ (monotonically increasing)
2. $x = \ln(y)$, valid for $y > 0$
3. $J = |\dfrac{dx}{dy}| = \dfrac{1}{y}$
4. $f_Y(y) = f_X(\ln y) \cdot \dfrac{1}{y} = \dfrac{1}{y\sigma\sqrt{2\pi}} \cdot \exp\left(-\dfrac{(\ln y - \mu)^2}{2\sigma^2}\right)$

This is the **log-normal distribution** — models many engineering quantities (shadowing in wireless, stock prices, component lifetimes with wear).

# 5.6 Engineering Application: Rayleigh from Gaussian

## Derivation of Rayleigh Distribution

In communications, the received signal has in-phase $(I)$ and quadrature $(Q)$ components:

- $I \sim N(0,\, \sigma^2)$, $Q \sim N(0,\, \sigma^2)$, independent

Envelope: $R = \sqrt{I^2 + Q^2}$

Using the transformation from $(I,\, Q) \to (R,\, \theta)$ with polar coordinates:

- $I = R \cdot \cos(\theta)$, $Q = R \cdot \sin(\theta)$
- Jacobian: $|\frac{\partial(I,Q)}{\partial(R,\theta)}| = R$

Joint PDF of $(R,\, \theta)$:

$$f_{R,\Theta}(r,\, \theta) = f_{I,Q}(r \cdot \cos\theta,\, r \cdot \sin\theta) \cdot r = \frac{r}{2\pi\sigma^2} \cdot e^{-r^2/(2\sigma^2)}$$

Marginalizing over $\theta \in [0,\, 2\pi)$:

$$f_R(r) = \frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)}, \quad r \geq 0$$

This is the **Rayleigh distribution** — fundamental to wireless fading channel modeling.

## 5.7 Practice Problems

**Problem 1**

Signal $X \sim \text{Uniform}[0,\,1]$. Output $Y = -2 \cdot \ln(X)$. Find the PDF of $Y$ and identify the distribution.

**Solution** CDF method: $F_Y(y) = P(-2\ln(X) \le y) = P(X \ge e^{-y/2}) = 1 - e^{-y/2}$ for $y \ge 0$. $f_Y(y) = \frac{1}{2}e^{-y/2} \to Y \sim \text{Exponential}(\beta = 2)$. (This is the inverse CDF method for generating exponentials!)

**Problem 2**

Noise voltage $X \sim N(0,\,\sigma^2)$ with $\sigma = 2\,\text{V}$. Power dissipated in $R = 50\,\Omega$: $P = \frac{X^2}{R}$.

(a) Find $E[P]$. (b) Find the PDF of $P$. (c) Find $P(P > 0.2\,\text{W})$.

**Solution**

(a) $E[P] = \dfrac{E[X^2]}{R} = \dfrac{\sigma^2}{R} = \dfrac{4}{50} = 0.08\,\text{W}$

(b) $P = \frac{X^2}{50}$. Let $W = X^2 \sim \sigma^2 \cdot \chi^2(1)$. Then $P = \frac{W}{50}$. $f_P(p) = 50 \cdot f_W(50p) = \frac{50}{2\sigma^2} \cdot \frac{50p}{\sigma^2}^{-1/2} \cdot e^{-50p/(2\sigma^2)}$ for $p > 0$.

(c) $P(P > 0.2) = P(\chi_1^2 > 2.5) = 1 - F_{\chi_1^2}(2.5) \approx 0.114$

**Problem 3**

$X \sim \text{Exponential}(\lambda = 1)$. $Y = \sqrt{X}$. Find $f_Y(y)$.

**Solution** $g(x) = \sqrt{x} \to x = y^2$, $\dfrac{dx}{dy} = 2y$.

$f_Y(y) = f_X(y^2) \cdot |2y| = e^{-y^2} \cdot 2y$ for $y \ge 0$. This is a Rayleigh distribution with $\sigma^2 = \frac{1}{2}$.

**Problem 4**

Distance $D \sim \text{Uniform}[1,\,10]$ km. Path loss: $L = 20 \cdot \log_{10}(D)$ dB. Find $E[L]$ and the PDF of $L$.

**Solution** $E[L] = E[20 \cdot \log_{10}(D)] = \displaystyle\int_1^{10} 20 \cdot \log_{10}(d) \cdot \frac{1}{9}\,dd = \frac{20}{9} \cdot \int_1^{10} \log_{10}(d)\,dd$

$= \frac{20}{9} \cdot \left[d \cdot \log_{10}(d) - \frac{d}{\ln(10)}\right]_1^{10} = \frac{20}{9} \cdot \left[10 - \frac{10}{\ln 10} + \frac{1}{\ln 10}\right] = \frac{20}{9} \cdot \left(10 - \frac{9}{\ln 10}\right) \approx 13.55\,\text{dB}$

PDF: $L$ ranges from 0 to 20 dB. $g^{-1}(l) = 10^{l/20}$, $|\frac{dg^{-1}}{dl}| = \frac{\ln 10}{20} \cdot 10^{l/20}$.

$f_L(l) = \frac{1}{9} \cdot \frac{\ln 10}{20} \cdot 10^{l/20}$ for $0 \le l \le 20$.

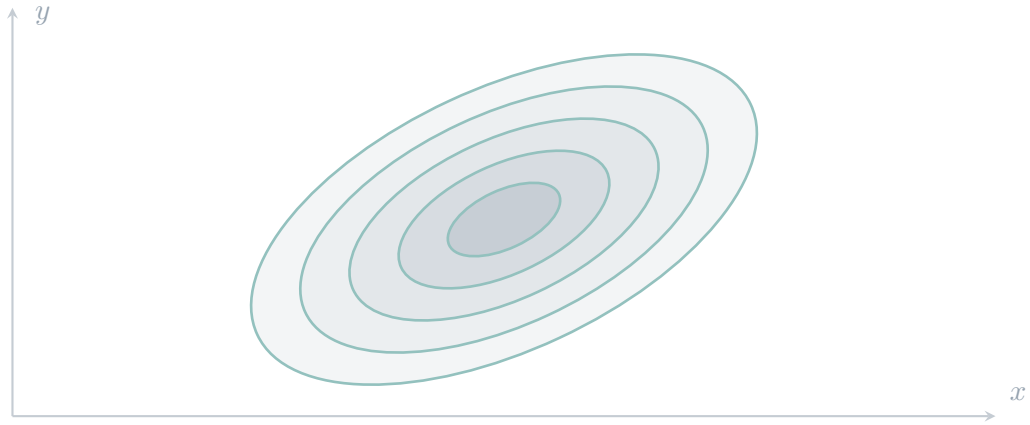
*Next Module: Module 6 — Two Random Variables*

Module 6

# Two Random Variables

دو متغیر تصادفی

6



## In This Module


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**Learning Objectives**

After completing this module, students will be able to:

1. Define and compute joint PMF, PDF, and CDF
2. Derive marginal distributions from joint distributions
3. Compute conditional distributions
4. Test independence of two random variables
5. Compute conditional expectations
6. Visualize and analyze joint distributions

## 6.1 Introduction: Why Study Joint Distributions?

Real engineering systems involve **multiple interacting uncertain quantities**:

- Voltage AND current in a noisy circuit
- Signal AND noise at a receiver
- Temperature AND resistance of a sensor
- I-channel AND Q-channel of a communication signal

Understanding the **joint behavior** tells us things marginals cannot — like correlation, dependence, and conditional behavior.

## 6.2 Joint PMF (Discrete Case)

**Definition**

For discrete RVs $X$, $Y$:

$$p_{X,Y}(x,y) = P(X = x, Y = y)$$

### Properties

1. $p_{X,Y}(x,\, y) \geq 0$
2. $\sum_x \Sigma_y p_{X,Y}(x,\, y) = 1$

### Visualization: Probability Table

**Example:** $X$ = number of retransmissions $(0,\, 1,\, 2)$, $Y$ = acknowledgment delay category $(1 = \text{fast},\, 2 = \text{slow})$

| | $Y = 1$ | $Y = 2$ | $p_X(x)$ |
|---|---|---|---|
| $X = 0$ | 0.30 | 0.10 | 0.40 |
| $X = 1$ | 0.15 | 0.20 | 0.35 |
| $X = 2$ | 0.05 | 0.20 | 0.25 |
| $p_Y(y)$ | 0.50 | 0.50 | 1.00 |

# 6.3 Joint PDF (Continuous Case)

**Definition**

For continuous RVs $X$, $Y$, the joint PDF $f_{X,Y}(x,\,y)$ satisfies:

$$P((X,Y)\in A)=\iint_A f_{X,Y}(x,y)\,dx\,dy$$

## Properties

1. $f_{X,Y}(x,\,y)\geq 0$
2. $\int\int f_{X,Y}(x,\,y)\,dx\,dy=1$

## Visualization

The joint PDF is a **surface** over the $(x,\,y)$ plane. Probability = volume under surface over a region.

**Engineering Example: Bivariate Gaussian**

Two correlated noise voltages $(X,\,Y)$ with correlation $\rho$:

$$f_{X,Y}(x,y)=\frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}\exp\left(-\frac{1}{2(1-\rho^2)}\left[\frac{x^2}{\sigma_X^2}-\frac{2\rho xy}{\sigma_X\sigma_Y}+\frac{y^2}{\sigma_Y^2}\right]\right)$$

## 6.4 Joint CDF

**Definition**

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

### Properties

1. $F_{X,Y}(-\infty,\, y) = 0,\; F_{X,Y}(x,\, -\infty) = 0$
2. $F_{X,Y}(\infty,\, \infty) = 1$
3. Non-decreasing in both arguments
4. $f_{X,Y}(x,\, y) = \dfrac{\partial^2 F_{X,Y}}{\partial x \partial y}$

# 6.5 Marginal Distributions

## Obtaining Marginals from Joint

**Discrete:**

$$p_X(x) = \sum_y p_{X,Y}(x,y), \quad p_Y(y) = \sum_x p_{X,Y}(x,y)$$

**Continuous:**

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx$$

## Visual Interpretation

The marginal PDF $f_X(x)$ is the **projection** (integral) of the joint surface onto the x-axis. Similarly $f_Y(y)$ projects onto the y-axis.

**Engineering Example**

Joint PDF: $f_{X,Y}(x,\,y) = 2e^{-x}e^{-2y}$, $x \geq 0$, $y \geq 0$
Marginals:

- $f_X(x) = \displaystyle\int_0^{\infty} 2e^{-x}e^{-2y}\,dy = 2e^{-x} \cdot \left[\frac{1}{2}\right] = e^{-x} \to X \sim \text{Exp}(1)$
- $f_Y(y) = \int_0^{\infty} 2e^{-x}e^{-2y}\,dx = 2e^{-2y} \cdot [1] = 2e^{-2y} \to Y \sim \text{Exp}(2)$

# 6.6 Conditional Distributions

## Discrete

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$$

## Continuous

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

## Interpretation

The conditional distribution describes $X$ when we **know** $Y = y$. It's a "slice" of the joint distribution at a fixed $y$ value, renormalized to integrate to 1.

**Engineering Example: Signal Given Channel State**

Received power $Y$ given channel state $X$:

- If channel is good ($X = 1$): $Y \sim N(10,\ 1)$
- If channel is poor ($X = 0$): $Y \sim N(2,\ 4)$
- $P(X = 1) = 0.7$, $P(X = 0) = 0.3$

Conditional PDFs describe receiver behavior in each state.

# 6.7 Independence of Two Random Variables

**Definition**

$X$ and $Y$ are independent if and only if:

$$f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y) \quad \text{for all } x, y$$

(Or equivalently for PMFs: $p_{X,Y}(x,\, y) = p_X(x) \cdot p_Y(y)$)

## Consequences of Independence

- $f_{X|Y}(x \mid y) = f_X(x)$ (knowing $Y$ doesn't change $X$)
- $E[XY] = E[X]E[Y]$
- $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$

## Testing Independence

Check: Does the joint factor into a product of a function of $x$ alone and $y$ alone?

- $f_{X,Y}(x,\, y) = 2e^{-x} \cdot e^{-2y} = [e^{-x}] \cdot [2e^{-2y}]$ ✔ Independent!
- $f_{X,Y}(x,\, y) = x + y$ for $0 \le x,\, y \le 1 \rightarrow$ cannot factor $\rightarrow$ NOT independent

**Engineering Example: Signal and Independent Noise**

Transmitted signal $S$ and channel noise $N$ are independent by physical assumption:
$f_{S,N}(s,\, n) = f_S(s) \cdot f_N(n)$
This independence assumption is fundamental to communication system design.

## 6.8 Conditional Expectation

**Definition**

$$E[X|Y=y] = \begin{cases} \sum_x x \cdot p_{X|Y}(x|y) & \text{discrete} \\ \int_{-\infty}^{\infty} x \cdot f_{X|Y}(x|y)\,dx & \text{continuous} \end{cases}$$

### Interpretation

$E[X \mid Y = y]$ is the **best prediction** of $X$ given that we observed $Y = y$ (in the mean-square sense).

**Engineering Example: Temperature-Dependent Resistance**

A resistor's value $R$ depends on temperature $T$. If $R \mid T \sim N(1000 + 0.5T,\, 4)$:
$E[R \mid T = 50°C] = 1000 + 0.5(50) = 1025\,\Omega$
Given temperature, we can predict resistance.

## 6.9 Practice Problems

**Problem 1**

Joint PDF: $f_{X,Y}(x,\,y) = 6(1-y)$ for $0 \le x \le y \le 1$, zero elsewhere.

(a) Verify normalization. (b) Find marginals. (c) Are X,Y independent? (d) Find $P(X < 0.5,\, Y < 0.5)$.

**Solution**

(a) $\int_0^1 \int_0^y 6(1-y)\,dx\,dy = \int_0^1 6y(1-y)\,dy = 6\left[\frac{y^2}{2} - \frac{y^3}{3}\right]_0^1 = 6\left(\frac{1}{2} - \frac{1}{3}\right) = 1$ ✔

(b) $f_X(x) = \int_x^1 6(1-y)\,dy = 6\left[\left(1 - \frac{y^2}{2}\right) - (y)\right]_x^1 = 3(1-x)^2$; $f_Y(y) = 6y(1-y)$

(c) $f_{X,Y} \neq f_X \cdot f_Y \to$ NOT independent (also, the support is triangular, not rectangular)

(d) $P(X < 0.5,\, Y < 0.5) = \displaystyle\int_0^{0.5}\int_0^{y} 6(1-y)\,dx\,dy = \int_0^{0.5} 6y(1-y)\,dy = 6\left[\frac{y^2}{2} - \frac{y^3}{3}\right]_0^{0.5} = 0.500$

**Problem 2**

Two sensors measure the same signal with independent noise. $X = S + N_1$, $Y = S + N_2$ where $S = 5\,\text{V}$, $N_1 \sim N(0,\,1)$, $N_2 \sim N(0,\,4)$. Are $X$ and $Y$ independent? Find $E[X]$, $E[Y]$, and explain why $\text{Cov}(X,\,Y) \neq 0$.

**Solution** $X$ and $Y$ are NOT independent because they share $S$ (though noise is independent). $E[X] = E[Y] = 5$. $\text{Cov}(X,\,Y) = \text{Cov}(S + N_1,\, S + N_2) = \text{Var}(S) + \text{Cov}(S,\, N_1) + \text{Cov}(N_2,\, S) + \text{Cov}(N_1,\, N_2)$. Since $S$ is constant, $\text{Var}(S) = 0$, so if $S$ were random they'd be correlated. With $S$ fixed, $X$ and $Y$ are actually independent in this case. If $S$ is random with $\text{Var}(S) = \sigma_S^2$, then $\text{Cov}(X,\,Y) = \sigma_S^2$.

**Problem 3**

Given joint PMF in the table of Section 6.2, find (a) $P(X \le 1,\, Y = 2)$, (b) $f_{X|Y}(x \mid Y = 1)$, (c) $E[X \mid Y = 1]$.

**Solution**

(a) $P(X \le 1,\, Y = 2) = p(0,\,2) + p(1,\,2) = 0.10 + 0.20 = 0.30$

(b) $p_{X|Y}(x \mid 1) = \frac{p(x,\,1)}{p_Y(1)}$: $p(0 \mid 1) = \frac{0.30}{0.50} = 0.60$, $p(1 \mid 1) = \frac{0.15}{0.50} = 0.30$, $p(2 \mid 1) = \frac{0.05}{0.50} = 0.10$

(c) $E[X \mid Y = 1] = 0(0.60) + 1(0.30) + 2(0.10) = 0.50$

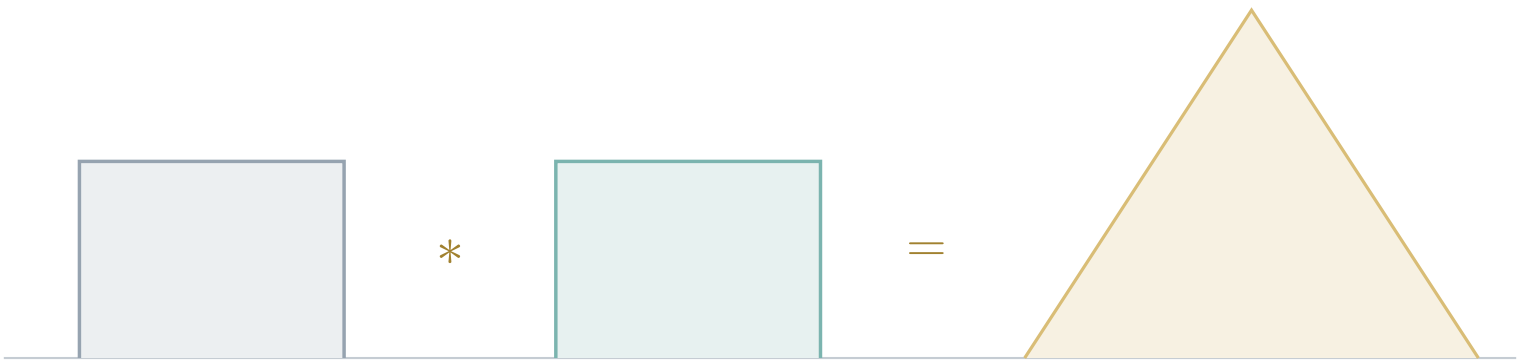
*Next Module: Module 7 — Functions of Two Random Variables*

Module 7

# Functions of Two Random Variables

توابع دو متغیر تصادفی

7



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**Learning Objectives**

After completing this module, students will be able to:

1. Derive the distribution of $Z = X + Y$ using convolution
2. Find distributions of differences, products, ratios, max, and min
3. Apply the bivariate Jacobian transformation method
4. Connect these results to engineering applications (signal+noise, reliability)

## 7.1 Introduction

Engineering systems combine random variables:

- **Received signal** = transmitted + noise: $R = S + N$
- **Total interference** = sum of multiple sources: $I = I_1 + I_2 + \ldots + I_n$
- **System lifetime** = minimum of component lifetimes: $T = \min(T_1, T_2)$
- **SNR** $= \dfrac{\text{signal power}}{\text{noise power}}$: ratio of RVs

We need methods to find distributions of **functions of two (or more) RVs**.

# 7.2 Sum of Two Random Variables: $Z = X + Y$

## CDF Method

$F_Z(z) = P(X + Y \le z) = \int\int_{x+y\le z} f_{X,Y}(x,\, y)\, dx\, dy$

## Convolution Formula (Independent Case)

If $X$ and $Y$ are **independent**:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z - x)\, dx = (f_X * f_Y)(z)$$

This is the **convolution** of $f_X$ and $f_Y$.

## Derivation

$F_Z(z) = P(X + Y \le z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_X(x) f_Y(y)\, dy\, dx$
$= \int_{-\infty}^{\infty} f_X(x) F_Y(z - x)\, dx$

Differentiating: $f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx$ ✔

## Important Special Cases

**Sum of two Gaussians (independent):**
$X \sim N(\mu_1,\, \sigma_1^2),\, Y \sim N(\mu_2,\, \sigma_2^2) \to Z = X + Y \sim N(\mu_1 + \mu_2,\, \sigma_1^2 + \sigma_2^2)$

**Sum of two exponentials (same rate):**
$X,\, Y \sim \text{Exp}(\lambda)$ independent $\to Z \sim \text{Gamma}\left(2,\, \frac{1}{\lambda}\right) = \text{Erlang}(2,\, \lambda)$

**Sum of two Poissons:**
$X \sim \text{Poisson}(\lambda_1),\, Y \sim \text{Poisson}(\lambda_2)$ independent $\to Z \sim \text{Poisson}(\lambda_1 + \lambda_2)$

**Sum of two uniforms:**
$X,\, Y \sim \text{Uniform}(0,\, 1)$ independent $\to Z$ has triangular distribution on [0,2]

**Engineering Example: Signal + Noise**

Transmitted signal $S \sim N(A,\, \sigma_s^2)$, noise $N \sim N(0,\, \sigma_n^2)$, independent.
Received: $R = S + N \sim N(A,\, \sigma_s^2 + \sigma_n^2)$
The received signal is still Gaussian — with increased variance (reduced SNR).

## 7.3 Difference: $Z = X - Y$

For independent $X$, $Y$:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(x - z)\, dx$$

**Gaussian case:** $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2) \to Z = X - Y \sim N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2)$

Note: variances still ADD (not subtract) because $\text{Var}(-Y) = \text{Var}(Y)$.

**Engineering Example: Differential Measurement**

Two sensors measure same quantity: $X_1 = \theta + N_1$, $X_2 = \theta + N_2$.
Difference: $X_1 - X_2 = N_1 - N_2 \sim N(0, \sigma_1^2 + \sigma_2^2)$
The difference eliminates the common signal but doubles the noise variance.

## 7.4 Product: $Z = XY$

For independent $X$, $Y$:

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{|x|} f_X(x) \cdot f_Y(z/x)\, dx$$

**Engineering Example: Power**

Voltage $V$ and current I in a linear circuit: $P = V \cdot I$.
If $V$ and I have known joint distribution, we can find the power distribution.

## 7.5 Ratio: $Z = \frac{X}{Y}$

For independent $X$, $Y$:

$$f_Z(z) = \int_{-\infty}^{\infty} |y| \cdot f_X(zy) \cdot f_Y(y)\, dy$$

**Engineering Example: SNR**

$$\text{SNR} = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{P_s}{P_n}$$

If both are chi-squared distributed (sum of squared Gaussians), the ratio follows an **F-distribution**.

## 7.6 Maximum: $Z = \max(X, Y)$

### CDF Approach

$F_Z(z) = P(\max(X, Y) \leq z) = P(X \leq z \text{ AND } Y \leq z)$

If independent: $F_Z(z) = F_X(z) \cdot F_Y(z)$

PDF: $f_Z(z) = f_X(z)F_Y(z) + F_X(z)f_Y(z)$

**Engineering Example: Parallel Redundancy**

System works if AT LEAST ONE component works. System fails only when ALL fail.
System lifetime $= \max(T_1, T_2)$ for parallel redundant components.
For $T_1, T_2 \sim \text{Exp}(\lambda)$ independent:
$F_Z(z) = (1 - e^{-\lambda z})^2$ for $z \geq 0$
$f_Z(z) = 2\lambda e^{-\lambda z}(1 - e^{-\lambda z})$
Mean lifetime: $E[\max] = \frac{3}{2\lambda} > \frac{1}{\lambda} = E[\text{single component}]$ (reliability improved!)

# 7.7 Minimum: $Z = \min(X, Y)$

## Survival Function Approach

$P(Z > z) = P(\min(X, Y) > z) = P(X > z \text{ AND } Y > z)$

If independent: $P(Z > z) = P(X > z) \cdot P(Y > z) = [1 - F_X(z)][1 - F_Y(z)]$

CDF: $F_Z(z) = 1 - [1 - F_X(z)][1 - F_Y(z)]$

**Engineering Example: Series System**

System fails when FIRST component fails. System lifetime $= \min(T_1, T_2)$.
For $T_1 \sim \text{Exp}(\lambda_1)$, $T_2 \sim \text{Exp}(\lambda_2)$ independent:
$P(Z > z) = e^{-\lambda_1 z} \cdot e^{-\lambda_2 z} = e^{-(\lambda_1+\lambda_2)z}$
Therefore: $\min(T_1, T_2) \sim \text{Exp}(\lambda_1 + \lambda_2)$

**Key result:** Failure rates ADD in series systems. Mean lifetime $= \frac{1}{\lambda_1+\lambda_2} < \min\left(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}\right)$.

# 7.8 General Bivariate Transformation (Jacobian Method)

## Setup

Given $(X,\, Y)$ with known joint PDF, find the joint PDF of $(U,\, V)$ where:

- $U = g_1(X,\, Y)$
- $V = g_2(X,\, Y)$

## Procedure

1. Solve for the inverse: $X = h_1(U,\, V)$, $Y = h_2(U,\, V)$
2. Compute the Jacobian determinant:

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

3. Apply:
$$f_{U,V}(u, v) = f_{X,Y}(h_1(u, v), h_2(u, v)) \cdot |J|$$
4. If only $U$ is needed, marginalize out $V$.

**Example: Sum and Difference**

$$U = X + Y,\, V = X - Y \rightarrow X = \frac{U+V}{2},\, Y = \frac{U-V}{2}$$
$$J = \left|\frac{\partial x}{\partial u} \cdot \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \cdot \frac{\partial y}{\partial u}\right| = \left|\frac{1}{2}\left(-\frac{1}{2}\right) - \frac{1}{2}\left(\frac{1}{2}\right)\right| = \frac{1}{2}$$
$$f_{U,V}(u,\, v) = \frac{1}{2} \cdot f_{X,Y}\left(\frac{u+v}{2},\, \frac{u-v}{2}\right)$$

## 7.9 Practice Problems

**Problem 1**

$X \sim \text{Exp}(1)$, $Y \sim \text{Exp}(1)$, independent. Find the PDF of $Z = X + Y$.

**Solution** Convolution: $f_Z(z) = \int_0^z e^{-x} \cdot e^{-(z-x)}\, dx = \int_0^z e^{-z}\, dx = ze^{-z}$, $z \geq 0$.

This is Gamma$(2,\, 1)$ = Erlang$(2,\, 1)$. $E[Z] = 2$, $\text{Var}(Z) = 2$.

**Problem 2**

Three components in series with failure rates $\lambda_1 = 0.001$, $\lambda_2 = 0.002$, $\lambda_3 = 0.003$ per hour.

(a) Find system MTTF. (b) Find $P(\text{system survives 100 hours})$.

**Solution**

(a) MTTF $= \frac{1}{0.001+0.002+0.003} = \frac{1}{0.006} = 166.7$ hours

(b) $P(T_{\text{sys}} > 100) = e^{-0.006\times 100} = e^{-0.6} = 0.549$

**Problem 3**

$X \sim N(5,\, 4)$, $Y \sim N(3,\, 9)$, independent. Find distribution and $P(X + Y > 12)$.

**Solution** $Z = X + Y \sim N(8,\, 13)$. $P(Z > 12) = P\left(Z' > \frac{12-8}{\sqrt{13}}\right) = Q(1.109) = 0.1337$.

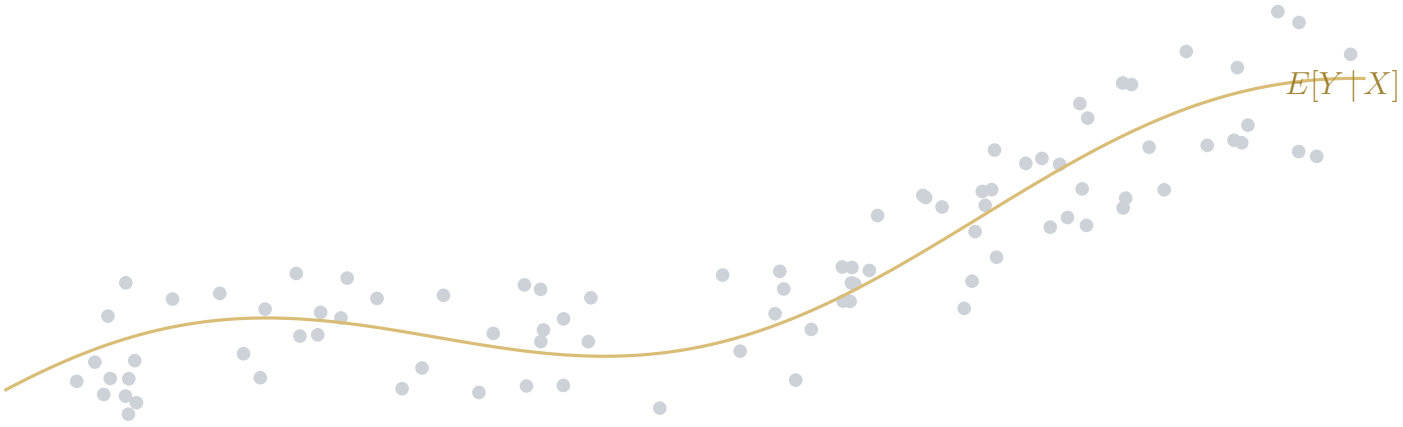
*Next Module: Module 8 — Conditional Expectation and Variance*

Module 8

# Conditional Expectation and Variance

امید ریاضی و واریانس شرطی

8



## In This Module

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**8.2** Conditional Expectation $E[X \mid Y = y]$ . . . 131
**8.3** Conditional Variance $\text{Var}(X \mid Y = y)$ . . . 132
**8.4** Law of Total Expectation (Tower Property) . . . 133
**8.5** Law of Total Variance . . . 134
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**Learning Objectives**

After completing this module, students will be able to:

1. Compute conditional expectations and variances
2. Apply the Law of Total Expectation
3. Apply the Law of Total Variance
4. Use these tools for engineering analysis across multiple operating conditions

## 8.1 Introduction

Many engineering systems operate under **different conditions** (channel states, modes, environments). We often need the **overall average performance** computed from conditional performance in each state.

- What is the average BER when the channel randomly switches between good and bad?
- What is the overall measurement variance when the sensor operates in different environments?

## 8.2 Conditional Expectation $E[X \mid Y = y]$

**Definition**

**Continuous:**

$$E[X|Y=y] = \int_{-\infty}^{\infty} x \cdot f_{X|Y}(x|y)\, dx$$

**Discrete:**

$$E[X|Y=y] = \sum_{x} x \cdot p_{X|Y}(x|y)$$

### Interpretation

$E[X \mid Y = y]$ is the **average value of** $X$ when we know $Y = y$. It is a function of $y$.

### $E[X \mid Y]$ as a Random Variable

When $Y$ is itself random, $E[X \mid Y]$ is a **random variable** — it takes the value $E[X \mid Y = y]$ when $Y = y$.

**Engineering Example**

Average received power given channel state:

- Good channel ($Y = 1$): $E[P \mid Y = 1] = 10$ mW
- Poor channel ($Y = 0$): $E[P \mid Y = 0] = 1$ mW
- $P(Y = 1) = 0.7$, $P(Y = 0) = 0.3$

## 8.3 Conditional Variance $\text{Var}(X \mid Y = y)$

**Definition**

$$\text{Var}(X|Y=y) = E[(X - E[X|Y=y])^2|Y=y] = E[X^2|Y=y] - (E[X|Y=y])^2$$

### Interpretation

The conditional variance measures the **remaining uncertainty in** $X$ after $Y$ is known.

**Engineering Example**

Measurement precision depends on temperature:

- At $T = 20°C$: $\text{Var}(X \mid T = 20) = 0.01$ (high precision)
- At $T = 80°C$: $\text{Var}(X \mid T = 80) = 0.09$ (low precision due to thermal effects)

# 8.4 Law of Total Expectation (Tower Property)

## Statement

$$E[X] = E[E[X|Y]]$$

**Discrete version:**

$$E[X] = \sum_y E[X|Y=y] \cdot P(Y=y)$$

**Continuous version:**

$$E[X] = \int_{-\infty}^{\infty} E[X|Y=y] \cdot f_Y(y)\, dy$$

## Proof (Discrete)

$E[E[X \mid Y]] = \sum_y E[X \mid Y=y] \cdot P(Y=y)$
$= \sum_y [\sum_x x \cdot P(X=x \mid Y=y)] \cdot P(Y=y)$
$= \sum_x \Sigma_y x \cdot P(X=x, Y=y)$
$= \sum_x x \cdot P(X=x) = E[X]$ ✔

## Interpretation

The **overall average** equals the **weighted average of conditional averages**, weighted by the probability of each condition.

**Engineering Example: Average BER Across Channel States**

A wireless system switches between three modes:

- Mode 1 (60% of time): BER $= 10^{-6}$
- Mode 2 (30% of time): BER $= 10^{-4}$
- Mode 3 (10% of time): BER $= 10^{-2}$

Overall BER $= E[\text{BER}] = 0.6(10^{-6}) + 0.3(10^{-4}) + 0.1(10^{-2})$
$= 0.6 \times 10^{-6} + 30 \times 10^{-6} + 10000 \times 10^{-6} \approx 1.003 \times 10^{-3}$
**Engineering insight:** The overall BER is dominated by the worst mode, even though it occurs only 10% of the time.

**Engineering Example: Average Lifetime with Defect Types**

Components are defect-free (90%) or have a hidden defect (10%):

- Defect-free: $E[T \mid \text{good}] = 10000$ hours
- With defect: $E[T \mid \text{defect}] = 500$ hours

$E[T] = 0.9(10000) + 0.1(500) = 9050$ hours

# 8.5 Law of Total Variance

## Statement

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y])$$

| Term | Name | Interpretation |
|---|---|---|
| $\text{Var}(X)$ | Total variance | Overall uncertainty in $X$ |
| $E[\text{Var}(X \mid Y)]$ | Expected conditional variance | Average "within-group" variability |
| $\text{Var}(E[X \mid Y])$ | Variance of conditional means | "Between-group" variability |

## Proof

$\text{Var}(X) = E[X^2] - (E[X])^2$

$E[X^2] = E[E[X^2 \mid Y]]$ (tower property)
$= E[\text{Var}(X \mid Y) + (E[X \mid Y])^2]$
$= E[\text{Var}(X \mid Y)] + E[(E[X \mid Y])^2]$

$(E[X])^2 = (E[E[X \mid Y]])^2$

$\text{Var}(X) = E[\text{Var}(X \mid Y)] + E[(E[X \mid Y])^2] - (E[E[X \mid Y]])^2$
$= E[\text{Var}(X \mid Y)] + \text{Var}(E[X \mid Y])$ ✔

## Interpretation: Decomposition of Uncertainty

Total uncertainty = Average uncertainty within each condition + Variability of the mean across conditions

**Engineering Example: Sensor in Two Environments**

A sensor operates indoors ($Y = 1$, prob 0.6) and outdoors ($Y = 2$, prob 0.4):

- Indoor: $E[X \mid Y = 1] = 25°C$, $\text{Var}(X \mid Y = 1) = 1$
- Outdoor: $E[X \mid Y = 2] = 15°C$, $\text{Var}(X \mid Y = 2) = 16$

**$E[\text{Var}(X \mid Y)] * * = 0.6(1) + 0.4(16) = 7.0$ (average within-condition variance)
**$\text{Var}(E[X \mid Y]) * * = E[(E[X \mid Y])^2] - (E[E[X \mid Y]])^2$
$= 0.6(25^2) + 0.4(15^2) - [0.6(25) + 0.4(15)]^2 = 375 + 90 - (21)^2 = 465 - 441 = 24$
**$\text{Var}(X) * * = 7.0 + 24.0 = 31.0$
Most variance comes from the **between-condition** difference (24 out of 31).

## 8.6 Practice Problems

**Problem 1**

A digital link operates over two channel types:

- Type $A$ (prob 0.7): $E[\text{delay} \mid A] = 5\,\text{ms}$, $\text{Var}(\text{delay} \mid A) = 2$
- Type $B$ (prob 0.3): $E[\text{delay} \mid B] = 20\,\text{ms}$, $\text{Var}(\text{delay} \mid B) = 25$

Find (a) $E[\text{delay}]$, (b) $\text{Var}(\text{delay})$.

**Solution**

(a) $E[D] = 0.7(5) + 0.3(20) = 3.5 + 6 = 9.5$ ms

(b) $E[\text{Var}(D \mid \text{type})] = 0.7(2) + 0.3(25) = 1.4 + 7.5 = 8.9$

$\text{Var}(E[D \mid \text{type}]) = 0.7(5^2) + 0.3(20^2) - 9.5^2 = 17.5 + 120 - 90.25 = 47.25$

$\text{Var}(D) = 8.9 + 47.25 = 56.15$

**Problem 2**

A random number $N$ of components are tested. $N \sim \text{Poisson}(10)$. Each has independent failure probability $p = 0.05$. Let $X$ = number of failures. Find $E[X]$ and $\text{Var}(X)$.

**Solution** $X \mid N \sim \text{Binomial}(N, 0.05)$. $E[X \mid N] = 0.05\,\text{N}$. $\text{Var}(X \mid N) = N(0.05)(0.95) = 0.0475\,\text{N}$.

$E[X] = E[E[X \mid N]] = E[0.05\,\text{N}] = 0.05(10) = 0.5$

$\text{Var}(X) = E[\text{Var}(X \mid N)] + \text{Var}(E[X \mid N]) = E[0.0475\,\text{N}] + \text{Var}(0.05\,\text{N}) = 0.0475(10) + 0.05^2(10) = 0.475 + 0.025 = 0.5$

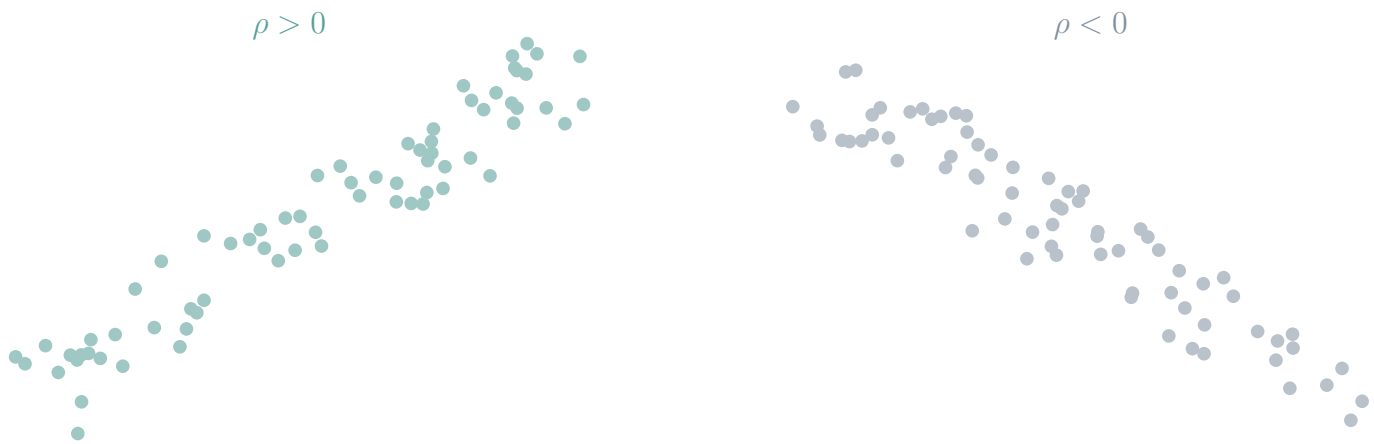
*Next Module: Module 9 — Covariance and Correlation*

Module 9

# Covariance and Correlation

کوواریانس و همبستگی

9



## In This Module


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**Learning Objectives**

After completing this module, students will be able to:

1. Compute and interpret covariance
2. Compute and interpret the correlation coefficient
3. Distinguish between independence and uncorrelatedness
4. Construct and interpret correlation matrices
5. Apply correlation analysis to engineering problems

## 9.1 Introduction: Measuring Linear Relationships

When two random variables are not independent, we need to quantify **how they relate**. Covariance and correlation measure the **linear relationship** between two RVs.

Engineering examples:

- Correlated noise in adjacent sensor channels
- Signal measurements at two antennas (MIMO correlation)
- Voltage and current in a circuit (related through impedance)
- Temperature and device performance (thermal dependence)

## 9.2 Covariance

**Definition**

$$\text{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)] = E[XY] - E[X]E[Y]$$

### Properties

1. $\text{Cov}(X,\, X) = \text{Var}(X)$
2. $\text{Cov}(X,\, Y) = \text{Cov}(Y,\, X)$ (symmetric)
3. $\text{Cov}(\text{aX} + b,\, \text{cY} + d) = \text{ac} \cdot \text{Cov}(X,\, Y)$
4. If $X,\, Y$ independent $\rightarrow \text{Cov}(X,\, Y) = 0$
5. $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\,\text{Cov}(X,\, Y)$
6. $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2\,\text{Cov}(X,\, Y)$

### Sign Interpretation

| $\text{Cov}(X,\, Y)$ | Meaning |
|---|---|
| $> 0$ | $X$ and $Y$ tend to increase together |
| $< 0$ | When $X$ increases, $Y$ tends to decrease |
| $= 0$ | No linear relationship (uncorrelated) |

**Engineering Example: Amplifier Gain and Output**

If gain $G$ varies randomly and input is fixed at $V_{\text{in}}$:

- Output: $V_{\text{out}} = G \cdot V_{\text{in}}$
- $\text{Cov}(G,\, V_{\text{out}}) = \text{Cov}(G,\, G \cdot V_{\text{in}}) = V_{\text{in}} \cdot \text{Var}(G) > 0$

Higher gain $\rightarrow$ higher output (positive covariance).

## 9.3 Correlation Coefficient

**Definition**

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

### Properties

1. **Bounded:** $-1 \leq \rho \leq 1$
2. $\rho = 1$**:** Perfect positive linear relationship ($Y$ = aX $+b$, $a > 0$)
3. $\rho = -1$**:** Perfect negative linear relationship ($Y$ = aX $+b$, $a < 0$)
4. $\rho = 0$**:** Uncorrelated (no linear relationship)
5. **Dimensionless:** Unlike covariance, $\rho$ doesn't depend on units

### Interpretation Scale

| | $\rho$ |
|---|---|
| 0.0 – 0.2 | Very weak/negligible |
| 0.2 – 0.4 | Weak |
| 0.4 – 0.6 | Moderate |
| 0.6 – 0.8 | Strong |
| 0.8 – 1.0 | Very strong |

**Engineering Example: Correlated Fading**

Two antenna paths with correlation $\rho = 0.3$:

- Low correlation $\rightarrow$ diversity works well (independent fading)
- High correlation $\rightarrow$ diversity gain is reduced

MIMO system design requires: $\rho < 0.5$ for effective spatial multiplexing.

## 9.4 Cross-Correlation

**Definition**

$$R_{XY} = E[XY]$$

### Relationship to Covariance

$$\text{Cov}(X, Y) = R_{XY} - \mu_X \mu_Y$$

If either $X$ or $Y$ has zero mean: $\text{Cov}(X, Y) = E[XY] = R_{XY}$

### Engineering Significance

Cross-correlation is fundamental in:

- Matched filters (correlation receivers in communications)
- Radar signal processing (detecting time delay)
- Signal detection (correlating received with known template)

# 9.5 Independence vs. Uncorrelatedness

## The Critical Distinction

| Statement | Always True? |
|---|---|
| Independent → Uncorrelated | ✔ YES |
| Uncorrelated → Independent | ✘ NO (in general) |
| Uncorrelated → Independent (jointly Gaussian) | ✔ YES |

## Why Independent → Uncorrelated

If $X$, $Y$ independent: $E[XY] = E[X]E[Y]$, so $\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0$.

## Counterexample: Uncorrelated but Dependent

Let $X \sim N(0, 1)$ and $Y = X^2$. Then:

- $\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = E[X^3] - 0 = 0$ (since $X$ is symmetric)
- But $Y$ is completely determined by $X$ → maximally dependent!

$\rho = 0$ **does NOT mean independent** (except for jointly Gaussian RVs).

## The Gaussian Exception

For **jointly Gaussian** random variables:

$$\text{Uncorrelated} \Leftrightarrow \text{Independent}$$

This is why the Gaussian distribution is so convenient in engineering — checking $\rho = 0$ suffices to establish independence.

## Engineering Implication

In communications with Gaussian noise:

- If noise samples are uncorrelated → they are independent
- This simplifies analysis enormously
- For non-Gaussian interference: uncorrelated does NOT guarantee independence

## 9.6 Correlation Matrix

**Definition for Random Vector $X = [X_1, X_2, \ldots, X_n]^T$**

The **covariance matrix** $\Sigma$:

$$\Sigma_{ij} = \text{Cov}(X_i, X_j)$$

$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots \\ \vdots & & \ddots \end{bmatrix}$$

The **correlation matrix** $R$ (normalized):

$$R_{ij} = \rho_{X_i X_j} = \frac{\Sigma_{ij}}{\sqrt{\Sigma_{ii}\Sigma_{jj}}}$$

### Properties of Covariance Matrix

1. Symmetric: $\Sigma = \Sigma^T$
2. Positive semi-definite: $x^T \Sigma x \geq 0$ for all $x$
3. Diagonal entries = variances
4. Off-diagonal entries = covariances

**Engineering Example: Three Sensors**

Three temperature sensors with:

- $\sigma_1 = \sigma_2 = \sigma_3 = 1°C$
- $\rho_{12} = 0.8$ (sensors 1,2 are nearby)
- $\rho_{13} = 0.3$ (sensor 3 is farther)
- $\rho_{23} = 0.4$

Correlation matrix:

$$R = \begin{bmatrix} 1.0 & 0.8 & 0.3 \\ 0.8 & 1.0 & 0.4 \\ 0.3 & 0.4 & 1.0 \end{bmatrix}$$

# 9.7 Variance of Linear Combinations

## General Formula

$$\text{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 \text{Var}(X_i) + 2\sum_{i<j} a_i a_j \text{Cov}(X_i, X_j)$$

In matrix form: $\text{Var}(a^T X) = a^T \Sigma a$

## Special Case: Sum of Two RVs

$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\,\text{Cov}(X,\,Y) = \sigma_X^2 + \sigma_Y^2 + 2\rho\sigma_X\sigma_Y$

| Correlation | $\text{Var}(X+Y)$ | Effect |
|---|---|---|
| $\rho = 1$ | $(\sigma_X + \sigma_Y)^2$ | Maximum — variances add constructively |
| $\rho = 0$ | $\sigma_X^2 + \sigma_Y^2$ | Independent addition |
| $\rho = -1$ | $(\sigma_X - \sigma_Y)^2$ | Minimum — cancellation |

## Engineering Application: Averaging Correlated Sensors

If $n$ sensors have equal variance $\sigma^2$ and pairwise correlation $\rho$:

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}[1 + (n-1)\rho]$$

- If $\rho = 0$ (independent): $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$ (full averaging benefit)
- If $\rho = 1$ (identical): $\text{Var}(\bar{X}) = \sigma^2$ (no benefit from averaging!)
- If $\rho = 0.5$, $n = 4$: $\text{Var}(\bar{X}) = \frac{\sigma^2(1+3\times0.5)}{4} = 0.625\sigma^2$ (reduced benefit)

## 9.8 Practice Problems

**Problem 1**

$X$ and $Y$ have: $E[X] = 2$, $E[Y] = 3$, $\text{Var}(X) = 4$, $\text{Var}(Y) = 9$, $E[XY] = 8$.

(a) Find $\text{Cov}(X, Y)$. (b) Find $\rho$. (c) Find $\text{Var}(X + Y)$. (d) Find $\text{Var}(2X - 3Y)$.

**Solution**

(a) $\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 8 - 2(3) = 2$

(b) $\rho = \dfrac{2}{\sqrt{4} \cdot \sqrt{9}} = \dfrac{2}{6} = \dfrac{1}{3}$

(c) $\text{Var}(X + Y) = 4 + 9 + 2(2) = 17$

(d) $\text{Var}(2X - 3Y) = 4(4) + 9(9) + 2(2)(-3)(2) = 16 + 81 - 24 = 73$

**Problem 2**

Four identical sensors ($\sigma^2 = 1$) have pairwise correlation $\rho = 0.5$. What is $\text{Var}(\bar{X})$?

**Solution** $\text{Var}(\bar{X}) = \dfrac{\sigma^2}{n} \cdot [1 + (n-1)\rho] = \dfrac{1}{4} \cdot [1 + 3(0.5)] = 0.25 \times 2.5 = 0.625$

Compare: if independent ($\rho = 0$): $\text{Var}(\bar{X}) = 0.25$. Correlation reduces the effectiveness of averaging.

**Problem 3**

Given $X \sim N(0, 1)$, define $Y = X$ when $|X| < 1$ and $Y = -X$ when $|X| \geq 1$. Show that $\text{Cov}(X, Y) \neq 0$ in this case but that $E[X \cdot Y]$ can be computed.

**Solution** $E[XY] = E[X^2 \cdot 1(|X| < 1)] + E[-X^2 \cdot 1(|X| \geq 1)] = E[X^2 \cdot 1(|X| < 1)] - E[X^2 \cdot 1(|X| \geq 1)]$

$= P(|X| < 1)E[X^2 || X\ | < 1] - P(|X| \geq 1)E[X^2 || X\ | \geq 1]$

This is nonzero since $E[X^2 || X\ | < 1] \neq E[X^2 || X\ | \geq 1]$, so $\text{Cov}(X, Y) = E[XY] - 0 \neq 0$.

$X$ and $Y$ are clearly dependent ($Y$ is defined from $X$), and in this case also correlated.

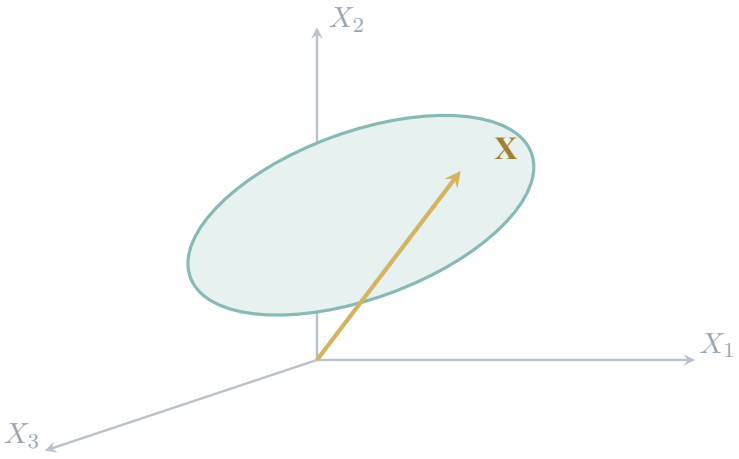
*Next Module: Module 10 — Multiple Random Variables*

Module 10

# Multiple Random Variables

چند متغیر تصادفی

10



## In This Module

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**Learning Objectives**

After completing this module, students will be able to:

1. Extend joint distributions to $N$ random variables
2. Define and work with random vectors
3. Compute mean vectors and covariance matrices
4. Apply the multivariate Gaussian distribution
5. Perform linear transformations of random vectors

## 10.1 Introduction: From 2 to $N$ Variables

Real systems often involve many uncertain quantities simultaneously:

- **Sensor array:** $N$ sensors each producing a noisy measurement
- **MIMO system:** $M \times N$ channel coefficients
- **Parameter estimation:** $p$ unknown parameters
- **Signal samples:** $N$ consecutive noise samples

We need a framework for **N-dimensional** random vectors.

## 10.2 Joint Distributions for $N$ Random Variables

### Joint PDF

For $X_1, X_2, \ldots, X_n$:

$$f_{X_1,\ldots,X_n}(x_1,\ldots,x_n) \geq 0, \quad \int \cdots \int f_{X_1,\ldots,X_n}\, dx_1 \cdots dx_n = 1$$

### Marginal Distributions

Integrate (or sum) out unwanted variables:

$$f_{X_1}(x_1) = \int \cdots \int f_{X_1,\ldots,X_n}(x_1,\ldots,x_n)\, dx_2 \cdots dx_n$$

### Conditional Distributions

$$f_{X_1|X_2,\ldots,X_n}(x_1|x_2,\ldots,x_n) = \frac{f_{X_1,\ldots,X_n}(x_1,\ldots,x_n)}{f_{X_2,\ldots,X_n}(x_2,\ldots,x_n)}$$

### Mutual Independence

$X_1, \ldots, X_n$ are mutually independent if:

$$f_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = \prod_{i=1}^{n} f_{X_i}(x_i)$$

## 10.3 Random Vectors

**Definition**

A **random vector** is an ordered collection of random variables:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix}$$

### Engineering Examples

| Random Vector | Components | Context |
|---|---|---|
| Sensor array output | $X_1, \ldots, X_n$ = sensor readings | Array processing |
| MIMO channel | $h_1, \ldots, h_n$ = channel gains | Wireless communications |
| Noise vector | $N_1, \ldots, N_n$ = noise samples | Signal processing |
| State vector | Position, velocity, acceleration | Kalman filtering |
| Feature vector | Extracted measurements | Pattern recognition |

## 10.4 Mean Vector

**Definition**

$$\boldsymbol{\mu} = E[\mathbf{X}] = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_n] \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix}$$

### Properties

- $E[AX + b] = \mathrm{AE[X]} + b = A\mu + b$
- Linearity extends to vectors and matrices

# 10.5 Covariance Matrix

**Definition**

$$\boldsymbol{\Sigma} = E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T]$$

Element $(i, j)$: $\Sigma_{ij} = \text{Cov}(X_i, X_j)$

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_n) \\ \text{Cov}(X_2, X_1) & \sigma_2^2 & \cdots & \text{Cov}(X_2, X_n) \\ \vdots & & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \cdots & \cdots & \sigma_n^2 \end{bmatrix}$$

## Properties

1. **Symmetric:** $\Sigma = \Sigma^T$
2. **Positive semi-definite:** $v^T \Sigma v \geq 0$ for all $v$
3. **Diagonal = variances:** $\Sigma_{ii} = \text{Var}(X_i)$
4. **Off-diagonal = covariances:** $\Sigma_{ij} = \text{Cov}(X_i, X_j)$
5. **Independent components $\rightarrow$ diagonal** $\Sigma$

## Alternative Formula

$$\boldsymbol{\Sigma} = E[\mathbf{X}\mathbf{X}^T] - \boldsymbol{\mu}\boldsymbol{\mu}^T$$

## 10.6 Multivariate Gaussian Distribution

**Definition**

$X \sim N(\mu, \Sigma)$ has PDF:

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{n/2}|\mathbf{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\mathbf{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$

where $|\Sigma|$ = determinant of $\Sigma$.

### Key Properties

1. **Marginals are Gaussian:** Any subset of components is jointly Gaussian
2. **Conditionals are Gaussian:** $X_1 \mid X_2 = x_2$ is Gaussian
3. **Linear transformations:** $Y = \mathrm{AX} + b \sim N(A\mu + b,\ A\Sigma A^T)$
4. **Uncorrelated = Independent** (unique to Gaussian!)
5. **Fully specified** by $\mu$ and $\Sigma$ (only 2 parameters needed per variable plus correlations)

### Why So Important?

- Central Limit Theorem: sums of many RVs $\rightarrow$ multivariate Gaussian
- Thermal noise vectors are multivariate Gaussian
- Analytically tractable (closed-form marginals, conditionals)
- Optimal filters (Kalman, Wiener) assume Gaussian

### $2D$ Case (Bivariate Gaussian)

For $n = 2$ with $\mu = [0,\ 0]^T$, $\sigma_1 = \sigma_2 = 1$:

$$f(x, y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right)$$

Contours are **ellipses** (tilted when $\rho \neq 0$).

# 10.7 Linear Transformations of Random Vectors

## Transformation $Y = AX + b$

If $X$ is a random vector with mean $\mu_X$ and covariance $\Sigma_X$:

$$E[\mathbf{Y}] = A\boldsymbol{\mu}_X + \mathbf{b}$$

$$\boldsymbol{\Sigma}_Y = A\boldsymbol{\Sigma}_X A^T$$

## Engineering Applications

**Beamforming:** Output $y = w^T X$ (weight vector applied to sensor array)

- $E[y] = w^T \mu$
- $\text{Var}(y) = w^T \Sigma w$

**Whitening:** Find matrix $W$ such that $W\Sigma W^T = \text{I}$ (decorrelates data)

- $W = \Sigma^{-1/2}$

**Principal Component Analysis:** Rotate to diagonalize $\Sigma$

## 10.8 Practice Problems

**Problem 1**

Random vector $X = [X_1,\ X_2,\ X_3]^T$ with $\mu = [1,\ 0,\ -1]^T$, and covariance matrix $\Sigma = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}$.

(a) Find $\text{Var}(X_1 + X_2 + X_3)$. (b) Find $\text{Cov}(X_1,\ X_2 + X_3)$. (c) Are $X_1$ and $X_3$ uncorrelated?

**Solution**

(a) $\text{Var}(X_1 + X_2 + X_3) = 1^T \Sigma 1 = 4 + 2 + 3 + 2(1) + 2(0) + 2(-1) = 9 + 0 = 9$
Wait: $= \sum_{ij}$ all entries $= 4 + 1 + 0 + 1 + 2 + (-1) + 0 + (-1) + 3 = 9$

(b) $\text{Cov}(X_1,\ X_2 + X_3) = \text{Cov}(X_1,\ X_2) + \text{Cov}(X_1,\ X_3) = 1 + 0 = 1$

(c) $\text{Cov}(X_1,\ X_3) = 0$, so yes, $X_1$ and $X_3$ are uncorrelated.

**Problem 2**

$X \sim N(0,\ \Sigma)$ with $\Sigma = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$. Find the conditional distribution of $X_1 \mid X_2 = x_2$.

**Solution** For bivariate Gaussian:

$$X_1 \mid X_2 = x_2 \sim N\left(\mu_1 + \rho\left(\frac{\sigma_1}{\sigma_2}\right)(x_2 - \mu_2),\ \sigma_1^2(1 - \rho^2)\right)$$

$$= N(\rho x_2,\ 1 - \rho^2)$$

The conditional mean is a linear function of $x_2$, and conditional variance is reduced by factor $(1 - \rho^2)$.

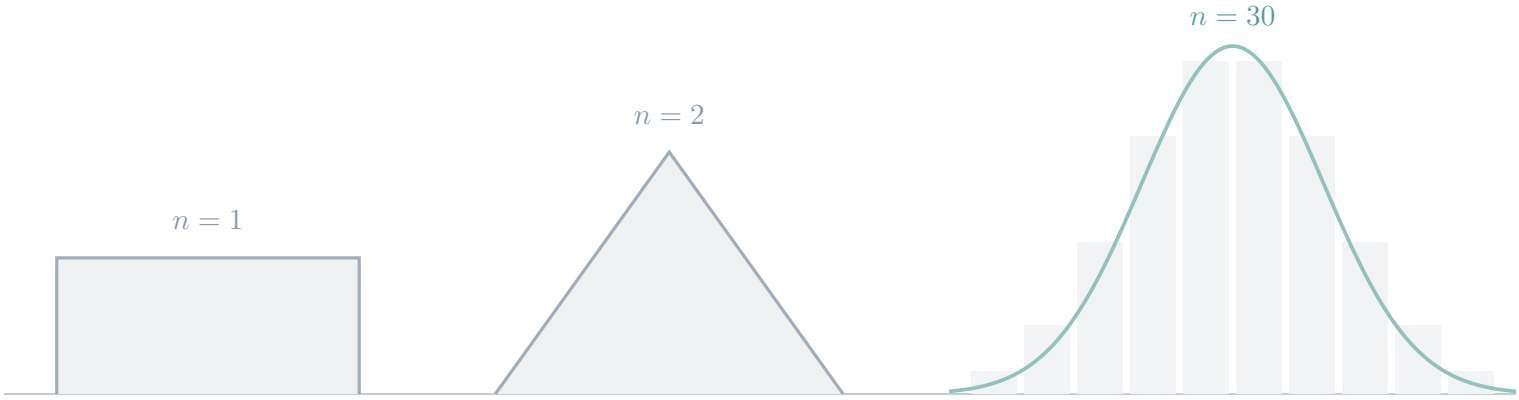
*Next Module: Module 11 — Central Limit Theorem*

Module 11

# Central Limit Theorem

قضیه حد مرکزی

11



## In This Module


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**Learning Objectives**

After completing this module, students will be able to:

1. State and explain the Law of Large Numbers
2. State and apply the Central Limit Theorem
3. Explain why Gaussian distributions appear so frequently in engineering
4. Use CLT for approximations
5. Explain the CLT through sampling distributions

## 11.1 Introduction: Why Is the Gaussian Everywhere?

The Gaussian distribution appears in:

- Thermal noise, measurement errors, manufacturing tolerances, aggregate interference, stock fluctuations

**The reason:** All these quantities are sums (or averages) of many small, independent random effects. The CLT guarantees convergence to Gaussian regardless of the underlying distribution.

## 11.2 Law of Large Numbers (LLN)

### Weak Law of Large Numbers

For $X_1, X_2, \ldots, X_n$ i.i.d. with mean $\mu$ and finite variance $\sigma^2$:

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{P} \mu \quad \text{as } n \to \infty$$

("Convergence in probability": $P(|\bar{X}_n - \mu| > \varepsilon) \to 0$ for any $\varepsilon > 0$)

### Strong Law of Large Numbers

$$\bar{X}_n \xrightarrow{a.s.} \mu \quad \text{as } n \to \infty$$

("Almost sure convergence": $P(\bar{X}_n \to \mu) = 1$)

### Engineering Significance

- **Sample average converges to true mean** as sample size grows
- Justifies estimating means from data
- Foundation of Monte Carlo simulation (simulate many times → average converges)
- Justifies: "If I measure enough times, my average will be close to the truth"

### How Fast?

By Chebyshev: $P(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}$

For 95% confidence that error $< \varepsilon$: need $n \geq \frac{\sigma^2}{0.05 \cdot \varepsilon^2}$

# 11.3 Central Limit Theorem (CLT)

## Formal Statement

For $X_1, X_2, \ldots, X_n$ i.i.d. with mean $\mu$ and finite variance $\sigma^2$:

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0,1)$$

## Practical Form

For large $n$:

$$\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$S_n = \sum_{i=1}^{n} X_i \approx N(n\mu, n\sigma^2)$$

## Conditions

1. $X_1, \ldots, X_n$ are independent
2. Identically distributed (can be relaxed with Lyapunov/Lindeberg CLT)
3. Finite mean $\mu$ and finite variance $\sigma^2$
4. $n$ is "large enough" (rule of thumb: $n \geq 30$, but depends on distribution shape)

## What CLT Does NOT Require

- Does NOT require $X_i$ to be Gaussian
- Does NOT require $X_i$ to be continuous
- Does NOT require any specific distribution

## 11.4 Rate of Convergence

How fast does the CLT "kick in"?

| Original Distribution | $n$ needed for good approximation |
|---|---|
| Symmetric (uniform) | $n \approx 5 - 10$ |
| Mildly skewed | $n \approx 20 - 30$ |
| Highly skewed (exponential) | $n \approx 30 - 50$ |
| Extremely skewed | $n \approx 50 - 100+$ |

**Rule:** More symmetric distributions $\rightarrow$ faster convergence.

## 11.5 Visual Demonstration

### Sum of Uniform RVs → Gaussian

- $n = 1$: Uniform (flat)
- $n = 2$: Triangular
- $n = 3$: Starts looking bell-shaped
- $n = 5$: Nearly Gaussian
- $n = 12$: Practically indistinguishable from Gaussian

### Sum of Exponential RVs → Gaussian

- $n = 1$: Exponential (heavily right-skewed)
- $n = 3$: Still skewed but less so (Gamma(3))
- $n = 10$: Nearly symmetric, bell-shaped
- $n = 30$: Essentially Gaussian

### Binomial → Normal Approximation

Binomial$(n, p) \approx N(np, np(1-p))$ for large $n$

This is CLT applied to sum of Bernoulli trials!

- Continuity correction: $P(X \leq k) \approx \Phi\left(\frac{k+0.5-np}{\sqrt{np(1-p)}}\right)$

## 11.6 Engineering Significance

### Why Thermal Noise is Gaussian

Thermal noise = aggregate effect of billions of electrons moving randomly. Each electron's contribution is tiny and independent → CLT applies → noise is Gaussian.

### Why Aggregate Interference is Gaussian

In a cellular network, total interference = sum of signals from many independent users. CLT → interference ≈ Gaussian (for many users).

### Why Measurement Errors are Gaussian

Total error = sum of many independent small error sources (calibration, quantization, thermal, vibration, etc.) → CLT → error ≈ Gaussian.

### Why Manufacturing Variations are Gaussian

Resistor value = nominal + sum of many small process variations (doping, etching, temperature, etc.) → CLT → value ≈ Gaussian around nominal.

# 11.7 Approximations Using CLT

## General Procedure

To approximate $P(S_n \leq x)$ where $S_n = X_1 + \ldots + X_n$:

1. Compute $\mu = E[X_i]$, $\sigma^2 = \text{Var}(X_i)$
2. $S_n \approx N(n\mu,\, n\sigma^2)$
3. $P(S_n \leq x) \approx \Phi\left(\dfrac{x - n\mu}{\sigma\sqrt{n}}\right)$

## Continuity Correction (for Discrete RVs)

When approximating a discrete sum by Gaussian:

- $P(X \leq k) \approx \Phi\left(\dfrac{k + 0.5 - n\mu}{\sigma\sqrt{n}}\right)$
- $P(X = k) \approx \Phi\left(\dfrac{k + 0.5 - n\mu}{\sigma\sqrt{n}}\right) - \Phi\left(\dfrac{k - 0.5 - n\mu}{\sigma\sqrt{n}}\right)$

**Example: Packet Errors**

1000 packets, each error probability 0.02. $X$ = total errors.
Exact: $X \sim \text{Binomial}(1000,\, 0.02)$
CLT approx: $X \approx N(20,\, 19.6)$
$P(X > 30) \approx 1 - \Phi\left(\dfrac{30.5 - 20}{\sqrt{19.6}}\right) = 1 - \Phi(2.37) = 0.0089$

## 11.8 Practice Problems

**Problem 1**

A factory produces bolts. Length $X \sim$ distribution with $\mu = 10\,\text{cm}$, $\sigma = 0.2\,\text{cm}$. A batch of 100 bolts is measured.

(a) What is the approximate distribution of $\bar{X}$? (b) $P(\bar{X} > 10.05)$? (c) $P(9.96 < \bar{X} < 10.04)$?

**Solution**

(a) $\bar{X} \sim N\left(10,\, \dfrac{0.04}{100}\right) = N(10,\, 0.0004)$, $\sigma_{\bar{X}} = 0.02$

(b) $P(\bar{X} > 10.05) = P(Z > 2.5) = 0.0062$

(c) $P(9.96 < \bar{X} < 10.04) = \Phi(2) - \Phi(-2) = 0.9545$

**Problem 2**

A communication link has 10000 bits with BER $= 0.001$. Approximate $P(\text{more than 15 errors})$ using CLT.

**Solution** $X \sim \text{Binomial}(10000,\, 0.001)$. $\mu = 10$, $\sigma^2 = 9.99$.

$$P(X > 15) \approx 1 - \Phi\left(\frac{15.5 - 10}{\sqrt{9.99}}\right) = 1 - \Phi(1.74) = 0.0409$$

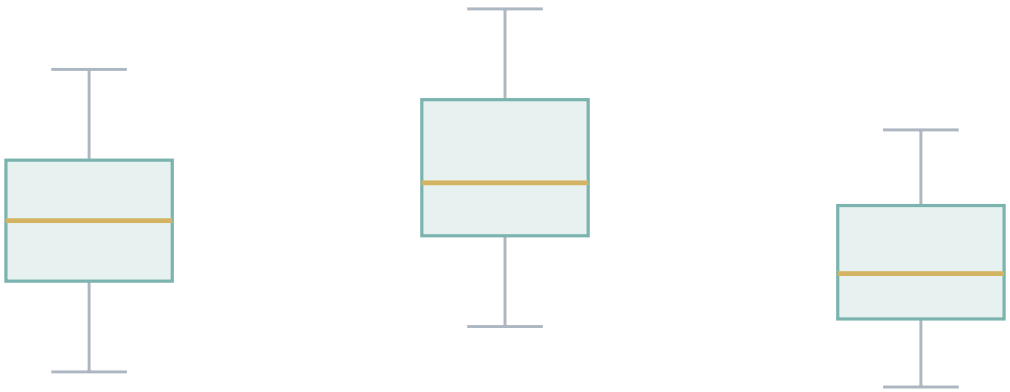
*Next Module: Module 12 — Introduction to Statistics*

Module 12

# Introduction to Statistics

مقدمه‌ای بر آمار

12



## In This Module


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**Learning Objectives**

After completing this module, students will be able to:

1. Distinguish between population and sample
2. Distinguish between parameter and statistic
3. Define and compute sample mean and sample variance
4. Explain sampling distributions and why they matter
5. Understand the transition from probability (model $\rightarrow$ predictions) to statistics (data $\rightarrow$ model)

# 12.1 Introduction: From Probability to Statistics

## The Two Directions

| Probability | Statistics |
|---|---|
| Model is KNOWN | Model is UNKNOWN |
| Predict outcomes | Infer model from data |
| Forward problem | Inverse problem |
| "Given distribution, what data?" | "Given data, what distribution?" |

**Probability** (Modules 1–11): Given a model $\rightarrow$ predict observations.
**Statistics** (Modules 12–16): Given observations $\rightarrow$ infer the model.

## Engineering Context

- **Probability:** "If noise is $N(0,\ 0.01)$, what is $P(\text{error})$?"
- **Statistics:** "I measured 1000 noise samples. What is $\sigma^2$?"

# 12.2 Population vs. Sample

## Population

The **complete** set of ALL items/measurements of interest.

| Engineering Context | Population |
|---|---|
| All chips ever produced by a fab | ALL chip delays |
| The noise process forever | ALL noise voltage values |
| All network packets ever sent | ALL packet delays |

Populations are usually **infinite or impractically large** — we cannot measure everything.

## Sample

A **subset** of the population that we actually observe/measure.

| Engineering Context | Sample |
|---|---|
| 50 chips tested from production line | Measured delays |
| 10000 noise voltage samples recorded | Recorded values |
| 500 packet delays measured | Observed delays |

## Why We Sample

- Cannot test every chip (destructive testing)
- Cannot record noise forever (finite memory)
- Cannot measure every packet (too many)

The goal: **Infer properties of the population from the sample.**

# 12.3 Parameter vs. Statistic

## Parameter

A **fixed** (but unknown) number that describes the population:

- $\mu$ = true population mean
- $\sigma^2$ = true population variance
- $p$ = true defect rate
- $\lambda$ = true failure rate

Parameters are **constants** — they have ONE true value (we just don't know it).

## Statistic

A **function of sample data** used to estimate or test parameters:

- $\bar{X}$ = sample mean (estimates $\mu$)
- $S^2$ = sample variance (estimates $\sigma^2$)
- $\hat{p}$ = sample proportion (estimates $p$)

⚠ **Key insight:** Statistics are RANDOM VARIABLES! Different samples give different values.

**Example**

True noise power: $\sigma^2 = 0.04\,\mathrm{V}^2$ (parameter — fixed, unknown)
Experiment 1: Collect 100 samples → compute $S^2 = 0.038$ (one realization)
Experiment 2: Collect 100 samples → compute $S^2 = 0.042$ (different realization)
Experiment 3: Collect 100 samples → compute $S^2 = 0.041$ (another realization)
Each $S^2$ is different because of randomness in sampling.

# 12.4 Random Sample

**Definition**

A **random sample** $X_1, X_2, \ldots, X_n$ is a set of $n$ independent, identically distributed $(i.i.d.)$ random variables from the population distribution.

## Requirements

1. **Independent:** Each observation doesn't influence others
2. **Identically distributed:** All come from the same population
3. **Random selection:** No systematic bias in choosing which items to measure

## When i.i.d. Applies

✔ Random selection of components from a large batch
✔ Independent measurements of a stable process
✔ Noise samples separated by more than the correlation time

## When i.i.d. Fails

✘ Consecutive samples of a correlated signal
✘ Measurements during a drifting process
✘ Non-random selection (e.g., only testing "suspicious" components)

# 12.5 Sampling Distributions

**Definition**

The **sampling distribution** of a statistic is the probability distribution of that statistic across all possible samples of size $n$.

## Why It Matters

A statistic (like $\bar{X}$) is a random variable. Its distribution tells us:

- How much will it vary from sample to sample?
- How close is it likely to be to the true parameter?
- How should we quantify our uncertainty?

This is the foundation for confidence intervals and hypothesis testing.

# 12.6 Sample Mean $\bar{X}$

**Definition**

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

## Properties

**Mean of $\bar{X}$:**

$$E[\bar{X}] = \mu$$

$\bar{X}$ is an **unbiased** estimator of $\mu$ (on average, it equals the truth).

**Variance of $\bar{X}$:**

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

More measurements $\rightarrow$ less variability $\rightarrow$ more precise estimate.

**Standard Error:**

$$\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

## Sampling Distribution of $\bar{X}$

- If population is Gaussian: $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ **exactly** for any $n$
- If population is NOT Gaussian: $\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$ **approximately** for large $n$ (by CLT)

## 12.7 Sample Variance $S^2$

**Definition**

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

### Why $n-1$? (Bessel's Correction)

Using $n$ in the denominator gives a **biased** estimator (systematically underestimates $\sigma^2$):

$$E\left[\frac{1}{n}\cdot\sum(X_i-\bar{X})^2\right] = \frac{n-1}{n}\sigma^2 \neq \sigma^2$$

Using $n-1$ corrects this:

$$E[S^2] = \sigma^2$$

**Intuition:** $\bar{X}$ is computed from the same data, so deviations from $\bar{X}$ are smaller than deviations from $\mu$. We "lose one degree of freedom" by estimating the mean.

### Degrees of Freedom

With $n$ data points and $\bar{X}$ estimated:

- The $n$ deviations $(X_i - \bar{X})$ sum to zero: $\sum(X_i - \bar{X}) = 0$
- Only $n-1$ are "free" — the last is determined
- $df = n-1$

### Sampling Distribution (Normal Population)

If population is $N(\mu, \sigma^2)$:

$$(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$$

## 12.8 Sample Standard Deviation

$$S = \sqrt{S^2} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2}$$

Note: $E[S] \neq \sigma$ (S is slightly biased for $\sigma$, though $S^2$ is unbiased for $\sigma^2$).

## 12.9 Connection to CLT

From Module 11: $\bar{X} \to N\left(\mu, \frac{\sigma^2}{n}\right)$ as $n \to \infty$.

This means the sampling distribution of $\bar{X}$ is approximately Gaussian. This is what allows us to:

1. Build confidence intervals (Module 14)
2. Perform hypothesis tests (Module 15)
3. Quantify uncertainty in estimates

## 12.10 Practice Problems

**Problem 1**

A population has $\mu = 50$ and $\sigma = 10$. A random sample of $n = 64$ is taken.

(a) What is $E[\bar{X}]$? (b) What is $\text{Var}(\bar{X})$? (c) What is $P(48 < \bar{X} < 52)$?

**Solution**

(a) $E[\bar{X}] = 50$

(b) $\text{Var}(\bar{X}) = \dfrac{100}{64} = 1.5625$, $\text{SE} = 1.25$

(c) $P(48 < \bar{X} < 52) = P(-1.6 < Z < 1.6) = 2\Phi(1.6) - 1 = 0.8904$

**Problem 2**

Why *do* we use $n - 1$ in the sample variance formula instead of $n$? Explain both mathematically and intuitively.

**Solution** Mathematically: $E\left[\frac{1}{n}\sum(X_i - \bar{X})^2\right] = \frac{n-1}{n} \cdot \sigma^2$, so dividing by $n - 1$ gives $E[S^2] = \sigma^2$. Intuitively: we use $\bar{X}$ from the same data (not the true $\mu$), so deviations from $\bar{X}$ are artificially small. One degree of freedom is "used up" estimating the mean.

**Problem 3**

An engineer measures noise power from 20 samples and gets $S^2 = 0.045\,\text{V}^2$. The true variance is $\sigma^2 = 0.04\,\text{V}^2$. Is this concerning? Use the sampling distribution to assess.

**Solution** Under $H_0 : \sigma^2 = 0.04$, $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(19)$. Observed: $\frac{19(0.045)}{0.04} = 21.375$. $P(\chi^2(19) > 21.375) \approx 0.31$. This is not unusual — about 31% of samples would give $S^2 \geq 0.045$ when $\sigma^2 = 0.04$. Not concerning.

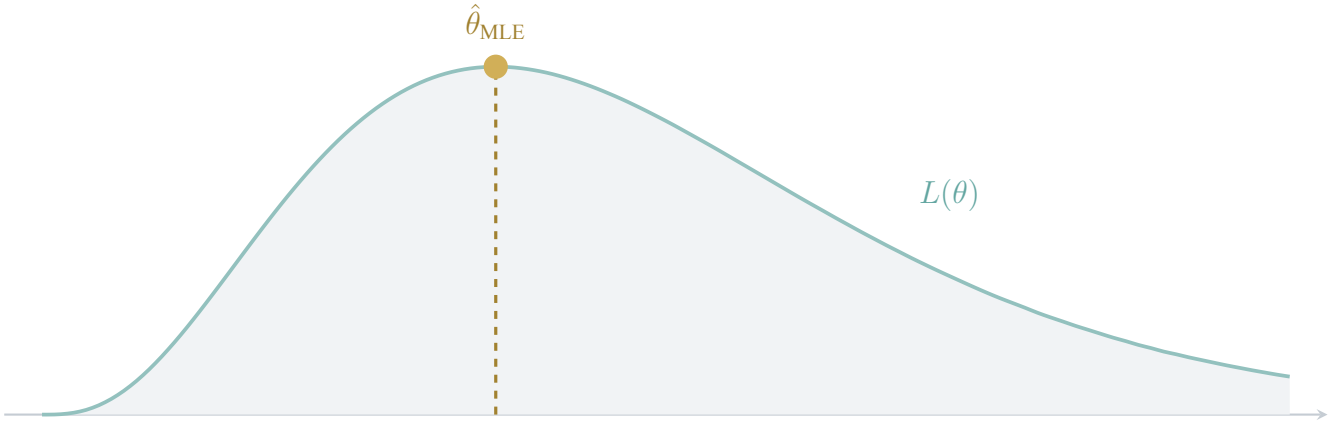
*Next Module: Module 13 — Parameter Estimation*

Module 13

# Parameter Estimation

برآورد پارامتر

13



## In This Module


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**Learning Objectives**

After completing this module, students will be able to:

1. Distinguish between estimator (rule) and estimate (number)
2. Evaluate estimator properties: bias, consistency, efficiency, MSE
3. Apply the Method of Moments
4. Apply Maximum Likelihood Estimation (MLE)
5. Compare estimators using MSE and bias-variance tradeoff

## 13.1 Introduction

We observe data $X_1, \ldots, X_n$ from a distribution with unknown parameter $\theta$. **Parameter estimation** gives us a method to infer $\theta$ from data.

Engineering examples:

- Estimate noise variance $\sigma^2$ from recorded samples
- Estimate failure rate $\lambda$ from component lifetimes
- Estimate channel gain from received signal strength
- Estimate mean signal level from measurements

# 13.2 Estimator vs. Estimate

## Definitions

| Term | Type | Definition | Example |
|---|---|---|---|
| **Estimator** $\hat{\theta}$ | Random variable (function of data) | The RULE applied to data | $\hat{\theta} = \bar{X} = \frac{1}{n}\sum X_i$ |
| **Estimate** | Number | The VALUE from a specific sample | $\hat{\theta} = 3.47$ (from one dataset) |

## Key Insight

The **estimator** is a random variable — it has a sampling distribution.
The **estimate** is one realization of that random variable.

**Analogy:** "Take the average" is the estimator (rule). "The average was 3.47" is the estimate (result).

# 13.3 Properties of Estimators

## Bias

$$B(\hat{\theta}) = E[\hat{\theta}] - \theta$$

- **Unbiased:** $B(\hat{\theta}) = 0$, i.e., $E[\hat{\theta}] = \theta$ (on average, correct)
- **Biased:** $E[\hat{\theta}] \neq \theta$ (systematic error)

Examples:

- $\bar{X}$ is unbiased for $\mu$: $E[\bar{X}] = \mu$ ✔
- $S^2$ (with $n-1$) is unbiased for $\sigma^2$: $E[S^2] = \sigma^2$ ✔
- $S$ (sample std dev) is biased for $\sigma$: $E[S] \neq \sigma$ (slight downward bias)

## Consistency

$\hat{\theta}_n$ is **consistent** if $\hat{\theta}_n \to \theta$ in probability as $n \to \infty$.

Practical meaning: with enough data, the estimator gets arbitrarily close to the truth.

Sufficient condition: If bias $\to 0$ and variance $\to 0$ as $n \to \infty$, then consistent.

## Efficiency

Among all unbiased estimators, the **efficient** one has the smallest variance.

The **Cramér-Rao Lower Bound** (CRLB) gives the minimum possible variance:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}$$

where $I(\theta)$ = Fisher information $= -E\left[\frac{\partial^2 \ln f(X;\theta)}{\partial \theta^2}\right]$.

An estimator achieving the CRLB is called **efficient** or **minimum variance unbiased (MVU)**.

## Mean Squared Error (MSE)

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + [B(\hat{\theta})]^2$$

$\text{MSE} = \text{Variance} + \text{Bias}^2$

This is the **bias-variance tradeoff**: sometimes a slightly biased estimator with much lower variance has smaller MSE than an unbiased one.

# 13.4 Method of Moments (MoM)

## Principle

Set population moments equal to sample moments, then solve for parameters.

- $k-$ th population moment: $\mu' k = E[X^k]$
- $k-$ th sample moment: $m' k = \frac{1}{n}\sum X_i^k$

## Procedure

1. Express parameters in terms of population moments
2. Replace population moments with sample moments
3. Solve for parameter estimates

**Example: Exponential Distribution**

Data from Exp($\lambda$). Population moment: $E[X] = \frac{1}{\lambda}$.
Set equal: $\frac{1}{n}\sum X_i = \frac{1}{\hat{\lambda}} \rightarrow \hat{\lambda}_{\text{MoM}} = \frac{1}{\bar{X}}$

**Example: Gaussian Distribution**

Data from $N(\mu, \sigma^2)$:

- 1st moment: $\hat{\mu} = \bar{X}$
- 2nd central moment: $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$ (note: biased, uses $n$ not $n-1$)

## Advantages/Disadvantages

✔ Simple, always applicable, closed-form solutions often available
✘ Not always efficient, can produce biased estimates, may give invalid parameter values

# 13.5 Maximum Likelihood Estimation (MLE)

## Principle

Choose the parameter value that makes the observed data **most probable**.

## Likelihood Function

Given data $x_1, \ldots, x_n$ from $f(x;\theta)$:

$$L(\theta) = \prod_{i=1}^{n} f(x_i;\theta)$$

## Log-Likelihood (usually easier)

$$\ell(\theta) = \ln L(\theta) = \sum_{i=1}^{n} \ln f(x_i;\theta)$$

## MLE Procedure

1. Write the log-likelihood $\ell(\theta)$
2. Differentiate: $\frac{d\ell}{d\theta} = 0$
3. Solve for $\hat{\theta}_{\text{MLE}}$
4. Verify it's a maximum $\left(\frac{d^2\ell}{d\theta^2} < 0\right)$

## Properties of MLE

1. **Consistent** (converges to true value)
2. **Asymptotically unbiased** (bias $\to 0$ as $n \to \infty$)
3. **Asymptotically efficient** (achieves CRLB for large $n$)
4. **Invariant:** If $\hat{\theta}$ is MLE of $\theta$, then $g(\hat{\theta})$ is MLE of $g(\theta)$

**Example: MLE for Gaussian Mean**

Data $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ with $\sigma^2$ known.

$$\ell(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$$

$$\frac{d\ell}{d\mu} = \frac{1}{\sigma^2}\sum(x_i - \mu) = 0 \to \hat{\mu}_{\text{MLE}} = \bar{X}$$

**Example: MLE for Exponential Rate**

Data $X_1, \ldots, X_n \sim \text{Exp}(\lambda)$.

$$\ell(\lambda) = n \cdot \ln(\lambda) - \lambda \cdot \sum x_i$$

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum x_i = 0 \to \hat{\lambda}_{\text{MLE}} = \frac{n}{\sum x_i} = \frac{1}{\bar{X}}$$

**Example: MLE for Gaussian Variance**

Data $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, $\mu$ known.

$$\ell(\sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$$

$$\frac{d\ell}{d(\sigma^2)} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum(x_i - \mu)^2 = 0$$

$$\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum(x_i - \mu)^2$$

Note: With $\mu$ unknown, $\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum(x_i - \bar{x})^2$ — biased! (Uses $n$, not $n-1$)

# 13.6 Engineering Examples

## Estimating Noise Variance

**Problem:** Measure $n$ samples of noise. Estimate noise power $\sigma^2$.

**Data:** $x_1, \ldots, x_n$ (noise voltage samples, assumed zero-mean)

**MLE:** $\hat{\sigma}^2 = \frac{1}{n}\sum x_i^2$ (biased by factor $\frac{n-1}{n}$)
**Unbiased:** $S^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$ or if $\mu = 0$ known: $\frac{1}{n}\sum x_i^2$ is actually unbiased for $E[X^2] = \sigma^2$

## Estimating Failure Rate

**Problem:** 20 components tested until failure. Lifetimes: $t_1, \ldots, t_{20}$.

**Model:** $T \sim \text{Exp}(\lambda)$, parameter $\lambda$ (failure rate).

**MLE:** $\hat{\lambda} = \frac{20}{\sum t_i} = \frac{1}{\bar{T}}$

If $\bar{T} = 500$ hours: $\hat{\lambda} = \frac{1}{500} = 0.002$ failures/hour.

## Estimating Channel Parameter

**Problem:** Rayleigh fading channel. Observed power samples: $p_1, \ldots, p_n$.

**Model:** Power $P \sim \text{Exp}\left(\frac{1}{\Omega}\right)$, where $\Omega = E[P]$ = average power.

**MLE:** $\hat{\Omega} = \bar{P} = \frac{1}{n}\sum p_i$

# 13.7 Comparing Estimators

## Bias-Variance Tradeoff

Two estimators for $\sigma^2$:

- $\hat{\theta}_1 = S^2$ (unbiased, variance $= \frac{2\sigma^4}{n-1}$)
- $\hat{\theta}_2 = \frac{1}{n}\sum(X_i - \bar{X})^2$ (biased by $-\frac{\sigma^2}{n}$, variance $= \frac{2\sigma^4(n-1)}{n^2}$)

$$\text{MSE}(\hat{\theta}_1) = \frac{2\sigma^4}{n-1}$$

$$\text{MSE}(\hat{\theta}_2) = \frac{2\sigma^4(n-1)}{n^2} + \frac{\sigma^4}{n^2} = \frac{\sigma^4(2n-1)}{n^2}$$

For any $n \geq 2$: $\text{MSE}(\hat{\theta}_2) < \text{MSE}(\hat{\theta}_1)$! The biased MLE actually has smaller total error.

## 13.8 Practice Problems

**Problem 1**

Data from $N(\mu,\,9)$: $\bar{x} = 4.2$, $n = 25$.

(a) Give the MLE of $\mu$. (b) What is $\text{Var}(\hat{\mu})$? (c) Is $\bar{X}$ efficient for $\mu$?

**Solution**

(a) $\hat{\mu}_{\text{MLE}} = \bar{X} = 4.2$

(b) $\text{Var}(\bar{X}) = \dfrac{\sigma^2}{n} = \dfrac{9}{25} = 0.36$

(c) CRLB for $N(\mu,\,\sigma^2)$: $\frac{1}{n \cdot I(\mu)} = \frac{\sigma^2}{n} = 0.36$. $\bar{X}$ achieves CRLB $\rightarrow$ yes, efficient.

**Problem 2**

Component lifetimes (hours): 120, 350, 200, 480, 90, 560, 310, 175, 420, 280.

(a) Estimate $\lambda$ (failure rate) by MLE assuming $\text{Exp}(\lambda)$. (b) Estimate MTTF.

**Solution**

(a) $\bar{X} = \dfrac{120 + 350 + 200 + 480 + 90 + 560 + 310 + 175 + 420 + 280}{10} = 298.5.$

$\hat{\lambda} = \frac{1}{298.5} = 0.00335$ failures/hour.

(b) $\text{MTTF} = \frac{1}{\hat{\lambda}} = \bar{X} = 298.5$ hours.

**Problem 3**

For Poisson($\lambda$) data: $x_1, \ldots, x_n$. Derive the MLE of $\lambda$.

**Solution** $\ell(\lambda) = \sum[x_i \ln(\lambda) - \lambda - \ln(x_i!)] = (\sum x_i)\ln(\lambda) - n\lambda - \sum \ln(x_i!)$

$\dfrac{d\ell}{d\lambda} = \dfrac{\sum x_i}{\lambda} - n = 0 \rightarrow \hat{\lambda}_{\text{MLE}} = \dfrac{1}{n}\sum x_i = \bar{X}.$

The MLE for Poisson rate is the sample mean.

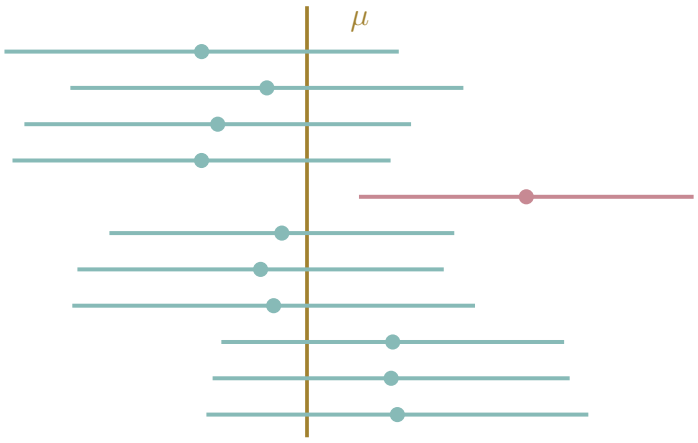
*Next Module: Module 14 — Confidence Intervals*

Module 14

# Confidence Intervals

بازه‌های اطمینان

14



## In This Module


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**Learning Objectives**

After completing this module, students will be able to:

1. Correctly interpret confidence intervals (frequentist interpretation)
2. Construct CIs for means (known and unknown variance)
3. Apply the t-distribution for small samples
4. Construct CIs for variance
5. Determine required sample sizes
6. Avoid common misconceptions about CIs

## 14.1 Introduction: Why Point Estimates Aren't Enough

A point estimate (e.g., $\bar{X} = 3.47$) gives no information about **precision**. We need to quantify uncertainty:

"The true mean is estimated to be 3.47, **give or take** some amount."

A **confidence interval** provides a range that, with a specified confidence level, captures the true parameter.

# 14.2 Correct Interpretation of Confidence Intervals

## ✔ CORRECT (Frequentist) Interpretation

> "If we repeat this experiment many times, each time computing a 95% CI, then approximately 95% of those intervals will contain the true parameter $\mu$."

The **procedure** has a 95% success rate. Each specific interval either contains $\mu$ or it doesn't.

## ✘ WRONG Interpretation

> ~~ "There is a 95% probability that $\mu$ is inside this particular interval $[2.1,\ 4.8]$." ~~

**Why this is wrong:** $\mu$ is a fixed (non-random) number. It's either in $[2.1, 4.8]$ or not — there is no probability about it. The **interval** is random (varies from sample to sample), not the parameter.

## Analogy

Think of CIs like a fishing net:

- Before you cast: 95% chance your net will catch the fish (the procedure works 95% of the time)
- After you cast: the fish is either caught or not — no probability anymore
- The fish $(\mu)$ doesn't move; your net (CI) does

# 14.3 CI for Mean — Known Variance (Z-interval)

## Setup

Data: $X_1, \ldots, X_n$ i.i.d. from $N(\mu, \sigma^2)$ with $\sigma^2$ **known**.

## Derivation

$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$, so $Z = \frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$

$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$

Solving for $\mu$:

$$P\left(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

## Formula

CI: $\bar{X} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$

## Common z-values

| Confidence Level | $\alpha$ | $z_{\alpha/2}$ |
|---|---|---|
| 90% | 0.10 | 1.645 |
| 95% | 0.05 | 1.960 |
| 99% | 0.01 | 2.576 |

**Example**

Noise measurements: $n = 36$, $\bar{X} = 0.52\,\text{V}$, $\sigma = 0.12\,\text{V}$ (known from equipment spec).
95% CI: $0.52 \pm 1.96 \times \frac{0.12}{\sqrt{36}} = 0.52 \pm 0.0392 = **[0.481, 0.559]V**$

# 14.4 CI for Mean — Unknown Variance (t-interval)

## The Problem

Usually $\sigma$ is unknown. We estimate it with $S$. But $\frac{\bar{X}}{\frac{S}{\sqrt{n}}}$ does NOT follow $N(0,\,1)$.

## Student's t-Distribution

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

The t-distribution:

- Symmetric about 0
- Heavier tails than $N(0,\,1)$ (more uncertainty because $S$ is estimated)
- Degrees of freedom: $\nu = n - 1$
- As $\nu \to \infty$: $t(\nu) \to N(0,\,1)$
- For $\nu \geq 30$: $t \approx z$ (practically)

## Formula

CI: $\bar{X} \pm t_{\alpha/2,n-1} \cdot \dfrac{S}{\sqrt{n}}$

**Example**

Component delay: $n = 15$, $\bar{X} = 2.34$ ns, $S = 0.45$ ns.
95% CI: $t_{0.025,14} = 2.145$
CI: $2.34 \pm 2.145 \times \frac{0.45}{\sqrt{15}} = 2.34 \pm 0.249 = **[2.09,\,2.59]$ ns**
(Wider than z-interval because of additional uncertainty in S.)

# 14.5 CI for Variance

## Setup

Data from $N(\mu,\, \sigma^2)$. Want CI for $\sigma^2$.

## Key Distribution

$$(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$$

## Formula

$$\left[\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}},\quad \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right]$$

**Example**

Noise power: $n = 20$, $S^2 = 0.045\,\mathrm{V}^2$.
95% CI for $\sigma^2$: $\chi^2_{0.025,19} = 32.852$, $\chi^2_{0.975,19} = 8.907$
CI: $\left[\frac{(19)(0.045)}{32.852},\, \frac{(19)(0.045)}{8.907}\right] = [0.026,\, 0.096]V^2$
95% CI for $\sigma$: $\left[\sqrt{0.026},\, \sqrt{0.096}\right] = [0.161,\, 0.310]V$

# 14.6 CI for Proportions

## Setup

$n$ trials, $\hat{x} = \frac{k}{n}$ successes. Estimate $p$ = true proportion.

## Large-Sample Formula (Wald Interval)

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

**Example: Bit Error Rate**

Transmitted 100,000 bits, observed 53 errors. $\hat{p} = \frac{53}{100000} = 5.3 \times 10^{-4}$.

95% CI: $5.3 \times 10^{-4} \pm 1.96\sqrt{5.3 \times 10^{-4} \times \frac{0.9995}{100000}} = 5.3 \times 10^{-4} \pm 1.43 \times 10^{-4}$

CI: $[3.9 \times 10^{-4},\ 6.7 \times 10^{-4}]$

# 14.7 Factors Affecting CI Width

| Factor | Effect on Width | Explanation |
|---|---|---|
| ↑ Sample size $n$ | Narrower | More data → more precision |
| ↑ Confidence level | Wider | More confidence → wider net |
| ↑ Variability $\sigma$ | Wider | More noise → harder to pin down $\mu$ |

## The Tradeoff

- Want narrow CI (precise) → need more data or accept lower confidence
- Want high confidence → accept wider CI or collect more data
- Want both narrow AND high confidence → need LOTS of data

## 14.8 Sample Size Determination

### Question: How many measurements *do* I need?

For CI width $\leq 2E$ (margin of error $E$):

$$n \geq \left(\frac{z_{\alpha/2} \cdot \sigma}{E}\right)^2$$

**Example**

Want to estimate mean delay within $\pm 0.5$ ms with 95% confidence. Expect $\sigma \approx 3$ ms.
$n \geq \left(1.96 \times \frac{3}{0.5}\right)^2 = (11.76)^2 = 138.3 \rightarrow$ need $n = 139$ **measurements**

## 14.9 Practice Problems

**Problem 1**

A sensor measures voltage. From 50 measurements: $\bar{X} = 3.32\,\text{V}$, $S = 0.15\,\text{V}$.

(a) Construct 95% CI for $\mu$. (b) Construct 99% CI for $\mu$. (c) Which is wider and why?

**Solution**

(a) $t_{0.025,49} \approx 2.010$. CI: $3.32 \pm 2.010\left(\frac{0.15}{\sqrt{50}}\right) = 3.32 \pm 0.043 = [3.277,\ 3.363]$

(b) $t_{0.005,49} \approx 2.680$. CI: $3.32 \pm 2.680\left(\frac{0.15}{\sqrt{50}}\right) = 3.32 \pm 0.057 = [3.263,\ 3.377]$

(c) 99% CI is wider — higher confidence requires a wider interval.

**Problem 2**

From 25 noise power measurements: $S^2 = 0.048\,\text{V}^2$. Construct 95% CI for $\sigma^2$.

**Solution** $\chi^2_{0.025,24} = 39.364$, $\chi^2_{0.975,24} = 12.401$

CI: $\left[\frac{24(0.048)}{39.364},\ \frac{24(0.048)}{12.401}\right] = [0.029,\ 0.093]V^2$

**Problem 3**

How many BER measurements needed to estimate BER within $\pm 20\%$ of its true value with 95% confidence, if expected BER $\approx 10^{-3}$?

**Solution** For proportion: $n \geq \frac{z^2 p(1-p)}{E^2}$ where $E = 0.2p = 2 \times 10^{-4}$.

$n \geq \frac{(1.96)^2(10^{-3})(0.999)}{(2\times10^{-4})^2} = 3.84 \times \frac{10^{-3}}{4\times10^{-8}} \approx 96{,}000$ bits.

Need to transmit approximately 96,000 bits (or observe $\sim 96$ errors).

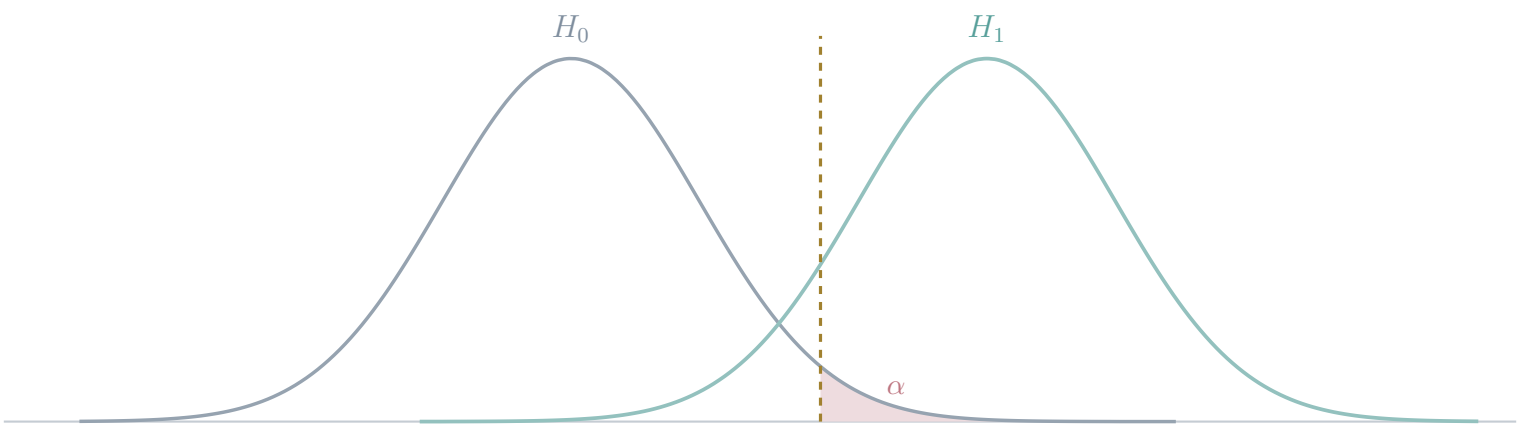
*Next Module: Module 15 — Hypothesis Testing*

Module 15

# Hypothesis Testing

آزمون فرض

15



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**Learning Objectives**

After completing this module, students will be able to:

1. Formulate null and alternative hypotheses for engineering problems
2. Identify Type I and Type II errors and their engineering consequences
3. Compute test statistics and p-values
4. Perform Z-tests and t-tests for means
5. Correctly interpret p-values (and avoid common misconceptions)
6. Choose between one-sided and two-sided tests

## 15.1 Introduction: Making Decisions from Data

**Engineering question:** “Is this sensor biased?” “Did the process change?” “Does the new algorithm perform better?”

We need a **systematic framework** to answer such questions from data, while controlling the probability of making errors.

## 15.2 Framework

### Null Hypothesis $H_0$

The **default assumption** — typically "nothing changed" or "no effect":

- $H_0 : \mu = \mu_0$ (sensor is not biased)
- $H_0 : \mu_1 = \mu_2$ (two methods perform equally)
- $H_0 : \sigma^2 = \sigma_0^2$ (variance hasn't changed)

### Alternative Hypothesis $H_1$

What we want to **detect** or prove:

- $H_1 : \mu \neq \mu_0$ (sensor IS biased — two-sided)
- $H_1 : \mu > \mu_0$ (performance improved — one-sided)
- $H_1 : \sigma^2 > \sigma_0^2$ (variance increased)

### Decision

Based on data, either:

- **Reject** $H_0 \rightarrow$ evidence supports $H_1$
- **Fail to reject** $H_0 \rightarrow$ insufficient evidence against $H_0$

> ⚠ "Fail to reject $H_0$" $\neq$ "$H_0$ is true". Absence of evidence $\neq$ evidence of absence.

# 15.3 Types of Errors

| | $H_0$ True (reality) | $H_1$ True (reality) |
|---|---|---|
| **Reject** $H_0$ (decision) | Type I Error $(\alpha)$ | Correct! (Power $= 1 - \beta$) |
| **Fail to reject** $H_0$ (decision) | Correct! | Type II Error $(\beta)$ |

## Type I Error (False Alarm) — probability $\alpha$

Rejecting $H_0$ when it's actually true. "Crying wolf."

Engineering analogy: Declaring a process has changed when it hasn't $\rightarrow$ unnecessary recalibration.

## Type II Error (Missed Detection) — probability $\beta$

Failing to reject $H_0$ when $H_1$ is actually true. "Missing the signal."

Engineering analogy: Failing to detect a sensor bias $\rightarrow$ biased measurements continue.

## Engineering Analogy: Radar Detection

| Statistical Term | Radar Equivalent |
|---|---|
| $H_0$: no target | No aircraft present |
| $H_1$: target present | Aircraft present |
| Type I error $(\alpha)$ | False alarm |
| Type II error $(\beta)$ | Missed detection |
| Power $(1 - \beta)$ | Probability of detection |

## The Tradeoff

Reducing $\alpha$ (fewer false alarms) $\rightarrow$ increases $\beta$ (more missed detections), and vice versa.

Fix $\alpha$ (significance level), then maximize power $= 1 - \beta$.

## 15.4 Significance Level $\alpha$

The **maximum acceptable Type I error rate**, chosen BEFORE looking at data.

Common choices: $\alpha = 0.05,\ 0.01,\ 0.10$

**Choosing $\alpha$:** Consider the cost of a false alarm:

- $\alpha = 0.01$: Very conservative (e.g., medical device safety)
- $\alpha = 0.05$: Standard in most engineering/science
- $\alpha = 0.10$: More tolerant (exploratory analysis)

## 15.5 Power of a Test

$$\text{Power} = 1 - \beta = P(\text{reject } H_0 | H_1 \text{ is true})$$

Power depends on:

1. **Effect size:** Larger true difference $\rightarrow$ easier to detect $\rightarrow$ more power
2. **Sample size $n$:** More data $\rightarrow$ more power
3. **Significance level $\alpha$:** Larger $\alpha \rightarrow$ more power (but more false alarms)
4. **Variability $\sigma$:** Less noise $\rightarrow$ more power

## 15.6 The p-value

**Definition**

The **p-value** is the probability of obtaining a test statistic **as extreme or more extreme** than the observed value, **assuming $H_0$ is true**.

$$p\text{-value} = P(\text{data this extreme or more} | H_0 \text{ true})$$

### Decision Rule

- If p-value $< \alpha \rightarrow$ Reject $H_0$
- If p-value $\geq \alpha \rightarrow$ Fail to reject $H_0$

### ✔ CORRECT Interpretation

> "The p-value is the probability of seeing data this extreme (or more) if $H_0$ were true. A small p-value means the data is unlikely under $H_0$."

### ✘ WRONG Interpretation

> ~~"The p-value is the probability that $H_0$ is true."~~

**Why wrong:** $H_0$ is either true or false (fixed). The p-value says nothing about $P(H_0 \text{ true})$. It measures how surprising the data is under $H_0$.

### Calibration

| p-value | Evidence against $H_0$ |
|---|---|
| $> 0.10$ | Weak or none |
| 0.05 – 0.10 | Marginal |
| 0.01 – 0.05 | Moderate |
| 0.001 – 0.01 | Strong |
| $< 0.001$ | Very strong |

# 15.7 One-Sided vs. Two-Sided Tests

## Two-Sided ($H_1$: $\mu \neq \mu_0$)

Used when deviation in **either** direction is important.

- Reject if $|T| > t_{\alpha/2,n-1}$
- p-value $= 2 \cdot P(T > |t_{\text{obs}}|)$

## One-Sided ($H_1$: $\mu > \mu_0$ or $H_1 : \mu < \mu_0$)

Used when only one direction of deviation matters.

- $H_1 : \mu > \mu_0$: Reject if $T > t_{\alpha,n-1}$
- $H_1 : \mu < \mu_0$: Reject if $T < -t_{\alpha,n-1}$

## When to Use Which

| Scenario | Test Type |
|---|---|
| Is the sensor biased (either direction)? | Two-sided |
| Does the new algorithm IMPROVE BER? | One-sided ($H_1$: BER $<$ BER$_{\text{old}}$) |
| Has noise power INCREASED? | One-sided ($H_1$: $\sigma^2 > \sigma_0^2$) |
| Did the process mean CHANGE? | Two-sided |

## 15.8 Tests for Means

### One-Sample Z-Test ($\sigma$ known)

Test: $H_0 : \mu = \mu_0$

Test statistic: $Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$

Under $H_0 : Z \sim N(0,\, 1)$

### One-Sample t-Test ($\sigma$ unknown)

Test: $H_0 : \mu = \mu_0$

Test statistic: $T = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}}$

Under $H_0 : T \sim t(n-1)$

### Two-Sample t-Test (comparing two means)

Test: $H_0 : \mu_1 = \mu_2$

Test statistic: $T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$

Approximate $df$ by Welch-Satterthwaite formula.

# 15.9 Test for Variance (Chi-Square Test)

## One-Sample Test

$H_0 : \sigma^2 = \sigma_0^2$

Test statistic: $\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$

Under $H_0 : \chi^2 \sim \chi^2(n-1)$

## Decision

- $H_1 : \sigma^2 > \sigma_0^2$: Reject if $\chi^2 > \chi^2_{\alpha,n-1}$
- $H_1 : \sigma^2 \neq \sigma_0^2$: Reject if $\chi^2 < \chi^2_{1-\alpha/2,n-1}$ or $\chi^2 > \chi^2_{\alpha/2,n-1}$

## 15.10 Step-by-Step Hypothesis Testing Procedure

1. **State hypotheses:** $H_0$ and $H_1$
2. **Choose significance level:** $\alpha$ (e.g., 0.05)
3. **Select test statistic:** $Z$, $t$, or $\chi^2$ depending on scenario
4. **Determine critical value(s)** or compute p-value
5. **Compute test statistic** from data
6. **Make decision:** Reject $H_0$ if test statistic in critical region (or $p < \alpha$)
7. **State conclusion** in engineering terms

## 15.11 Engineering Examples

**Example 1: Is a Sensor Biased?**

A sensor should read 0V with no input. 20 measurements give $\bar{X} = 0.012\,\text{V}$, $S = 0.03\,\text{V}$.
$H_0 : \mu = 0$ (no bias), $H_1 : \mu \neq 0$ (biased), $\alpha = 0.05$

$$T = \frac{0.012 - 0}{\frac{0.03}{\sqrt{20}}} = \frac{0.012}{0.00671} = 1.789$$

$t_{0.025,19} = 2.093$. Since $|T| = 1.789 < 2.093 \rightarrow$ **Fail to reject $H_0$.**
p-value $= 2 \cdot P(t(19) > 1.789) \approx 0.090$. Not significant at 5% level.
Conclusion: Insufficient evidence to conclude the sensor is biased.

**Example 2: Did Manufacturing Process Change?**

Before: $\mu_0 = 100\,\Omega$ (established). After modification: $n = 30$, $\bar{X} = 101.2\,\Omega$, $S = 3.5\,\Omega$.
$H_0 : \mu = 100$, $H_1 : \mu \neq 100$, $\alpha = 0.05$

$$T = \frac{101.2 - 100}{\frac{3.5}{\sqrt{30}}} = \frac{1.2}{0.639} = 1.878$$

$t_{0.025,29} = 2.045$. $|T| = 1.878 < 2.045 \rightarrow$ **Fail to reject $H_0$.**

**Example 3: Does New Algorithm Improve BER?**

Old algorithm: $\text{BER}_0 = 10^{-3}$. New algorithm tested: $n = 50000$ bits, 38 errors.

$$\text{BER}_{\text{new}} = \frac{38}{50000} = 7.6 \times 10^{-4}$$

$H_0 : p = 0.001$, $H_1 : p < 0.001$ (improvement), $\alpha = 0.05$

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.00076 - 0.001}{\sqrt{0.001 \times \frac{0.999}{50000}}} = -\frac{0.00024}{0.000141} = -1.70$$

p-value $= P(Z < -1.70) = 0.0446 < 0.05 \rightarrow$ **Reject $H_0$.**
Conclusion: Evidence supports that the new algorithm has lower BER.

**Example 4: Has Noise Power Increased?**

Historical: $\sigma_0^2 = 0.04\,\text{V}^2$. New measurements: $n = 25$, $S^2 = 0.058\,\text{V}^2$.
$H_0 : \sigma^2 = 0.04$, $H_1 : \sigma^2 > 0.04$, $\alpha = 0.05$

$$\chi^2 = \frac{(24)(0.058)}{0.04} = 34.8$$

$\chi^2_{0.05,24} = 36.415$. Since $34.8 < 36.415 \rightarrow$ **Fail to reject $H_0$.**

## 15.12 Practice Problems

**Problem 1**

A manufacturing spec requires mean resistance = 100Ω. A sample of 16 resistors gives $\bar{X} = 101.5\,\Omega$, $S = 3\,\Omega$. At $\alpha = 0.05$, is there evidence the process has drifted?

**Solution** $H_0 : \mu = 100$, $H_1 : \mu \neq 100$. $T = \frac{101.5-100}{\frac{3}{4}} = 2.0$. $t_{0.025,15} = 2.131$. $|T| = 2.0 < 2.131$ $\rightarrow$ Fail to reject. p-value $\approx 0.064 > 0.05$.

**Problem 2**

Average packet delay was 5ms. After network upgrade, 40 measurements give $\bar{X} = 4.6\,\text{ms}$, $S = 1.2\,\text{ms}$. Test if delay decreased at $\alpha = 0.01$.

**Solution** $H_0 : \mu = 5$, $H_1 : \mu < 5$. $T = \frac{4.6-5}{\frac{1.2}{\sqrt{40}}} = -\frac{0.4}{0.190} = -2.11$. $t_{0.01,39} \approx -2.426$. $T = -2.11 > -2.426 \rightarrow$ Fail to reject at 1% level. (Would reject at 5% since $t_{0.05,39} \approx -1.685$.)

**Problem 3**

Explain why "$p = 0.04$ means there is only a 4% chance $H_0$ is true" is WRONG.

**Solution** The p-value is $P(\text{data this extreme} \mid H_0 \text{ true})$, NOT $P(H_0 \text{ true} \mid \text{data})$. It conditions on $H_0$ being true and asks about data extremity. To get $P(H_0 \text{ true} \mid \text{data})$ would require Bayesian analysis with a prior. The p-value is a property of the data-generating process under $H_0$, not a posterior probability about $H_0$.

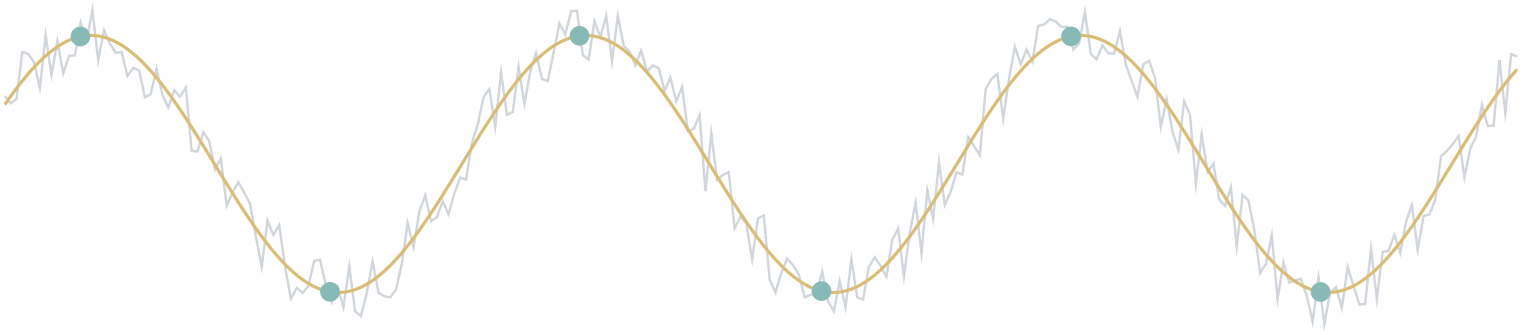
*Next Module: Module 16 — Engineering Statistical Applications*

Module 16

# Engineering Statistical Applications

کاربردهای آماری مهندسی

16



## In This Module

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**Learning Objectives**

This capstone module integrates ALL previous modules into real Electrical Engineering applications. Students will be able to:

1. Apply probabilistic modeling to communication systems
2. Analyze signal processing problems statistically
3. Model electronic component variations
4. Assess power system reliability
5. Handle measurement uncertainty in control systems
6. Connect probability/statistics to machine learning foundations

# 16.1 Communications

## 16.1.1 BER Estimation and Confidence Intervals

**Problem:** Estimate the Bit Error Rate of a communication link.

**Model:** Each bit is a Bernoulli trial with error probability $p$ (the BER).

- $n$ bits transmitted, $k$ errors observed
- $\hat{p} = \frac{k}{n}$ (MLE of BER)
- 95% CI: $\hat{p} \pm 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

## 16.1.2 AWGN Noise Modeling

Additive White Gaussian Noise is the fundamental noise model:

- **Distribution:** $N\left(0,\, \frac{N_0}{2}\right)$ per dimension
- **Why Gaussian?** CLT — thermal noise is sum of many electron contributions
- **Characterization:** Fully described by power spectral density $N_0$ or variance $\sigma^2 = N_0 B$ (B = bandwidth)

## 16.1.3 Rayleigh Fading Channel

**Model:** Channel coefficient $h = h_I + j \cdot h_Q$ where $h_I,\, h_Q \sim N(0,\, \sigma^2)$ independent.

- Envelope $|h| \sim$ Rayleigh$(\sigma)$
- Power $|h|^2 \sim$ Exponential $\left(\frac{1}{2\sigma^2}\right)$
- **Outage probability:** $P(|h|^2 < \gamma_{\text{th}}) = 1 - \exp\left(-\frac{\gamma_{\text{th}}}{2\sigma^2}\right)$

## 16.1.4 Rician Fading

When a line-of-sight (LOS) component exists:

- $h = h_{\text{LOS}} + h_{\text{scatter}}$
- $|h| \sim$ Rician$(\nu,\, \sigma)$ where $K = \frac{\nu^2}{2\sigma^2}$ is the K-factor

K-factor: $K = 0$ (Rayleigh) $\to$ $K = \infty$ (no fading, AWGN)

## 16.1.5 SNR Statistics

**Instantaneous SNR:** $\gamma = |h|^2 \times \frac{E_s}{N_0}$

For Rayleigh: $\gamma \sim$ Exponential$(\bar{\gamma})$ where $\bar{\gamma}$ = average SNR

**Average BER over fading:**

$$\bar{P}_e = \int_0^\infty P_e(\gamma) \cdot f_\gamma(\gamma)\, d\gamma$$

For BPSK over Rayleigh: $\bar{P}_e = \frac{1}{2}\left(1 - \sqrt{\frac{\bar{\gamma}}{1+\bar{\gamma}}}\right)$

# 16.2 Signal Processing

## 16.2.1 Measurement Noise Characterization

**Problem:** Characterize noise in measured signal $x(t) = s(t) + n(t)$.

**Approach:**

1. Collect noise-only samples (signal absent)
2. Estimate noise statistics: $\hat{\mu}_n$, $\hat{\sigma}_n^2$
3. Test for Gaussianity (histogram, QQ-plot)
4. Compute SNR: SNR $= \frac{P_{\text{signal}}}{\hat{\sigma}_n^2}$

## 16.2.2 Correlation-Based Detection

**Problem:** Detect a known signal in noise.

**Method:** Correlate received signal with known template. Under $H_0$ (noise only), the correlator output has known distribution $\rightarrow$ set threshold.

# 16.3 Electronics

## 16.3.1 Component Tolerance Modeling

**Problem:** Resistors have nominal value $R_0$ with manufacturing tolerance.

**Model:** $R \sim N(R_0, \sigma_R^2)$ where $\sigma_R$ depends on tolerance class.

For $\pm 5\%$ tolerance (99.7% within bounds): $3\sigma = 0.05R_0 \rightarrow \sigma = 0.0167R_0$

## 16.3.2 Monte Carlo Circuit Analysis

**Problem:** A voltage divider uses $R_1$ and $R_2$. What is the distribution of output voltage?

$$V_{\text{out}} = V_{\text{in}} \times \frac{R_2}{R_1 + R_2}$$

With $R_1$, $R_2$ random: $V_{\text{out}}$ is a random variable!

## 16.3.3 Measurement Uncertainty Propagation

If $Z = f(X, Y)$ and $X$, $Y$ have small uncertainties:

$$\sigma_Z^2 \approx \left(\frac{\partial f}{\partial X}\right)^2 \sigma_X^2 + \left(\frac{\partial f}{\partial Y}\right)^2 \sigma_Y^2$$

**Example:** Power $P = \frac{V^2}{R}$. $\sigma_P^2 \approx \frac{2\text{V}}{R}^2 \sigma_V^2 + \frac{V^2}{R^2}^2 \sigma_R^2$

# 16.4 Power Systems

## 16.4.1 Load Uncertainty

**Problem:** Peak load is uncertain. Model: $L \sim N(\mu_L, \sigma_L^2)$

Design question: What capacity $C$ ensures $P(\text{load} > \text{capacity}) < 0.01$?

$C = \mu_L + 2.326\sigma_L$ (for 1% exceedance probability)

## 16.4.2 System Reliability

**Series system:** All components must work.

$R_{\text{sys}} = R_1 \times R_2 \times \ldots \times R_n$ (for independent components)

**Parallel system:** At least one must work.

$R_{\text{sys}} = 1 - (1 - R_1)(1 - R_2)\ldots(1 - R_n)$

# 16.5 Control Systems

## 16.5.1 Sensor Noise Effects

**Problem:** A control system uses noisy sensor readings.

Measurement: $y(k) = x(k) + n(k)$ where $n \sim N(0, \sigma_n^2)$

Controller acts on $y(k) \rightarrow$ tracking error includes noise contribution.

**Solution approaches:**

- Averaging (low-pass filter): reduces noise by $\sqrt{n}$
- Kalman filter: optimal fusion of model prediction and measurement

## 16.5.2 Parameter Uncertainty

System model: $G(s) = \frac{K}{\tau s+1}$ where $K$ and $\tau$ are uncertain.

Monte Carlo approach: sample $K$ and $\tau$ from their distributions, simulate system response many times, compute statistics of output.

# 16.6 Machine Learning / AI Connections

## 16.6.1 Sampling and Data Collection

- Training data = sample from population
- Test data = independent sample for validation
- i.i.d. assumption underlies most ML theory

## 16.6.2 Estimation as Foundation of ML

- Linear regression = MLE under Gaussian noise assumption
- Neural network training = MLE (or MAP) for model parameters
- Loss function minimization = maximum likelihood principle

## 16.6.3 Probability Distributions in ML

- Generative models: learn $P(X)$ directly
- Gaussian Mixture Models: weighted sum of Gaussians
- Variational Autoencoders: latent variables with Gaussian priors

## 16.6.4 Bias-Variance Tradeoff

$$\text{Test Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Noise}$$

This is exactly the MSE decomposition from estimation theory (Module 13)!

- High bias: underfitting (too simple model)
- High variance: overfitting (too complex model)
- Optimal: balance between the two

## 16.6.5 Confidence and Uncertainty in Predictions

- Confidence intervals for model predictions
- Bayesian neural networks: posterior over weights $\rightarrow$ prediction uncertainty
- Calibration: predicted probabilities should match observed frequencies

## 16.7 Key Integration Points

| Module | Application in EE |
|---|---|
| 1. Probability | Event probabilities, fault diagnosis |
| 2. Random Variables | Modeling uncertain quantities |
| 3. Distributions | Choosing noise/fading/failure models |
| 4. Moments | Signal power, noise power, SNR |
| 5. Transformations | dB conversion, power from voltage |
| 6 – 7. Joint/Functions | Signal+noise, system combinations |
| 8. Conditional | Performance under different states |
| 9. Correlation | MIMO channels, sensor arrays |
| 10. Multiple RVs | Vector processing, beamforming |
| 11. CLT | Why noise is Gaussian, averaging |
| 12. Statistics | Estimating system parameters |
| 13. Estimation | MLE for channel/noise parameters |
| 14. Confidence | Quantifying measurement uncertainty |
| 15. Hypothesis | Quality control, performance comparison |

*End of course modules. The modules conclude with engineering applications and a summary of key concepts.*